THE FRACTIONAL DE GIORGI CONJECTURE IN DIMENSION EIGHT NEAR THE HALF-LAPLACIAN

Abstract. We prove that every bounded entire solution of $(-\Delta)^s u = u - u^3$ in $\mathbb{R}^8$ with $\partial_{x_8} u > 0$ is one-dimensional whenever $s_0 < s < 1/2$, for some $s_0 \in (0, 1/2)$. No limits in the monotone direction are prescribed. The geometric argument first establishes separation of neighboring stable interfaces at scale $\sqrt{1-2s}$. Averaged stability then passes to classical minimal laminations, and the stable Bernstein theorem in $\mathbb{R}^7$ gives uniform curvature bounds. Compactness of cone links and a homogeneous gap inequality yield regular stable-cone rigidity in ambient dimensions four through seven. This classifies the stable end profiles of the eight-dimensional solution and makes it a global minimizer. Its directed minimizing blow-downs are half-spaces, producing the oriented phase trapping required by improvement of flatness. The threshold is obtained by compactness and is not quantified.

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1. Introduction

The fractional De Giorgi problem asks whether monotonicity in one direction forces a bounded entire Allen–Cahn solution to have parallel hyperplanes as its level sets. We consider the equation

$$(-\Delta)^s u = u - u^3 \quad \text{in } \mathbb{R}^8, \qquad u \in L^\infty(\mathbb{R}^8), \qquad \partial_{x_8} u > 0, \qquad 0 < s < \frac{1}{2}. \tag{1.1}$$

The operator is normalized to have Fourier symbol $|\xi|^{2s}$.

1.1. Main results.

An increasing one-dimensional layer q_s is a solution of

$$(-\Delta)^s q_s = q_s - q_s^3 \quad \text{in } \mathbb{R}, \qquad q_s' > 0, \qquad \lim_{t\to-\infty} q_s(t) = -1, \quad \lim_{t\to+\infty} q_s(t) = 1.$$

Our main result is the following symmetry theorem.

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Theorem 1.1 (Eight-dimensional symmetry near the half-Laplacian)**.** *There is $s_0 \in (0,1/2)$ such that for every $s \in (s_0,1/2)$, every bounded classical solution of (1.1) is one-dimensional. More precisely, there are $e \in S^7$ with $e_8 > 0$, a constant $c \in \mathbb{R}$, and a strictly increasing one-dimensional layer $q_s : \mathbb{R} \to (-1,1)$ such that*

$$u(x) = q_s(e \cdot x - c) \qquad (x \in \mathbb{R}^8).$$

No limiting assumption as $x_8 \to \pm\infty$ is imposed. The threshold s_0 depends only on the dimension and on the fixed cubic potential, not on the solution.

The argument gives no numerical value of s_0 and does not address all exponents below $1/2$.

We also obtain the stable De Giorgi result in dimension seven. A classical solution v in $\mathbb{R}^7$ is stable if

$$\int_{\mathbb{R}^7} \phi(-\Delta)^s \phi \, dx + \int_{\mathbb{R}^7} (3v^2 - 1)\phi^2 \, dx \geq 0 \qquad \text{for every } \phi \in C_c^\infty(\mathbb{R}^7).$$

Theorem 1.2 (Stable De Giorgi classification in dimension seven near the half-Laplacian)**.** *There is $s_0 \in (0,1/2)$ such that, for every $s \in (s_0,1/2)$, every bounded classical solution $v : \mathbb{R}^7 \to \mathbb{R}$ of*

$$(-\Delta)^s v = v - v^3 \qquad \text{in } \mathbb{R}^7$$

that is stable is either the constant 1, the constant -1, or an increasing one-dimensional layer. More precisely, in the nonconstant case there are $e \in S^6$, $c \in \mathbb{R}$, and a strictly increasing one-dimensional layer $q_s : \mathbb{R} \to (-1,1)$ such that

$$v(x) = q_s(e \cdot x - c) \qquad (x \in \mathbb{R}^7).$$

No monotonicity or limiting assumption is imposed on v. The threshold depends only on the dimension and the fixed cubic potential, not on the solution.

By enlarging the two thresholds if necessary, we may use a common $s_0 \in (0,1/2)$ in Theorems 1.1 and 1.2.

The relevant geometric order is

$$p = 2s, \qquad \sigma = 1 - p.$$

Thus the limit $s \uparrow 1/2$ is the local-perimeter limit $p \uparrow 1$. A cone is a measurable phase invariant under positive dilations. It is nontrivial when both phases have positive measure. Regularity means that the entire genuine boundary, taken from the density-one representative of the phase, is an embedded C^2 hypersurface away from the vertex, with the phase occupying one side locally. Stationarity and stability refer to zero first variation and nonnegative second variation of the p-perimeter under smooth compactly supported ambient flows away from the vertex. Here the perimeter kernel is $|X - Y|^{-m-p}$ in ambient dimension m; its order p is twice the exponent s in the equation. The precise conventions are given in Definitions 2.2 and 2.4. Our geometric result is the following near-local rigidity theorem.

Theorem 1.3 (Regular stable cone rigidity near the local limit)**.** *For every $m \in \{4,5,6,7\}$ there is $p_m \in (0,1)$ such that every nontrivial regular stationary stable p-minimal cone in $\mathbb{R}^m$ is a half-space, up to a null set, whenever $p_m < p < 1$.*

Nonlocal minimal surfaces provide the interface problem [3]. The stable-solution theory of Cabré–Cinti–Serra connects phase-transition blow-downs to stable nonlocal cones and reduces stable solution classification to regular stable-cone rigidity [2]. Near-local rigidity in dimensions three and four was established in [1, 6]. We develop the geometric argument through ambient dimension seven, the dimension needed to classify end profiles of (1.1). Throughout this argument, regularity refers to the entire genuine phase boundary away from the cone vertex; no assertion about arbitrary singular stable cones is required.

1.2. **Outline of the paper.** The proof is developed in the order of its prerequisites. Theorem 1.3 is proved in Section 6. Section 7 first proves Theorem 1.2 and then uses it to classify the end profiles; Theorem 1.1 is proved at the end of that section.

(i) Section 2 fixes the phase conventions, perimeter variations and local-limit normalization, including the signed exterior flux used in the subsequent estimates.
(ii) Section 3 constructs complete clipped graph charts and proves the graph equation, fixed-source bounds and regularity estimates under a critical-tube hypothesis.
(iii) Section 4 derives the matching adjacent-gap equation and stability inequalities. Their inverse-moment consequence gives critical sheet separation without an a priori bound on the number of sheets.
(iv) Section 5 passes averaged packet stability to classical minimal leaves in rescaled sequences with bounded curvature. Their complete two-sided stable covers allow the stable Bernstein theorem to exclude curvature blow-up and give a uniform curvature bound.
(v) Section 6 obtains compact spherical links, controls their heights and multiplicity, and excludes multiple sheets by the homogeneous gap inequality. This proves the cone rigidity theorem.
(vi) Section 7 proves stable-solution classification in dimension seven, applies it to the end profiles to obtain minimization, classifies directed blow-downs, and derives oriented phase trapping. Improvement of flatness then completes the eight-dimensional symmetry theorem.

1.3. **The forward geometric argument.** The geometric estimates must remain uniform when the number of nearby interfaces is unbounded. Lemma 2.7 combines the exterior Dirichlet and potential interactions into a signed normal flux with a volume representation, so their control requires no area bound. The complete clipped charts of Lemma 3.3 retain every interface in the interior region. In these charts, the phase outside a tube supplies the fixed source estimated in Lemma 3.4. The graph regularity estimates in Section 3 are conditional on that tube and on a local curvature bound.

Section 4 establishes the required tube width. Stationarity and stability give adjacent-gap inequalities with a matching positive coefficient. Lemma 4.5 then permits inverse-moment exponents $2 < q < 4 + 2\sqrt{2}$; in horizontal dimension $d \le 6$, one can choose $q > d$. The point-selection argument in Theorem 4.8 turns a failure of $\sqrt{\sigma}$ separation into a positive entire normalized gap, which the inverse-moment estimate excludes. Point selection supplies the critical tube before the conditional graph estimates are used. The constants depend on the local curvature bound, but not on the number of sheets.

For sequences with bounded curvature, separation now supplies uniform graph regularity and classical minimal limits. Prescribing one scalar normal speed on all sheets and averaging by the packet degree retains the classical Jacobi inequality, as proved in Lemma 5.2. On expanding domains, Proposition 5.4 produces complete two-sided stable covers of the limiting leaves. These hypotheses are obtained in a curvature blow-up by point selection and rescaling; the uniform curvature estimate is the conclusion of this argument. Theorem 5.5 rules out the normalized nonflat limit using Hong–Li–Wang's stable Bernstein theorem in $\mathbb{R}^7$ and its lower-dimensional corollary [13, Theorem 1.1 and Corollary 1.2].

The resulting annular curvature bounds give compact spherical links by Lemma 6.1. Averaged stability passes to the classical cone over the limiting link; Lemma 6.2 forces that link to be an equator. This yields the equatorial convergence in Proposition 6.3, without yet bounding multiplicity. Proposition 6.5 places all sheets in one common rotated frame. On the unit horizontal sphere their heights are $O(\sqrt{\sigma})$ and their adjacent gaps are bounded below by $c\sqrt{\sigma}$; hence their number is bounded. A remaining multiple-sheet sequence would give a positive one-homogeneous gap, excluded by Lemma 6.6 for $3 \le d \le 6$. Finally, Lemma 6.7 makes the single stationary conical graph planar. The complete cone argument appears in Section 6.

1.4. **From end profiles to the eight-dimensional solution.** Vertical monotonicity defines the end profiles $v^{\pm}(x') = \lim_{t\to\pm\infty} u(x',t)$, for $x' \in \mathbb{R}^7$. Lemma 7.7 proves that these are bounded classical stable solutions in seven dimensions: local compactness passes the equation to translated limits, and long vertical cutoffs pass stability to the lower dimension. Theorem 1.2 therefore makes each profile a constant well or an increasing one-dimensional layer. These are precisely the end profiles required by Cabré–Cinti–Serra's criterion [2, Proposition 7.9], whose application in Corollary 7.9 makes u a global energy minimizer.

Lemma 7.13 supplies a nontrivial minimizing conical phase limit along a subsequence of every prescribed sequence of dilation scales $R_j \to \infty$. Vertical monotonicity survives as $E + te_8 \subset E$ for $t > 0$. Near $p = 1$, minimizing-cone rigidity in dimension seven gives punctured regularity for these eight-dimensional cones [4, 11]. The directed-cone classification in Lemma 7.12 then makes each limit a half-space. Its proof treats boundary poles and finite vertical fibers separately; the latter case uses the measurable-subgraph theorem of Farina–Valdinoci [11, Theorem 1.3].

Improvement of flatness requires oriented trapping at every sufficiently large radius. Lemma 7.15 obtains this by excluding any prescribed sequence of bad radii, using phase convergence for both u and $-u$. For a fixed admissible well-neighborhood parameter, it supplies at each large radius R a unit normal e_R and a slab $|e_R \cdot x| \le a(R)R$, with $a(R)$ nonincreasing and tending to zero. The corresponding negative and positive well neighborhoods occupy the two outer sides in B_R. The normal e_R may depend on R. This is the hypothesis of Theorem 7.14, the Dipierro–Serra–Valdinoci improvement-of-flatness theorem [9]. The final proof in Section 7 obtains one-dimensionality and uses inherited stability to identify the two end limits of the increasing profile as -1 and 1.

2. Variational foundations and the local limit

The geometric estimates use the perimeter order $p \in (0,1)$, $\sigma = 1 - p$, ambient dimension m, and horizontal dimension $d = m - 1$. Their signs and local-limit constants are fixed here before any graph comparison is made. Phases are treated by their density-one representatives, so every regular interface separates an occupied side from an unoccupied side.

2.1. Phases and genuine interfaces.

Definition 2.1 (Geometric phases, curvatures and normal forms)**.** Let $m \ge 2$, $d = m-1$, $0 < p < 1$, $\sigma = 1-p$, and $c_d = |S^{d-1}|/d = |B_1^d|$. Write B_r^d for a horizontal Euclidean ball and $Q_r = B_r^d \times (-r, r)$. A phase is a measurable subset $E \subset \mathbb{R}^m$, identified up to Lebesgue null sets. For all topological and geometric constructions, use its canonical density-one representative

$$E^{(1)} = \left\{ X : \lim_{r\downarrow 0} \frac{|E \cap B_r(X)|}{|B_r|} = 1 \right\}.$$

Write E for this representative when taking a topological boundary or geometric data. A regular embedded boundary in an open set means that its entire topological boundary there is a relatively closed embedded C^2 hypersurface without hypersurface boundary there, and that locally $E^{(1)}$ is exactly one of the two open sides. Equivalent regular phases therefore have the same boundary, outward normal and curvatures. Complementation means complementation of the phase class followed by taking its density-one representative; on a regular patch this is the opposite open side, with the same boundary and reversed outward normal. Dilation and translation act on the phase class and then on this canonical representative. Its outward normal is ν_E, its second fundamental form is A, and $|A|$ is its Euclidean tensor norm. The classical convention is $H_{\mathrm{cl}} = \operatorname{div}_{\partial E} \nu_E$.

For disjoint measurable sets, and for an open set U, define

$$L_p(A,B) = \int_A \int_B |X - Y|^{-m-p}\, dY\, dX, \qquad \mathrm{Per}_p(E;U) = L_p(E \cap U, E^c) + L_p(E \setminus U, E^c \cap U).$$

At a regular boundary point define nonlocal curvature, whenever the indicated principal value exists, by

$$H_p[E](X) = \lim_{\delta\downarrow 0} \int_{\mathbb{R}^m\setminus B_\delta(X)} (\chi_{E^c} - \chi_E)(Y)|X-Y|^{-m-p}\,dY.$$

For a globally defined graph h, put

$$B_h = \{(x,t) : t < h(x)\}, \quad X_h(x) = (x, h(x)),$$
$$F_h(x) = \sigma H_p[B_h](X_h(x)), \qquad \mathcal{M}(h) = \operatorname{div} \frac{\nabla h}{\sqrt{1+|\nabla h|^2}}.$$

For a compactly supported scalar normal test ϕ on $\Sigma = \partial E$, define

$$Q_p[\phi] = \mathcal{Q}_{p,E}[\phi] = \frac{1}{2}\iint_{\Sigma\times\Sigma} \frac{(\phi(X)-\phi(Y))^2}{|X-Y|^{m+p}}\,d\mathcal{H}^d_Y\,d\mathcal{H}^d_X - \int_\Sigma \phi(X)^2 \int_\Sigma \frac{1-\nu_E(X)\cdot\nu_E(Y)}{|X-Y|^{m+p}}\,d\mathcal{H}^d_Y\,d\mathcal{H}^d_X.$$

The integrals are taken separately when they converge. If the exterior boundary is not regular, the remote ordered Dirichlet and potential terms are first combined: where $\phi(Y) = 0$ their sum is the pairing of $\phi(X)^2\nu_E(X)$ with the remote vector flux. This flux is defined distributionally by $-D\chi_E$, namely

$$\int_{\partial E} f(Y)\nu_E(Y)\,d\mathcal{H}^d_Y := \int_E \nabla_Y f(Y)\,dY$$

for smooth compactly supported functions f avoiding the singular evaluation point. Compact variations are understood through their finite localized perimeter differences: a flow supported in a compact subset of U may be evaluated in any smaller regular domain containing that support. This convention also defines variations when unrelated contributions to $\mathrm{Per}_p(E;U)$ are infinite.

Proof. A phase class must determine the geometric objects used in its variations. Null changes leave every quantity $|E\cap B_r(X)|$ unchanged, so they also leave $E^{(1)}$ unchanged. Testing density along rational radii proves measurability; Lebesgue differentiation gives $E^{(1)} = E$ almost everywhere. For a local open graph side, interior points have density one and exterior points density zero. At a point of the graph, rescaling a ball and using the tangent plane gives density $1/2$, because the relative volume between the graph and its tangent half-space tends to zero. Thus the canonical representative recovers precisely the open side, its boundary and its orientation. In particular all geometric data above are independent of null modifications.

Density is also covariant under similarities. For $\lambda > 0$, an orthogonal map O, $a \in \mathbb{R}^m$ and $X \in \mathbb{R}^m$, change of variables in balls gives

$$\frac{|(\lambda OE + a)\cap B_r(\lambda OX + a)|}{|B_r|} = \frac{|E\cap B_{r/\lambda}(X)|}{|B_{r/\lambda}|}.$$

Consequently $(\lambda OE + a)^{(1)} = \lambda OE^{(1)} + a$. In particular, dilation invariance of a phase class up to null sets gives exact dilation invariance of its canonical representative.

Nonnegative measurable integrands define L_p and Per_p, including their possible value $+\infty$. Both H_p and the graph quantity F_h are used only where the curvature principal value exists. Likewise the remote pairing for a compact smooth f is already a finite, null-invariant volume integral. It remains to extend this pairing to the noncompact kernel that occurs after localization.

For the noncompact remote kernel, fix a compact set S of evaluation points in the regular region and a smooth local cutoff η, compactly supported there and equal to one near S. Put

$$f_X(Y) = (1-\eta(Y))|X-Y|^{-m-p}, \qquad \mathcal{F}(X) = \int_E \nabla_Y f_X(Y)\,dY.$$

For each $X \in S$, f_X vanishes near X, has a bounded gradient on bounded regions, and satisfies $|\nabla f_X(Y)| \le C_X |Y|^{-m-p-1}$ at infinity. Hence its gradient is integrable and the displayed volume extension is finite and null-invariant. Changing the local cutoff from η_1 to η_2 changes f_X by $(\eta_1 - \eta_2)|X - Y|^{-m-p}$, a compact test avoiding X; the remote term changes by that compact pairing and the corresponding local term by its negative. Thus the full paired form is independent of its local decomposition. This construction uses no unsigned exterior area integral.

□

Definition 2.2 (Stationary and stable phases)**.** For the perimeter in Definition 2.1, stationarity in U means that every smooth flow compactly supported in U has zero first perimeter derivative. A stationary phase is stable in U when all such flows have nonnegative second derivative. The flows are arbitrary compact variations; no volume constraint is imposed and they need not be translations or rotations.

Definition 2.3 (Genuine local graph boundaries)**.** A canonical phase as in Definition 2.1 has a genuine $C^{k,\alpha}$ boundary near X, where $k \ge 1$ and $0 < \alpha \le 1$, if a rigid coordinate change identifies its entire local boundary with one $C^{k,\alpha}$ graph and identifies the phase with exactly one of that graph's two open sides. The same graph-side condition defines genuine C^k boundaries. A genuine $C^{1,1}$ boundary has a C^1 graph with locally Lipschitz gradient. A compact uniformly $C^{1,1}$ family of patches has common positive chart radius and common bounds for the gradients and their Lipschitz constants. These local notions retain the global relative-closedness and embeddedness convention of Definition 2.1 whenever a regular boundary is asserted.

Definition 2.4 (Regular stationary stable cones)**.** Using phases and genuine regular boundaries of 2.1, a regular cone $C \subset \mathbb{R}^m$ satisfies $tC = C$ for all $t > 0$ and has a relatively closed embedded C^2 boundary away from zero, locally separating its two phases. Its entire spherical link $\partial C \cap S^{m-1}$ is then an embedded compact C^2 hypersurface. Conversely this link regularity, with genuine local phase separation, describes the same regularity of the cone away from zero. It is nontrivial if both phases have positive measure. Stationarity and stability, in the sense of 2.2, concern compact variations supported in $\mathbb{R}^m \setminus \{0\}$.

Proof. Dilation invariance puts the radial line through every nonzero boundary point in the boundary. Its radial vector is tangent to the boundary and transverse to the unit sphere. The implicit-function theorem therefore intersects the regular C^2 boundary with the sphere in an embedded C^2 hypersurface. This is the entire link: relative closedness away from zero makes it closed in the sphere, hence compact. Conversely, for a compact embedded link, the map

$$(r, \omega) \longmapsto r\omega, \qquad r > 0,$$

is a C^2 embedding onto the cone boundary away from its vertex. Its inverse is $X \mapsto (|X|, X/|X|)$, and its differential $(a, v) \mapsto a\omega + rv$ is injective because $v \perp \omega$ and $r > 0$. These maps provide the C^2 charts. The phase-side condition passes through these charts in either direction. It is needed: a topological interface created by deleting a null set does not define a regular phase boundary. □

Definition 2.5 (Globally minimizing fractional phases)**.** A phase E is globally p-minimal if the perimeter of Definition 2.1 satisfies $\mathrm{Per}_p(E; U) \le \mathrm{Per}_p(F; U)$ for every bounded open U and every phase F that agrees with E outside U. A minimizing cone is a globally p-minimal phase invariant, up to null sets, under every positive dilation. It is nontrivial when both phases have positive measure. Half-space conclusions for directed cones are always equalities up to null sets.

2.2. **Curvature and compact normal variations.**

Lemma 2.6 (Local curvature cancellation)**.** *At a genuine $C^{1,1}$ phase boundary point the principal value H_p of 2.1 exists for $0 < p < 1$. On compact uniformly $C^{1,1}$ boundary patches the diagonal truncation error is bounded by $C\delta^{1-p}/(1-p)$, uniformly at the evaluation points. The tail beyond radius R is bounded by $|S^{m-1}|R^{-p}/p$.*

Proof. The cancellation is local. At the evaluation point, choose tangent coordinates and write the genuine boundary of Definitions 2.1 and 2.3 as $t = h(x)$, with $h(0) = 0$, $Dh(0) = 0$ and gradient Lipschitz constant L. On a convex chart,

$$h(x) = \int_0^1 Dh(tx) \cdot x \, dt, \qquad |h(x)| \le \frac{L}{2}|x|^2.$$

The matching tangent half-space has zero signed curvature integral on each centered annulus. By genuine one-sided occupancy, the two phases can differ in a small ball only in the strip $|t| \le (L/2)|x|^2$; their signed characteristic functions differ by at most two. With $r = |x|$, the omitted diagonal contribution is at most

$$C \int_0^\delta r^{d-1} r^2 r^{-m-p} \, dr = C \int_0^\delta r^{-p} \, dr = \frac{C\delta^{1-p}}{1-p}.$$

The horizontal projection of an ambient ball of radius δ lies in $\{r < \delta\}$, so this also controls the stated ambient truncation.

Uniformity requires a common tangent chart. In an original chart about x_0, write

$$Z(z) = (z, h(x_0 + z) - h(x_0)), \qquad |z| < \rho,$$

where $|Dh| \le M$ and $\mathrm{Lip}(Dh) \le L$. Restrict a finite covering to inner charts so that ρ is a common positive margin. Let P project orthogonally onto the tangent plane at $Z(0)$, identify that plane with $\mathbb{R}^d$, and put $F = PZ$, $D_0 = DF(0)$. Since $DZ(0)w$ is tangent,

$$|D_0 w| = \sqrt{|w|^2 + |Dh(x_0)w|^2} \ge |w|, \qquad \|D_0^{-1}\| \le 1, \qquad \|DF(z) - D_0\| \le L|z|.$$

Take $r \le \rho/2$ with $Lr \le 1/4$, omitting the latter restriction if $L = 0$. In $F(z) = D_0 z + R(z)$, the remainder has Lipschitz constant at most $1/4$. Hence

$$|F(z) - F(w)| \ge \tfrac{3}{4}|z - w|, \qquad |R(z)| \le \tfrac{L}{2}|z|^2.$$

For $|y| < r/2$, the map $z \mapsto D_0^{-1}(y - R(z))$ is a contraction of $\overline{B}_r$ into itself. Its fixed point defines $F^{-1}(y)$; the inverse has derivative norm at most $4/3$. Applying $A^{-1} - B^{-1} = A^{-1}(B - A)B^{-1}$ to its derivatives gives

$$\mathrm{Lip}(D(F^{-1})) \le (4/3)^3 L.$$

For the tangent unit normal n, set $G(z) = n \cdot Z(z)$. Then $DG(0) = 0$, $\|DG\| \le \sqrt{1 + M^2}$, and $\mathrm{Lip}(DG) \le L$. The height $g = G \circ F^{-1}$ satisfies

$$Dg(0) = 0, \qquad \mathrm{Lip}(Dg) \le (4/3)^2 L + \sqrt{1 + M^2}(4/3)^3 L.$$

The regraph remains in the original patch because it is exactly $Z(F^{-1}(y))$. A further common ambient restriction, depending only on ρ, r, M, excludes the outside boundary and preserves one-sided occupancy. Thus the cancellation radius and discrepancy constant are uniform.

Finally, the absolute value of the signed phase is at most one. Polar integration outside B_R gives

$$\int_{|Y|>R} |Y|^{-m-p} \, dY = \frac{|S^{m-1}|}{p} R^{-p}.$$

The diagonal and tail bounds prove existence of the principal value and both errors. □

Lemma 2.7 (Combined remote normal flux)**.** *For the phases and quadratic form of 2.1, the remote contribution at an evaluation point X is a signed vector flux, not an unsigned exterior area integral. If χ is a smooth compactly supported cutoff whose support avoids X, then*

$$\int_{\partial E} \chi(Y) \nu_E(Y) |X - Y|^{-m-p} \, d\mathcal{H}^d_Y = \int_E \nabla_Y \big(\chi(Y)|X - Y|^{-m-p}\big) \, dY.$$

For rough exterior phases the left side is the distributional pairing defined in 2.1, and the right side is absolutely defined and bounded by its integral over all of $\mathbb{R}^m$. More generally, let U be an

open set in which E has regular boundary, and fix a compact set $S \subset U$ of evaluation points. Choose $\eta \in C_c^\infty(U)$ equal to one near S. Set $f_X(Y) = (1-\eta(Y))|X-Y|^{-m-p}$. The noncompact remote flux is

$$\mathcal{F}(X) = \int_E \nabla_Y f_X(Y)\, dY, \qquad \sup_{X\in S} |\mathcal{F}(X)| < \infty.$$

For any smooth outer cutoffs $0 \le \eta_R \le 1$, equal to one on B_R and zero outside B_{2R}, with $|\nabla \eta_R| \le C/R$, one has, for all sufficiently large R,

$$\sup_{X\in S} \left| \int_E \nabla_Y (f_X \eta_R)(Y)\, dY - \mathcal{F}(X) \right| \le C_S R^{-1-p}.$$

These bounds depend only on the fixed support, local cutoff, m, p and outer-cutoff gradient bound, independently of E. In particular every such exhaustion gives the same remote pairing.

For a regular cone of 2.4 and its translates and positive dilates, all remote surface integrals in the displayed normal quadratic form converge separately; no uniform bound over different cones is asserted.

Proof. The cancellation in Definition 2.1 fixes what must be estimated. For a remote point with $\phi(Y) = 0$, the two ordered Dirichlet terms supply $\phi(X)^2 K(X-Y)$, whereas the potential subtracts $\phi(X)^2(1-\nu_E(X)\cdot\nu_E(Y))K(X-Y)$. Their sum is

$$\phi(X)^2 \nu_E(X)\cdot\nu_E(Y) K(X-Y).$$

For a regular boundary, componentwise divergence applied to $f(Y) = \chi(Y)K(X-Y)$ gives the identity with positive sign. Distributionally this is the identity $D\chi_E = -\nu_E \mathcal{H}^d \lfloor \partial E$. For an arbitrary measurable phase the volume side defines the pairing directly and is bounded by $\int_{\mathbb{R}^m} |\nabla(\chi K)|$, finite since the compact support avoids X.

The noncompact remote kernel pairing is the unique cutoff limit of the compact pairings already defined. Fix a compact set S of evaluation points in the regular region and a smooth local cutoff η equal to one on a neighborhood of S, with compact support in that region. For $X \in S$ put

$$K_X(Y) = |X-Y|^{-m-p}, \qquad f_X(Y) = (1-\eta(Y))K_X(Y).$$

This function is smooth and vanishes near X; its gradient is integrable over all of $\mathbb{R}^m$. Since $\eta = 1$ on a neighborhood of the compact set S, the removed diagonal has uniform positive separation. The gradient is uniformly bounded on bounded regions and has a common integrable majorant at infinity. Thus $\sup_{X\in S} \int_{\mathbb{R}^m} |\nabla f_X| < \infty$, which gives the claimed uniform bound for the volume flux, independently of E. Choose smooth outer cutoffs η_R equal to one on B_R, zero outside B_{2R}, with $0 \le \eta_R \le 1$ and $|\nabla \eta_R| \le C/R$. The functions $f_X \eta_R$ are smooth compactly supported tests avoiding X. Define the remote vector pairing by

$$\lim_{R\to\infty} \int_E \nabla_Y (f_X\eta_R)(Y)\, dY = \int_E \nabla_Y f_X(Y)\, dY.$$

To prove the equality and cutoff independence, take R larger than twice a ball containing $S \cup \operatorname{supp}\eta$. On $|Y| > R$, uniformly for $X \in S$, $|f_X(Y)| \le C_S|Y|^{-m-p}$ and $|\nabla_Y f_X(Y)| \le C_S|Y|^{-m-p-1}$. Hence

$$\|\nabla(f_X\eta_R) - \nabla f_X\|_{L^1(\mathbb{R}^m)} \le \int_{|Y|>R} |\nabla f_X(Y)|\, dY + \frac{C}{R}\int_{R<|Y|<2R} |f_X(Y)|\, dY \le C_S R^{-p-1}.$$

Both radial estimates converge for every $p > 0$. The same calculation holds for any family of outer cutoffs with comparable radii and a uniform C/R gradient bound, so all such exhaustions give the same limit. Multiplication by χ_E, bounded by one, preserves this L^1 convergence; no exterior

surface regularity or area bound is used. The limit is null-invariant and therefore defines a unique extension of the remote flux to this localized kernel.

For the cones of Definition 2.4, the compact C^2 link has finite area. Polar parameterization bounds boundary area in radius-R annuli by CR^d. At infinity the remote surface integral is therefore bounded by

$$C\int_R^\infty r^{d-1}r^{-m-p}\,dr = C\int_R^\infty r^{-2-p}\,dr < \infty.$$

Near the vertex, evaluation points in a compact set away from zero have bounded kernel, and the remaining area is bounded by $C\int_0^a r^{d-1}\,dr$. Translations and positive dilations leave convergence intact. These separate-convergence constants need not be uniform across different links. □

Lemma 2.8 (Compact normal tests and approximation)**.** *On a regular embedded boundary of 2.1 in U, every compactly supported C^1 scalar normal test extends to the normal trace of a compactly supported C^1 ambient vector field in U. It is approximable by normal traces of smooth compactly supported fields, with convergence in C^1 on the boundary. The normal quadratic form of 2.1, with the remote interpretation of 2.7, is continuous under this approximation for each fixed $p < 1$.*

Proof. The regular boundary in Definition 2.1 has C^1 normals and local tubular projections. Cover the compact trace support by finitely many tubular charts. Extend $\phi\nu_E$ constantly along each normal fiber, multiply by compact cutoffs, and combine with a C^1 partition of unity. This gives a compact C^1 ambient field in U. Disjoint sheet supports on the compact set have positive separation and can be treated in separate neighborhoods. Mollification with sufficiently small radii retains one fixed compact ambient support and converges in C^1. Since ν_E is C^1, the normal traces converge in the same topology.

On a local sheet the squared trace differences and squared normal differences are bounded by $C|X-Y|^2$. Surface coordinates give the integrable radial majorant Cr^{-p} for each fixed $p < 1$, so dominated convergence handles the diagonal and compact off-diagonal terms.

For the exterior, combine the terms as in Lemma 2.7. Let S be a common compact trace support and choose $\eta \in C_c^\infty(U)$ equal to one near S. Lemma 2.7, applied to $f_X(Y) = (1-\eta(Y))|X-Y|^{-m-p}$, gives uniform convergence of the outer-cutoff fluxes to

$$\mathcal{F}(X) = \int_E \nabla_Y f_X(Y)\,dY, \qquad X \in S,$$

with error at most $C_S R^{-1-p}$. The same lemma gives $\sup_{X\in S}|\mathcal{F}(X)| < \infty$. A finite embedded-chart cover gives $\mathcal{H}^d(S) < \infty$, so the difference of the remote quadratic pairings is at most

$$\mathcal{H}^d(S)\|\phi_j^2 - \phi^2\|_\infty \sup_{X\in S}|\mathcal{F}(X)| \longrightarrow 0.$$

The existing uniform cutoff error permits taking this trace limit before or after removing the cutoff. Together with local domination, this proves continuity of the full paired form for the fixed surface. □

Proposition 2.9 (First variation and stationary curvature equation)**.** *For a genuine $C^{1,1}$ phase boundary in U (in particular a regular embedded C^2 boundary of 2.1), the first perimeter derivative along a compact smooth flow with normal speed ϕ is*

$$\frac{d}{dt}\operatorname{Per}_p(E_t;U)\Big|_{t=0} = \int_{\partial E}\phi(X)H_p[E](X)\,d\mathcal{H}^d_X.$$

Here the perimeter and curvature are those of 2.1. For every bounded smooth $W \Subset U$, the localized perimeter $\operatorname{Per}_p(E;W)$ is finite; a flow supported in W is evaluated by its finite localized perimeter difference. Consequently stationarity in the sense of 2.2 is equivalent to $H_p[E] = 0$ on $\partial E \cap U$.

Proof. Fix a smooth flow Ψ_t supported in $K \Subset U$, write $E_t = \Psi_t(E)$, and choose a bounded smooth domain W with $K \Subset W \Subset U$. We work with the finite localized differences from Definition 2.1. Near $\overline{W}$, finitely many genuine $C^{1,1}$ charts in Definition 2.3 cover the boundary. For small $|t|$, bounded first and second derivatives of the flow and its inverse, positive Jacobians and the implicit graph theorem preserve common inner radii and gradient-Lipschitz bounds. Surface Jacobians and normal velocities are bounded and continuous. All nonzero velocities are supported on patches with common compact support $S \Subset W$ and uniformly bounded areas.

We next prove finiteness for every bounded smooth $W \Subset U$. Put $W_a = \{X : \operatorname{dist}(X, W) < a\}$ and choose $\rho_0 > 0$ such that $\overline{W_{2\rho_0}} \Subset U$. The set $\Gamma_W = \partial E \cap \overline{W_{\rho_0}}$ is compact and has a finite cover by inner genuine $C^{1,1}$ graph charts, each contained in a larger graph chart. These charts give constants $C, r_0 > 0$ such that

$$|\{X : \operatorname{dist}(X, \Gamma_W) \le r\}| \le Cr, \qquad 0 < r < r_0.$$

Indeed, in each chart the graph has a bounded Lipschitz constant M on a bounded horizontal base. For small r, every point within distance r of its inner graph patch has horizontal projection in the larger base and lies in the vertical strip $|X_m - h(X')| \le (1 + M)r$. The strip has volume at most Cr; summing the finite cover proves the bound. If Γ_W is empty the bound is immediate. Choose $0 < \rho < \min\{\rho_0, r_0\}$. If $X \in W$, $|z| < \rho$ and $\chi_E(X + z) \neq \chi_E(X)$, the segment from X to $X + z$ meets the topological boundary inside W_{ρ_0}. Thus X belongs to the $|z|$-tube of Γ_W. With

$$A(z) = \int_W |\chi_E(X + z) - \chi_E(X)|\, dX,$$

we obtain $A(z) \le C|z|$ for $|z| < \rho$, and $A(z) \le |W|$ for all z. Tonelli's theorem and the change of variables $Y = X + z$ give, as identities of nonnegative extended-real integrals,

$$\begin{aligned} I &:= \int_{\mathbb{R}^m} |z|^{-m-p} A(z)\, dz \\ &= L_p(E \cap W, E^c) + L_p(E^c \cap W, E) \\ &= \operatorname{Per}_p(E; W) + L_p(E \cap W, E^c \cap W). \end{aligned}$$

Consequently

$$\operatorname{Per}_p(E; W) \le I \le C|S^{m-1}|\frac{\rho^{1-p}}{1-p} + |W||S^{m-1}|\frac{\rho^{-p}}{p} < \infty.$$

Only the regular boundary in the fixed neighborhood of W enters the small-translation bound. The second term uses only the bounded characteristic functions and so requires no exterior regularity. The transported charts have uniform bounds of the same type for small $|t|$, so the argument also gives $\operatorname{Per}_p(E_t; W) < \infty$.

Fix smooth profiles $0 \le \chi, \eta \le 1$ on $[0, \infty)$, with $\chi = 0$ on $[0, 1]$, $\chi = 1$ on $[2, \infty)$, $\eta = 1$ on $[0, 1]$ and $\eta = 0$ on $[2, \infty)$. Set

$$K_{\delta,R}(X, Y) = \chi(|X - Y|/\delta)\eta(|X - Y|/R)|X - Y|^{-m-p}.$$

These kernels are smooth, radial and even, bounded by the original kernel, and converge to it away from the diagonal as $\delta \downarrow 0$, $R \uparrow \infty$.

For this smooth integrable kernel, transport gives

$$\operatorname{Per}_{\delta,R}(E_t; W) - \operatorname{Per}_{\delta,R}(E_0; W) = \int_0^t \int_{\partial E_\tau} \phi_\tau H_{\delta,R}[E_\tau]\, d\mathcal{H}^d\, d\tau.$$

Indeed, changing variables through Ψ_τ and differentiating the Jacobian gives, for a compact smooth g,

$$\frac{d}{d\tau}\int_{E_\tau} g = \int_{E_\tau} \operatorname{div}(gV_\tau) = \int_{\partial E_\tau} g\phi_\tau \, d\mathcal{H}^d.$$

The divergence theorem here follows in each graph chart by vertical slicing and the fundamental theorem of calculus, then a partition of unity. Only the support of the velocity is involved. Fubini therefore permits differentiation of both phase variables: the moving E_τ side contributes $\int \phi_\tau \int_{E_\tau^c} K_{\delta,R}$, and the moving complement contributes $-\int \phi_\tau \int_{E_\tau} K_{\delta,R}$. Every changed point belongs to W, so the localization counts all changed interactions. This establishes the displayed sign and normalization. The same computation appears in Dávila–del Pino–Wei [7], before (1.5) and in Appendix B, with opposite signed-curvature convention and order denoted there by s.

We justify the error for these smooth cutoffs explicitly. The common $C^{1,1}$ charts, regraphed at their tangent planes as in 2.6, give a common radius $r_* > 0$ and a constant M such that at each $X \in S \cap \partial E_\tau$ the local graph g satisfies $g(0) = Dg(0) = 0$ and $|g(z')| \le M|z'|^2$. Let T_X be its occupied tangent half-space. Genuine phase separation implies that the mismatch with T_X in $B_{r_*}(X)$ is contained between this graph and its tangent plane. For $0 < a < r_*$, Tonelli and $|(z', z_m)| \ge |z'|$ give

$$\int_{B_a(X)} \frac{|\chi_{E_\tau}(Y) - \chi_{T_X}(Y)|}{|X-Y|^{m+p}}\, dY \le \int_{|z'|<a} |g(z')|\,|z'|^{-m-p}\, dz' \le \frac{M|S^{d-1}|}{1-p} a^{1-p}, \qquad d = m-1.$$

This is an absolute mismatch estimate, so it remains valid after multiplication by any radial weight between zero and one. The signed tangent half-space integral vanishes for every radial cutoff. For $2\delta < r_*$, decompose the removed weight as

$$1 - \chi(|X-Y|/\delta)\eta(|X-Y|/R) = (1-\chi(|X-Y|/\delta)) + \chi(|X-Y|/\delta)(1-\eta(|X-Y|/R)).$$

The first term is supported in $B_{2\delta}(X)$ and is bounded by the absolute mismatch estimate, with a factor two for the signed phase. The second is supported where $|X-Y| > R$; the signed mismatch is at most two, and its integral is at most $2|S^{m-1}|R^{-p}/p$. The common chart constants are independent of X and small $|\tau|$. Thus, uniformly on these transported velocity supports,

$$|H_{\delta,R}[E_\tau] - H_p[E_\tau]| \le C\frac{\delta^{1-p}}{1-p} + C\frac{R^{-p}}{p}.$$

Bounded speeds and areas let us remove both cutoffs in the time integral. On the left, dominated convergence applies to each of the two finite local perimeters. We have obtained the untruncated integrated identity before differentiating it.

Pull its surface integrand back to ∂E_0. At fixed δ, R, the flow, normal velocity and Jacobian are continuous in time; phase indicators converge locally in L^1, and the smooth truncated kernel and uniform integrable tail make the surface integral continuous. The residual bound makes the untruncated integral its uniform limit. The fundamental theorem of calculus now gives the claimed derivative at zero. The same localized difference evaluates the variation in U, even if unrelated terms there are infinite; its remote terms can also be expressed through Lemma 2.7.

Uniform truncation convergence also proves continuity of H_p on compact regular patches. Therefore $H_p = 0$ implies stationarity in Definition 2.2. Conversely, zero first variation gives zero pairing with every compact scalar normal test: for C^2 boundaries use Lemma 2.8; for $C^{1,1}$ boundaries extend $\phi\nu_E$ in graph charts and mollify compact fields uniformly, using the locally Lipschitz normal. The curvature is bounded on the common compact support, so uniform trace convergence suffices. Hence $\int \phi H_p = 0$ for all such tests, and continuity gives $H_p = 0$ pointwise. No volume constraint enters this argument. □

Proposition 2.10 (Paired second perimeter variation)**.** *Let E have genuine regular boundary in U and be stationary there in the sense of 2.2. For a compact smooth flow with scalar normal speed ϕ, its second perimeter derivative is $2Q_p[\phi]$, with the normal form and conventions of 2.1 and remote cancellation of 2.7. In particular stability is equivalent to $Q_p[\phi] \geq 0$ for every compact normal test. Compact C^1 normal tests are admissible.*

Proof. Use the sign and perimeter normalization of Definition 2.1.

To justify that the cutoff calculation computes an actual second derivative, fix the smooth flow Ψ_t and a compact set $K \Subset U$ outside which it is the identity, and choose a bounded smooth W with $K \Subset W \Subset U$. Proposition 2.9 applies to the present C^2 boundary and gives $\mathrm{Per}_p(E; W) < \infty$. Write $J_t = \det D\Psi_t$ and

$$T_t(X, Y) = J_t(X)J_t(Y)|\Psi_t(X) - \Psi_t(Y)|^{-m-p}.$$

Since the flow is the identity outside $K \Subset W$, it preserves W. Its uniform Jacobian and bi-Lipschitz bounds for small $|t|$ give $T_t(X, Y) \leq C|X - Y|^{-m-p}$. Changing variables separately in the nonnegative localized integrals therefore yields $\mathrm{Per}_p(E_t; W) \leq C\,\mathrm{Per}_p(E; W) < \infty$.

Changing variables in the finite localized difference, and using $\Psi_t(W) = W$, yields the absolutely convergent fixed-domain identity

$$D(t) := \mathrm{Per}_p(E_t; W) - \mathrm{Per}_p(E; W) = \int_E \int_{E^c} (T_t(X, Y) - T_0(X, Y))\, dY\, dX.$$

For $|t|$ in one fixed small interval the flow is uniformly bi-Lipschitz. Its velocity and acceleration differences are bounded by $C|X - Y|$, the first two time derivatives of its Jacobians are bounded, and all these derivatives vanish when both points lie outside K. Consequently, for $j = 1, 2$,

$$|\partial_t^j T_t(X, Y)| \leq C|X - Y|^{-m-p}(\mathbf{1}_K(X) + \mathbf{1}_K(Y)).$$

This majorant is integrable on $E \times E^c$: its integral is at most $2C\,\mathrm{Per}_p(E; W)$. Thus D is C^2 by dominated differentiation. Choose the radial diagonal and tail cutoffs with $|D^j K_{\delta,R}(Z)| \leq C|Z|^{-m-p-j}$ for $j = 0, 1, 2$, with C independent of δ, R; fixed smooth profiles at scales δ and R have these bounds. The same majorant applies after the change of variables to their first two time derivatives. For each $X \neq Y$, the cutoff kernels and those derivatives converge to their untruncated counterparts, uniformly for t in this small interval. Dominated convergence therefore gives $D_{\delta,R} \to D$ in C^2. In particular the limit of $D''_{\delta,R}(0)$ equals $D''(0)$, rather than merely a formal limiting quadratic expression. No differentiability or perimeter finiteness of the remote exterior is used in this argument.

For the local shape identity use Chan–Dipierro–Serra–Valdinoci [6, Appendix A, Proposition A.1, formula (A.3) and Remark A.2], in the arXiv:2308.06328v2 version. This localization requires finite perimeter in the variation domain and a $C^{1,1}$ interface there. Our phase is an open graph side locally in W; its possibly merely measurable exterior enters only through volume integrals of its bounded characteristic function. The fixed-domain differentiation above and Lemma 2.7 justify the same computation in this setting. Insert the smooth even diagonal and tail cutoffs first. Whole-boundary expressions below are the combined local and remote pairings of Definition 2.1. The stationary part of the differentiated first variation is

$$\begin{aligned} &2\int_\Sigma \phi(X) \int_\Sigma (\phi(X) - \phi(Y))K_{\delta,R}(X - Y)\, d\mathcal{H}^d_Y d\mathcal{H}^d_X \\ &\quad - 2\int_\Sigma \phi(X)^2 \int_\Sigma (1 - \nu_E(X) \cdot \nu_E(Y))K_{\delta,R}(X - Y)\, d\mathcal{H}^d_Y d\mathcal{H}^d_X. \end{aligned}$$

Write $V = V_0$ for the Eulerian velocity, $V^\top$ for its tangential part and $\dot{V}_0 = \partial_t V_t|_{t=0}$. The remaining term is

$$\mathcal{R}_{\delta,R} = \int_\Sigma H_{\delta,R}[E]((\operatorname{div} V)\phi - \operatorname{div}_\Sigma(\phi V^\top) + \dot V_0 \cdot \nu_E)\, d\mathcal{H}^d.$$

For an autonomous flow the last term is zero, as in the cited formula. For a time-dependent flow it is the additional derivative of the velocity in the first variation. The coefficient in parentheses is bounded on one fixed compact support, independently of the kernel cutoffs. Tangential integration by parts has placed the curvature itself, rather than its derivatives, in this remainder. Symmetrize the compact regular part, where the truncated integrals are finite, and combine the remote Dirichlet and potential contributions before integration through the signed flux of Lemma 2.7. This gives twice the paired form of Definition 2.1. If the whole boundary is regular and the terms converge separately, this is the usual identity $2\int \phi(X)\int(\phi(X)-\phi(Y))K_{\delta,R} = \iint(\phi(X)-\phi(Y))^2 K_{\delta,R}$. No separate unsigned exterior integral is used for a rough phase.

The local squared trace differences and normal differences are $O(|X-Y|^2)$, so the surface radial majorant is Cr^{-p}. Dominated convergence removes the diagonal cutoff. Before removing the tail cutoff, combine the remote Dirichlet and potential pieces as the volume-controlled flux of Lemma 2.7. Proposition 2.9 identifies stationarity with $H_p = 0$, and Lemma 2.6 bounds the remaining regularized-curvature terms by $C\delta^{1-p}/(1-p) + CR^{-p}/p$ on the compact support. They vanish for each fixed p. The surviving expression is exactly $2Q_p[\phi]$.

Lemma 2.8 extends the identity and its nonnegativity consequence to compact C^1 normal traces by form continuity; no pointwise Jacobi principal value is required for such a trace. Stability in Definition 2.2 implies $Q_p \geq 0$. Conversely, this inequality applied to the trace of every compact smooth flow, together with the identity, gives stability. □

2.3. **Normalization of the local limit.**

Lemma 2.11 (Complement and similarity invariance)**.** *Complementation, translations, rotations and positive dilations preserve stationarity and stability of 2.2, with corresponding transformed domains. For the conventions of 2.1, dilation by r^{-1} multiplies H_p by r^p and classical curvature by r; complementing changes the signs of oriented curvatures but preserves Q_p.*

Proof. In the perimeter of Definition 2.1, the change $(X,Y) \mapsto (\lambda X, \lambda Y)$ contributes the factor λ^{m-p}. Rotations and translations leave it unchanged, while complementing exchanges the two phases. Conjugation gives a bijection of the compact flows in the corresponding domains. Their first and second derivatives therefore retain the zero and nonnegative signs in Definition 2.2.

The volume curvature scales by λ^{-p}. For the classical scaling, let $\Sigma_\lambda = \lambda\Sigma$. Its outward normal satisfies $\nu_\lambda(\lambda X) = \nu_E(X)$. If γ is a surface curve with $\gamma(0) = X$, $\gamma'(0) = \tau$ and $|\tau| = 1$, then $\gamma_\lambda(t) = \lambda\gamma(t/\lambda)$ has the same unit tangent at λX. Differentiating the normal identity yields

$$D_\tau \nu_\lambda(\lambda X) = \lambda^{-1} D_\tau \nu_E(X).$$

For corresponding orthonormal tangent frames, every second-fundamental-form entry therefore gains λ^{-1}; so do its eigenvalues, norm and trace. In particular

$$H_{\mathrm{cl},\lambda}(\lambda X) = \sum_{i=1}^{d} \langle D_{\tau_i}\nu_\lambda, \tau_i\rangle = \lambda^{-1} H_{\mathrm{cl}}(X).$$

Translations preserve the derivatives, and rotations act on the normal and frame by the same orthogonal map. Taking $\lambda = r^{-1}$ gives the stated factors r^p and r.

Complementation negates the signed phase and the outward normal, hence both oriented curvatures. Normal scalar products and local Q_p are unchanged. For the exterior pairing use Definition 2.1 and Lemma 2.7. The identity $D\chi_{E^c} = -D\chi_E$ reverses every compact remote vector flux, while its evaluation normal also reverses. Their scalar pairing is unchanged. The cutoff convergence in Lemma 2.7 passes this identity to the noncompact remote pairing. Canonicalization changes only a null representative, so the identities remain valid. □

Proposition 2.12 (Classical curvature and same-sheet Jacobi normalization)**.** *Fix $0 < \alpha \le 1$. For the graph and curvature conventions of 2.1, the upward normal is $(-\nabla h, 1)/\sqrt{1+|\nabla h|^2}$ and its classical curvature is $-\mathcal{M}(h)$. For a globally Lipschitz graph with $C^{2,\alpha}$ regularity near the evaluation points, $\sigma H_p[B_h] \to -c_d \mathcal{M}(h)$ locally as $p \uparrow 1$.*

For a compact normal test on a single uniformly regular sheet the corresponding local quadratic-form limit is

$$(1-p)Q_p[\phi] \longrightarrow \frac{c_d}{2}\int_\Sigma (|\nabla_\Sigma \phi|^2 - |A_\Sigma|^2\phi^2)\, d\mathcal{H}^d$$

when its normalized remote contribution vanishes. More precisely, this holds for $p_j \uparrow 1$, sheets converging in C^2 on a neighborhood of common compact supports with a uniform $C^{2,\alpha}$ bound, and test functions converging in C^1 on those sheets, when all remaining contributions to $(1-p_j)Q_{p_j}$ tend to zero.

Proof. The subgraph in Definition 2.1 has upward normal $(-Dh, 1)/\sqrt{1+|Dh|^2}$. Taking its surface divergence gives $H_{\mathrm{cl}} = -\mathcal{M}(h)$.

For the fractional limit, rotate each evaluation point onto its tangent plane, with upward outward normal. Uniform local $C^{2,\alpha}$ regularity supplies, in the resulting coordinates,

$$h(0) = Dh(0) = 0, \quad a(z) = h(z) - h(0), \quad |a(z)| \le C|z|^2, \quad a(z) = \tfrac{1}{2}D^2h(0)[z,z] + O(|z|^{2+\alpha}).$$

Replace the graph outside a fixed smaller chart by a smooth extension. The altered phase lies at fixed distance and contributes $O(\sigma)$, by Lemma 2.6. Subtracting the tangent half-space before integrating vertically gives the exact local term

$$-2\sigma \int_{|z|<a_0} \int_0^{a(z)} (|z|^2 + t^2)^{-(d+1+p)/2}\, dt\, dz.$$

The inner integral is oriented when $a(z) < 0$. Lemma 2.6 removes the ambient diagonal omission for each fixed p. Set $r = |z|$, $\beta = (d+1+p)/2$. The mean-value theorem gives

$$|(r^2+t^2)^{-\beta} - r^{-2\beta}| \le \beta r^{-2\beta-2}t^2.$$

After vertical integration, its error is at most $Cr^{-d-1-p-2}|a(z)|^3$. The normalized radial bound is

$$C\sigma \int_0^{a_0} r^{2-p}\, dr = C\sigma \frac{a_0^{3-p}}{3-p} \longrightarrow 0.$$

The Taylor remainder separately costs $C\sigma \int_0^{a_0} r^{\alpha-p}\, dr \to 0$. The leading term is

$$-\sigma \int_{|z|<a_0} D^2h(0)[z,z]|z|^{-d-1-p}\, dz.$$

Since

$$\sigma \int_0^{a_0} r^{-p}\, dr = a_0^{1-p} \to 1, \qquad \int_{S^{d-1}} \omega \otimes \omega\, d\omega = c_d I,$$

it converges to $-c_d \operatorname{tr} D^2h(0) = c_d H_{\mathrm{cl}}$. The fixed-distance tail is $O(\sigma)$. Uniform chart bounds make the convergence locally uniform, and the value is invariant under the tangent rotation.

For the form limit, fix a compact collar of the common test supports inside the converging regular patch, with positive distances both from the test supports to the complement of the collar and from the collar to the edge of that patch. Integrate X over this collar and choose the tangent radius smaller than both distances. This includes near-diagonal pairs with X outside the support and Y inside. Pairs outside the collar with a nonzero test value are fixed-distance remainders covered by the stated vanishing hypothesis. Use tangent coordinates $z = r\omega$ on the sheet. The Jacobian is $1 + O(r^2)$ and the ambient distance is $r(1 + O(r^2))$. The converging tests and sheets satisfy, uniformly for X in this collar, with the tests extended by zero outside their supports,

$$(\phi(X) - \phi(Y))^2 = r^2(\nabla_\Sigma \phi(X)\cdot\omega)^2 + o(r^2), \quad 1 - \nu(X)\cdot\nu(Y) = \tfrac{1}{2}r^2|A_X\omega|^2 + o(r^2).$$

For each prescribed $\varepsilon > 0$, choose a common small radius where each remainder divided by r^2 is at most ε. The normalized radial mass is bounded by $\sigma \int_0^{a_0} r^{-p}\, dr \le \max\{1, a_0\}$, so these remainders are uniformly small. Fixed annuli vanish after multiplication by σ. The angular identity gives $(c_d/2) \int |\nabla_\Sigma \phi|^2$ for the Dirichlet part, with its prefactor $1/2$, and $(c_d/2) \int |A|^2 \phi^2$ for the potential. Compact C^2 sheet and C^1 test convergence passes the spatial integrals to the limit. By hypothesis the remaining normalized contributions vanish, with the exterior interpreted through Lemma 2.7. Subtracting the two parts proves the claimed Jacobi normalization. □

Proposition 2.13 (Uniform local elliptic estimates used in the gap argument)**.** *On fixed nested Euclidean balls, a positive solution of $-\operatorname{div}(a\nabla v) = f$ with uniformly elliptic uniformly Lipschitz coefficients satisfies an interior Harnack estimate with an additive bounded-forcing term. In particular, for $B_1 \Subset B_2$ and $f \in L^\infty(B_2)$,*

$$\sup_{B_1} v \le C\big(\inf_{B_1} v + \|f\|_{L^\infty(B_2)}\big).$$

For every fixed $1 < q < \infty$ the corresponding interior estimate is

$$\|v\|_{W^{2,q}(B_1)} \le C_q\big(\|v\|_{L^q(B_2)} + \|f\|_{L^q(B_2)}\big).$$

The constants depend only on the dimension, nested balls, ellipticity and coefficient Lipschitz bounds (and q). In particular they are uniform for coefficient families with uniform local $C^{1,\alpha}$ bounds. The same statements hold on the compact sphere in fixed nested smooth coordinate charts.

Proof. Interpret a weak solution as $v \in W^{1,2}_{\mathrm{loc}}$ satisfying the divergence equation in distributions. Gilbarg–Trudinger [12, Theorems 8.17 and 8.18] give respectively local supremum and weak Harnack estimates with forcing. Take their vector sources equal to zero and their scalar source equal to $-f$. For $d \ge 2$ fix $Q > d$, so bounded f belongs to the required $L^{Q/2}$ space. On a ball $B_{4r}(y) \subset B_2$, choose $1 < \theta < d/(d-2)$ if $d > 2$, and any fixed $\theta > 1$ if $d = 2$. The two estimates give

$$\operatorname*{ess\,sup}_{B_r(y)} v \le C\left(\left(\frac{1}{|B_{2r}(y)|}\int_{B_{2r}(y)} v^\theta\right)^{1/\theta} + \|f\|_\infty\right) \le C'\left(\operatorname*{ess\,inf}_{B_r(y)} v + \|f\|_\infty\right).$$

Here r is fixed by the nested balls, so the source factors are absorbed in their constants. A finite chain of overlapping balls joins any two points of B_1 inside B_2; iterating this bound gives the asserted estimate, with constants controlled only by the stated data. If $d = 1$, lift to $V(x_1, x_2) = v(x_1)$ on the radius-two disk, with matrix $\operatorname{diag}(a(x_1), 1)$ and forcing $f(x_1)$. Integration of a compact test in x_2 verifies the weak equation. The coefficient bounds remain controlled, and the essential extrema on the unit disk equal those on $(-1, 1)$. The two-dimensional estimate therefore gives the one-dimensional claim. Local regularity proved below supplies a continuous representative, so the essential extrema equal the stated pointwise ones.

The interior strong-solution estimate in the same reference, Theorem 9.11, says that, for $D' \Subset D$ and every $1 < q < \infty$,

$$\|w\|_{W^{2,q}(D')} \le C_q\big(\|w\|_{L^q(D)} + \|Lw\|_{L^q(D)}\big).$$

It requires continuous uniformly elliptic principal coefficients with controlled modulus and bounded drift. Expanding the present divergence equation produces principal matrix a and drift $\partial_i a_{ij}$, bounded almost everywhere by the Lipschitz norm. We establish local $W^{2,q}$ membership first, so that this strong estimate can legitimately be used for the given weak solution.

Choose a ball $D \Subset B_2$ containing $\overline{B_1}$, and a smooth ζ compact in B_2, equal to one near $\overline{D}$. Mollify a to smooth matrices a_ε on D, with the same ellipticity, a common Lipschitz bound and uniform convergence to a. Put $f_\varepsilon = (\zeta f) * \rho_\varepsilon$; its $L^q(D)$ norm is bounded by $\|f\|_{L^q(B_2)}$. The original weak equation gives

$$\zeta f = -\operatorname{div}(\zeta a \nabla v) + (\nabla \zeta) \cdot a \nabla v \in H^{-1}(\mathbb{R}^d),$$

because the coefficients in this expression are compactly supported L^2 functions. Hence $f_\varepsilon \to f$ in $H^{-1}(D)$, as well as locally in L^q.

Lax–Milgram supplies the solution $v_\varepsilon - v \in H^1_0(D)$ of $-\operatorname{div}(a_\varepsilon \nabla v_\varepsilon) = f_\varepsilon$. Subtracting the weak equations and testing with their difference yields

$$\|\nabla(v_\varepsilon - v)\|_{L^2(D)} \le C\big(\|f_\varepsilon - f\|_{H^{-1}(D)} + \|a_\varepsilon - a\|_{L^\infty(D)}\|\nabla v\|_{L^2(D)}\big) \longrightarrow 0.$$

Poincaré's inequality gives strong $H^1(D)$ convergence. Interior difference quotients, iterated on smaller balls for each smooth approximating equation, make v_ε smooth there. The strong estimate is therefore available with constants controlled uniformly by the common ellipticity and Lipschitz bounds.

If $1 < q < 2$, strong L^2 convergence gives strong L^q convergence. Applying the estimate directly bounds v_ε in local $W^{2,q}$; weak compactness and distributional derivative identification give the same membership for v. This argument has used no L^2 condition on f.

For $q \ge 2$, both v_ε and f_ε are locally bounded in L^2, so start with an interior $W^{2,2}$ bound. On successive nested balls, Sobolev embedding improves a $W^{2,r}$ bound to an $L^{r'}$ bound for $r' \le dr/(d-2r)$ if $2r < d$, and for any finite r' if $2r \ge d$. At stages with $r < q$, choose $r' > r$, $r' \le q$, strictly below a critical endpoint when necessary. The forcing is bounded in every exponent at most q, so the strong estimate on the next ball yields $W^{2,r'}$. Stop immediately if $r = q$, including the initial case $q = 2$. When $r < q$ and $2r < d$, choose

$$r' = \min\left\{q, \left(\frac{1}{r} - \frac{1}{d}\right)^{-1}\right\}.$$

This satisfies $r < r' \le q$ and $1/r' \ge 1/r - 1/d > 1/r - 2/d$, so it lies strictly below the Sobolev endpoint. A capped step reaches q and ends the iteration; every uncapped step decreases $1/r$ by $1/d$. Thus after finitely many uncapped steps either q has been reached or $2r \ge d$. In the latter case, if $r < q$, the embedding permits $r' = q$ on the next ball. Dimensions one and two are already in this latter range at the initial exponent $r = 2$. Hence the iteration always terminates at the prescribed q after finitely many balls. Weak compactness again gives the membership of v.

Now apply the strong estimate to v itself on $B_1 \Subset D \Subset B_2$. Its divergence expansion holds almost everywhere, and the norms over D are bounded by the displayed B_2 norms. This gives the quantitative assertion with no approximation-dependent term. On the sphere, a fixed finite smooth atlas introduces fixed smooth density and metric factors; they preserve uniform ellipticity and bounded drift. The same estimates on nested charts prove the spherical version. □

3. Graph equations and regularity at the critical tube scale

A local graph sees both its own curvature and the phase outside its chart. We first express the graph curvature exactly, then estimate the exterior phase as a fixed volume source. The regularity statements below assume a tube of the indicated width; they do not yet assert that neighboring interfaces have that separation.

3.1. **The graph equation and complete charts.**

Definition 3.1 (The graph curvature nonlinearity)**.** For $d \ge 1$ and $0 < p < 1$ define

$$G_p(z) = \int_0^z (1 + r^2)^{-(d+1+p)/2}\, dr.$$

It is odd and smooth, and $G_p'(z) = (1 + z^2)^{-(d+1+p)/2}$ is even and positive. On each bounded interval the derivatives needed below are bounded uniformly for p in $[p_0, 1]$, for any $p_0 > 0$.

Proof. The integrand is even and smooth in its real variable. Its integral from zero is consequently odd, finite on every bounded interval and smooth, with the displayed derivative by the fundamental theorem of calculus. Positivity and evenness of G_p' follow from the same formula. Every derivative used here is continuous in (p,z); compactness of $[p_0,1]\times[-M,M]$ therefore gives its uniform bound. □

Lemma 3.2 (Exact volume-normalized graph equation)**.** *Let h be globally Lipschitz on $\mathbb{R}^d$ and locally $C^{1,1}$ on an open set Ω. For every $0<p<1$, the curvature of its subgraph, in the conventions of 2.1, satisfies for $x\in\Omega$*

$$F_h(x)=2(1-p)\,\mathrm{PV}\int_{\mathbb{R}^d} G_p\Big(\frac{h(x)-h(y)}{|x-y|}\Big)\,|x-y|^{-d-p}\,dy,$$

where G_p is defined in 3.1. The horizontal radial principal value agrees with the ambient-ball curvature principal value. In particular, if this subgraph is stationary near $X_h(x)$, then the displayed expression vanishes by 2.9.

Proof. The subgraph in Definition 2.1 is a genuine local $C^{1,1}$ phase boundary in Definition 2.3. Fix $x\in\Omega$ and subtract its tangent half-space, whose boundary is $t=h(x)+\nabla h(x)\cdot(y-x)$. The signed half-space integral vanishes under centrally symmetric truncations. The remaining phase discrepancy has thickness $O(|y-x|^2)$, so Lemma 2.6 bounds its integral over $|y-x|<\delta$ by $C\delta^{1-p}/(1-p)$. Both the ambient-ball and horizontal-cylinder omissions therefore vanish and give the same principal value; each cancels the tangent contribution separately.

Write $r=|y-x|$ and shift the vertical coordinate by $h(x)$. Subtraction of the horizontal half-space through the point gives

$$2\int_0^{h(x)-h(y)}(r^2+v^2)^{-(d+1+p)/2}\,dv=2r^{-d-p}G_p\Big(\frac{h(x)-h(y)}{r}\Big),$$

with G_p from Definition 3.1. The tangent contribution in this expression is odd in the horizontal direction. The remainder near zero has radial majorant Cr^{-p}. At infinity, the global Lipschitz bound keeps the argument of G_p bounded, and $\int_1^\infty r^{-1-p}\,dr<\infty$. These bounds justify Fubini for the subtracted truncated integrals and the removal of their truncations. Multiplication by $\sigma=1-p$ proves the exact equation. If the phase is stationary in Definition 2.2, Proposition 2.9 makes the curvature, and hence this expression, zero. □

Lemma 3.3 (A clipped chart containing every interior interface)**.** *Using the regular boundary convention of 2.1, let $M=\partial E$ be a regular embedded boundary in $B_2\subset\mathbb{R}^{d+1}$, and suppose*

$$0\in M,\qquad T_0M=\mathbb{R}^d\times\{0\},\qquad |A_M|\le 1\quad \text{in } B_2.$$

There is an absolute constant $c_*>0$ with the following property. For $0<\varepsilon\le 1/100$, put

$$r=c_*\varepsilon,\qquad b=\frac{\varepsilon r}{10}.$$

There are finitely many ordered C^2 functions $g_1<\cdots<g_K$ on B^d_{2r} such that:

1. *their graphs are contained in M and satisfy*

$$|g_i|<3b,\qquad |\nabla g_i|\le\varepsilon/100,\qquad |D^2g_i|\le 2;$$

2. *$(0,g_i(0))$, $1\le i\le K$, are exactly the intersections of M with $\{0\}\times[-2b,2b]$;*
3. *they contain every interface in the smaller cylinder:*

$$M\cap\big(B^d_r\times(-b,b)\big)=\Big(\bigcup_{i=1}^K \operatorname{graph} g_i\Big)\cap\big(B^d_r\times(-b,b)\big);$$

4. consecutive graphs are actual adjacent interfaces on the whole common base:

$$M \cap \{(x,t) : x \in B^d_{2r}, g_i(x) < t < g_{i+1}(x)\} = \varnothing \qquad (1 \le i < K).$$

Consequently the phase alternates across consecutive graphs. The conclusion permits these graph patches to cross the caps of the smaller cylinder and permits further interfaces beyond the packet in the larger cylinder.

Proof. Use the genuine regular boundary and curvature conventions of 2.1. We give a proof using only the curvature bound and embeddedness.

Step 1: a single branch has a graph patch of fixed size. Fix $P \in M \cap B_{1/2}$. Intrinsic closed balls about P of radius at most $1/4$ are compact. To see this, their ambient positions lie in $\overline{B}_{3/4} \subset B_2$. Every sequence there has an ambiently convergent subsequence, whose limit lies in M by relative closedness. A local embedded graph neighborhood of the limit contains all sufficiently late points of the subsequence. Within that neighborhood, ambient convergence implies convergence in intrinsic distance. The intrinsic distance bound therefore persists at the limit. This proves compactness. For Q at intrinsic distance at most $1/8$ from P, choose curves from P to Q whose lengths decrease to that distance. Parametrize them with constant speed on $[0,1]$. After discarding finitely many curves, their lengths are less than $1/4$, so they remain in the compact intrinsic ball of radius $1/4$. Arzelà–Ascoli and lower semicontinuity of length give a minimizing curve.

An arclength parametrized minimizing curve is C^2 even though the induced metric is only C^1. In local coordinates write that metric as (G_{ij}). The curve locally minimizes energy as well as length, and its weak Euler–Lagrange equation is

$$\frac{d}{dt}(G_{ij}(x)\dot{x}^j) = \frac{1}{2}\partial_i G_{jk}(x)\dot{x}^j\dot{x}^k.$$

The curve has bounded coordinate speed. Thus the right-hand side is bounded, its momenta are Lipschitz, and multiplication by $G^{-1}(x(t))$ shows that $\dot{x}$ is Lipschitz. The right-hand side is now continuous, so the equation gives $x \in C^2$. Initial-value geodesics exist by Peano's theorem for the continuous Christoffel coefficients. Their speed is constant by the geodesic equation. The Gauss formula and the curvature bound give $|\gamma''| \le 1$ for unit speed, so position and velocity have limits at every finite endpoint in this compact interior region. The limiting velocity is a unit tangent vector, and a new Peano solution extends the old curve. Neither uniqueness of geodesics nor an injectivity-radius bound is needed.

If $\gamma : [0, L] \to M$ is a unit-speed minimizing geodesic starting at P, with $L \le R_0 := 1/8$, then

$$|\gamma''| \le 1, \qquad |\gamma(L) - P - L\gamma'(0)| \le L^2/2, \qquad |\nu(\gamma(L)) - \nu(P)| \le L.$$

The last estimate uses a continuous choice of unit normal along the curve; a global orientation is unnecessary. Projection to $T_P M$ is a local diffeomorphism on the intrinsic ball of radius R_0. On its intrinsic boundary the projected distance from P is at least $R_0 - R_0^2/2 > R_0/2$. The component containing P of the inverse image, inside the intrinsic open R_0 ball, of the tangent ball of radius $R_0/4$ projects properly and locally diffeomorphically to that tangent ball. Indeed, a sequence above a compact subset of the tangent ball has a limit in the closed intrinsic R_0 ball. The boundary-distance bound excludes intrinsic distance R_0 at that limit. Components are closed in the inverse image, so the limit remains in the chosen component. Its image is therefore both open and closed. It is a connected covering of a simply connected ball, and hence a diffeomorphism.

The same proof applies to horizontal projection if the unoriented normal at P makes angle at most $1/10$ with the vertical. Indeed, the lower bound on the projected distance becomes $R_0\cos(1/10) - R_0^2/2 > R_0/2$. The resulting horizontal graph is defined on $B^d_{r_0}(P')$, where $r_0 = R_0/4$, has slope at most $1/2$, and has Hessian bounded by 2. For the Hessian estimate, the graph formula for the second fundamental form gives

$$|D^2 g| \le (1+|\nabla g|^2)^{3/2}|A_M| \le 2.$$

Applied at $P=0$, this produces a reference graph g_* on $B^d_{r_0}$ satisfying

$$g_*(0)=0, \quad \nabla g_*(0)=0, \quad |g_*(x)| \le |x|^2, \quad |\nabla g_*(x)| \le 2|x|, \quad |D^2 g_*| \le 2.$$

Step 2: all branches in a thin cylinder have nearly horizontal tangent planes. There is an absolute C_0 such that, whenever $a, b_0 > 0$ are sufficiently small, every $Q \in M \cap (B^d_a \times (-b_0, b_0))$ satisfies

$$|\operatorname{proj}_{T_Q M} e_m| \le C_0(\sqrt{b_0} + a).$$

Choose a unit tangent $T \in T_Q M$ and a unit-speed intrinsic geodesic with $\gamma(0) = Q$ and $\gamma'(0) = T$. For a fixed absolute $L_0 > 0$, this curve exists on $[-L_0, L_0]$, stays in B_2, and has horizontal projection inside $B^d_{r_0}$; for example first require $a < r_0/8$ and then take $L_0 = r_0/8$. Let $\pi(x,t) = x$ be horizontal projection and define

$$f(t) = \gamma_m(t) - g_*(\pi\gamma(t)).$$

The estimates in the preceding geodesic and reference-graph estimates give $|f''| \le 4$. Whenever $f(t_0) = 0$, embeddedness says that M has the same tangent plane as the reference patch at $\gamma(t_0)$. Therefore $f'(t_0) = 0$. In particular f cannot have a transverse zero.

If $f(0) = 0$, the tangent estimate follows directly from the reference-graph estimates just established. Otherwise put $B = |f(0)|$ and $A = |f'(0)|$, and reverse T if necessary so that $\operatorname{sign}(f(0))f'(0) = -A$. On $[0, \min\{A/8, L_0\}]$ the signed derivative is at most $-A/2$. If

$$B < \frac{A}{2}\min\{A/8, L_0\},$$

the function would have a transverse zero. It follows that

$$A \le C(\sqrt{B} + B/L_0).$$

Since $B \le b_0 + a^2$, this bounds the tangential projection of $(-\nabla g_*(Q'), 1)$ by $C(\sqrt{b_0}+a)$. Its distance from e_m is at most $2a$. Taking the supremum over unit tangent vectors proves the angle estimate $|\operatorname{proj}_{T_Q M} e_m| \le C_0(\sqrt{b_0}+a)$.

Apply the angle estimate $|\operatorname{proj}_{T_Q M} e_m| \le C_0(\sqrt{b_0}+a)$ with $a = 4r$ and $b_0 = 4b$. Since $r = c_*\varepsilon$ and $b = c_*\varepsilon^2/10$, we may choose c_* so small that every tangent plane in $B^d_{4r} \times (-4b, 4b)$ has horizontal graph slope at most $\varepsilon/1000$. The graph patch of Step 1 through a point in this cylinder with horizontal position in B^d_{2r} contains the common base B^d_{2r} after restriction. On that common base its slope is at most $\varepsilon/1000 + 8r$, using the Hessian bound 2 and horizontal distance at most $4r$ from its center. Choose c_* so that this is at most $\varepsilon/100$. This controls the full restricted patch even if it leaves the thinner cylinder.

Step 3: enumerate every relevant branch and prove adjacency. The intersection of M with the compact segment $\{0\} \times [-2b, 2b]$ is compact, and Step 2 makes every intersection transverse. It is consequently a finite set. List its heights in increasing order and take the graph patch of Step 2 through each intersection. These are the functions $g_1, \dots, g_K$. Their heights satisfy

$$|g_i(x)| \le 2b + (\varepsilon/100)|x| \le 2b + b/5 < 3b \quad (x \in B^d_{2r}).$$

If two of these graphs met, local embeddedness would make them equal near that point. Their coincidence set would then be both open and closed in the connected common base, so they would agree everywhere, contrary to their different axis heights. They are therefore disjoint and retain their order.

Every point of $M \cap (B_r^d \times (-b,b))$ has a graph patch whose height at the axis differs by at most $\varepsilon r/100 = b/10$. That axis height lies in $(-2b, 2b)$, so this graph is one of the listed graphs. This proves coverage of the smaller cylinder.

Finally, a hypothetical extra boundary point between two successive listed graphs over B_{2r}^d has height in $(-3b, 3b)$, so Step 2 supplies its graph over the same base. This graph cannot meet either of the two bounding graphs. Its axis height lies strictly between their axis heights and must also occur in the enumeration, a contradiction. This proves adjacency. The set has a fixed phase on each connected intervening slab and changes phase at every regular interface, which proves the final assertion. □

3.2. **Forcing from a fixed phase source.**

Lemma 3.4 (Tubular agreement and a fixed-source estimate)**.** *Use the graph curvature and subgraph definitions of 2.1 and stationarity of 2.2. Let $p \in [p_0, 1)$, where $p_0 > 0$. Let h be a globally defined $C^{1,1}$ graph, let V be a relatively open region of its graph, and let $\rho > 0$. Suppose each $X \in V$ has an ambient open neighborhood on which E is stationary. After one fixed choice between E and E^c, denote the chosen phase by E and suppose*

$$\chi_E = \chi_{B_h} \quad \text{in } B_\rho(X) \quad \text{for every } X \in V.$$

At graph points in V the following ordinary integral represents the normalized graph curvature:

$$\sigma H_p[B_h](X) = 2\sigma \int_{\mathbb{R}^m} (\chi_E - \chi_{B_h})(Y)|X - Y|^{-m-p}\, dY.$$

In particular,

$$|\sigma H_p[B_h](X)| \le C(m, p_0)\sigma\rho^{-p},$$
$$|\nabla_{\text{graph}\, h}(\sigma H_p[B_h])(X)| \le C(m, p_0)\sigma\rho^{-p-1}.$$

The right-hand side of the displayed fixed signed-source integral also defines a smooth ambient function in a neighborhood of V. On any horizontal convex region whose graph lies in V and has slope at most M_0, the pulled-back function F_h consequently satisfies

$$[F_h]_{C^{0,1}} \le C(m, p_0)(1 + M_0)\sigma\rho^{-p-1}.$$

Proof. Complementation preserves stationarity by 2.11, so E denotes the single chosen phase throughout. For $X \in V$, let O_X be the ambient stationary neighborhood in the hypothesis, with stationarity understood as in 2.2. Restrict O_X to a smaller ball contained in $B_\rho(X)$. The tubular phase agreement identifies the chosen phase with the lower local side of the $C^{1,1}$ graph of h and identifies its boundary with the graph. This is a genuine $C^{1,1}$ boundary as defined in 2.3. The first-variation proposition therefore applies on this neighborhood at X.

By 2.9, stationarity gives $H_p[E](X) = 0$. The local $C^{1,1}$ graph and genuine tubular agreement give existence of both principal values by 2.6. In the conventions of 2.1, subtract $\sigma H_p[E](X) = 0$ from $\sigma H_p[B_h](X)$. The difference of the phase functions is $2(\chi_E - \chi_{B_h})$. This signed source vanishes in $B_\rho(X)$, so no principal value is needed in the difference. The kernel tail is exactly

$$\int_{|Y-X|\ge\rho} |X - Y|^{-m-p}\, dY = \frac{|S^{m-1}|}{p}\rho^{-p},$$

which proves the asserted $C\sigma\rho^{-p}$ size bound.

Keep the signed source $\chi_E - \chi_{B_h}$ fixed while differentiating the evaluation point. Its support stays away from each $X \in V$, and dominated differentiation gives

$$\left|\nabla_X\left(2\sigma\int(\chi_E-\chi_{B_h})(Y)|X-Y|^{-m-p}\,dY\right)\right|\le 2\sigma(m+p)\frac{|S^{m-1}|}{p+1}\rho^{-p-1}.$$

Every higher derivative exists on the union of the smaller balls $B_{\rho/2}(X)$, $X\in V$, by the same argument. Restriction of this ambient derivative to the graph proves the asserted $C\sigma\rho^{-p-1}$ tangential gradient bound. The derivative of the graph parametrization has norm at most $\sqrt{1+M_0^2}$, which gives the asserted pulled-back Lipschitz bound. No derivative of a moving characteristic function has been taken. □

Corollary 3.5 (Fixed-source bounds at the critical tube radius)**.** *Under the hypotheses of 3.4, set $\rho=c\sqrt{\sigma}$ for a fixed $c>0$. Then the normalized graph curvature of 2.1 satisfies $|F_h|\le C\sqrt{\sigma}$. On a horizontal convex region of bounded slope whose graph lies in the given region V, its Lipschitz seminorm is at most C. The constants depend on m,p_0,c and the fixed slope bound, not on p or the number of other interfaces of the phase.*

Proof. Insert $\rho=c\sqrt{\sigma}$ into Lemma 3.4. In the range $0<\sigma=1-p<1$ of Definition 2.1,

$$\sigma(\sqrt{\sigma})^{-p}=\sqrt{\sigma}\,\sigma^{(1-p)/2}\le\sqrt{\sigma},\qquad \sigma(\sqrt{\sigma})^{-p-1}=\sigma^{(1-p)/2}\le 1.$$

The factors c^{-p} and c^{-p-1} stay bounded for $p\in[p_0,1)$. This gives small forcing and a uniform Lipschitz bound on the same convex graph regions. The source has absolute value at most one and involves one fixed phase difference, so the constants use neither an interface count nor exterior area. □

3.3. **Uniform graph regularity.**

Definition 3.6 (Weak normalized operators and Fall's kernel class)**.** Let $d\ge 1$, $0<t<1$, $0<\alpha<1$, and $\kappa>0$. Write $Q_2^{\mathrm{pol}}=B_2^d\times[0,2)$, to distinguish this parameter domain from the spatial cylinder Q_r in 2.1. A measurable symmetric kernel belongs to $\mathcal{K}^t(\kappa,\alpha,Q_2^{\mathrm{pol}})$ if it satisfies the global bound $|K(x,y)|\le\kappa^{-1}|x-y|^{-d-2t}$, the lower bound $K(x,y)\ge\kappa|x-y|^{-d-2t}$ for $x,y\in B_2^d$, and

$$A_K(x,r,\omega)=r^{d+2t}K(x,x+r\omega)$$

extends to $B_2^d\times[0,2)\times S^{d-1}$ with $C^{0,\alpha}$ norm at most κ^{-1} and with $A_K(x,0,\omega)=A_K(x,0,-\omega)$. Define $L^t(\mathbb{R}^d)$ by the weighted norm $\|w\|_{L^t}=\int_{\mathbb{R}^d}|w(y)|(1+|y|)^{-d-2t}dy$. For $w\in H^t(B_2^d)\cap L^t(\mathbb{R}^d)$ define the weak operator as a distribution on B_2^d, on tests $\psi\in C_c^\infty(B_2^d)$, by

$$\langle L_Kw,\psi\rangle=\frac{1}{2}\iint_{\mathbb{R}^d\times\mathbb{R}^d}(w(x)-w(y))(\psi(x)-\psi(y))K(x,y)\,dx\,dy.$$

The operator normalization used below is $(1-t)L_K$.

Proof. It remains to justify that the weak operator in this construction is a distribution, including continuity in the test-function topology. Fix a compact set $S\Subset B_2^d$, and let $\psi\in C_c^\infty(B_2^d)$ have support in S, extended by zero outside B_2^d. Constants C_S below may depend on d,t,κ and the distance of S from the exterior of B_2^d; no uniformity as $t\uparrow 1$ is asserted in this construction.

On $B_2^d\times B_2^d$, the upper bound on K and Cauchy–Schwarz give

$$\iint_{B_2^d\times B_2^d}|w(x)-w(y)||\psi(x)-\psi(y)||K(x,y)|\,dx\,dy\le\kappa^{-1}[w]_{H^t(B_2^d)}[\psi]_{H^t(B_2^d)}.$$

The zero extension of the smooth compact test is globally Lipschitz with Lipschitz constant at most $\|\nabla\psi\|_\infty$. Therefore

$$[\psi]^2_{H^t(B_2^d)} \le \|\nabla\psi\|^2_\infty \iint_{B_2^d\times B_2^d} |x-y|^{2-d-2t}\,dx\,dy \le C(d,t)\|\nabla\psi\|^2_\infty,$$

where the last radial integral is finite because $t<1$.

For the two ordered contributions with one variable outside B_2^d, symmetry reduces the absolute bound to a constant times

$$\|\psi\|_\infty \int_S \int_{(B_2^d)^c} (|w(x)|+|w(y)|)|x-y|^{-d-2t}\,dy\,dx.$$

Write $\delta = \operatorname{dist}(S,(B_2^d)^c)>0$. Uniformly for $x\in S$, the integral of the kernel over this exterior is bounded by $C(d,t)\delta^{-2t}$. Moreover, uniformly for $x\in S$ and $y\notin B_2^d$,

$$|x-y|^{-d-2t} \le C_S(1+|y|)^{-d-2t}.$$

For $2\le|y|\le 4$ this follows from $|x-y|\ge\delta$, and for $|y|>4$ from $|x-y|\ge|y|/2$. The exterior contribution is thus bounded by $C_S\|\psi\|_\infty(\|w\|_{L^1(B_2^d)}+\|w\|_{L^t})$. The local L^1 norm is at most $|B_2^d|^{1/2}\|w\|_{L^2(B_2^d)}$ and is controlled by the full H^t norm. If both variables are outside the test support, the test difference vanishes. These estimates prove absolute convergence and the explicit continuity bound

$$|\langle L_K w,\psi\rangle| \le C_S(\|w\|_{H^t(B_2^d)}+\|w\|_{L^t(\mathbb{R}^d)})\|\psi\|_{C^1(\mathbb{R}^d)}.$$

Linearity follows from the bilinear integral. For any sequence of tests with common compact support S converging to zero in the smooth test-function topology, the last C^1 seminorm tends to zero, so the action tends to zero. This establishes its distributional meaning. The kernel-class conditions are the explicitly stated construction; no existence assertion about a kernel is needed here. □

Proposition 3.7 (Order-uniform normalized Fall estimate)**.** *Let $t\in[15/16,1)$ and $K\in\mathcal{K}^t(\kappa,1/2,Q_2^{\mathrm{pol}})$ of 3.6. If $w\in H^t(B_2^d)\cap L^t(\mathbb{R}^d)$ and $(1-t)L_Kw=f$ weakly on B_2^d, with $f\in L^{4d}(B_2^d)$, then*

$$\|w\|_{C^{1,1/2}(B_1^d)} \le C(d,\kappa)(\|w\|_{L^2(B_2^d)}+\|w\|_{L^t(\mathbb{R}^d)}+\|f\|_{L^{4d}(B_2^d)}),$$

where the constant is independent of t in this range.

Proof. Apply Fall [10], *Regularity results for nonlocal equations and applications*, arXiv:1806.09139v3, Definition 1.1 and Theorem 1.3(ii), with dimension d, potential zero, $\alpha=1/2$ and $P=4d$. The kernel and weak normalization are those of 3.6. Here

$$\frac{d}{P}=\frac14,\qquad 2t-1\ge\frac78>\frac12,\qquad \min\{2t-d/P-1,\alpha\}=\frac12.$$

The paragraph following that theorem asserts stability of its estimates as $t\uparrow1$ after replacing L_K by $(1-t)L_K$, with $P>d/(2t_0)$. This condition holds for $t_0=15/16$. The robustness assertion is for the coefficient class with its dimension, ellipticity and Hölder bounds fixed, not for one individually chosen kernel. Thus its bound and a near-one threshold depend only on d,κ here.

We justify uniformity on the remaining orders by a varying-order version of the source's proof. Fix $I=[a,b]\subset[15/16,1)$ with $b<1$, and put $g=f/(1-t)$. Then $\|g\|_{4d}\le(1-b)^{-1}\|f\|_{4d}$. Write

$$D=\|w\|_{L^2(B_2^d)}+\|w\|_{L^t(\mathbb{R}^d)}+\|g\|_{L^{4d}(B_2^d)}.$$

The case $D=0$ is immediate. All constants in the compact-order argument may depend on d,κ,a,b, but not on $t\in I$. The elementary radial bounds used below are

$$\int_0^1 r^{1-2t}dr \le \frac{1}{2(1-b)}, \qquad \int_1^\infty r^{-1-2t}dr \le \frac{1}{2a}, \qquad \int_1^\infty r^{\alpha-2t}dr \le \frac{1}{2a-1-\alpha}, \quad \alpha = \frac{1}{2}.$$

Here are the cutoff reductions with their bounds. Fix $x_0 \in B^d_{3/2}$, translate it to the origin, and take $R = 1/16$. Write $\varphi_s(x) = \varphi(x/s)$ for a fixed smooth function $0 \le \varphi \le 1$, equal to one on B^d_1 and supported in B^d_2. Translations by these x_0 change the weighted tail norm by only a fixed factor, since $1 + |y - x_0|$ and $1 + |y|$ are uniformly comparable. Set $\mu_t(x, y) = |x - y|^{-d-2t}$ and

$$\overline{K}(x, y) = \varphi_{2R}(x)\varphi_{2R}(y)K(x, y) + (2 - \varphi_R(x) - \varphi_R(y))\mu_t(x, y).$$

Where the first product is nonzero, both points lie in $B^d_{4R}(x_0) \subset B^d_2$ and $|x - y| < 8R < 2$. Thus $\overline{K}$ is symmetric and globally positive elliptic with uniform constants: if both points are in B^d_{2R} the first product is one, and if either is outside this ball the second coefficient is at least one. Its polar coefficient is

$$A_{\overline{K}}(x, r, \omega) = \varphi_{2R}(x)\varphi_{2R}(x + r\omega)A_K(x, r, \omega) + 2 - \varphi_R(x) - \varphi_R(x + r\omega).$$

The first product is supported strictly inside the region where A_K is controlled, and is extended by zero outside that region. The formula gives a uniform $C^{0,1/2}$ bound in (x, r), with L^∞ angular norm, on the whole polar domain; its trace at $r = 0$ is even. These are the global anisotropic coefficient conditions of Fall's Definition 1.8 used in Theorem 4.3. On each bounded polar cylinder it also has the angular Hölder bound. In particular,

$$|A_{\overline{K}}(x, r, \omega) - A_{\overline{K}}(0, 0, \omega)| \le C \min\{(|x| + r)^{1/2}, 1\},$$

so the coefficient-freezing radius is common to all $t \in I$.

For $x \in B^d_{R/4}$ define the ordinary distant integral

$$G_{v,J,s}(x) = \int_{\mathbb{R}^d} v(y)(1 - \varphi_s(y))J(x, y)\, dy,$$

and put $\overline{V} = G_{1,K,2R} - G_{1,\mu_t,R}$ and $\overline{f} = G_{w,K,2R} - G_{w,\mu_t,R}$. Direct subtraction of the weak pairings, as in Fall's Lemma 6.1, gives

$$L_{\overline{K}}w + \overline{V}w = g + \overline{f} \quad \text{in } B^d_{R/4}.$$

In these integrals $|x - y| \ge 3R/4$ wherever $1 - \varphi_R(y) \ne 0$, and $|x - y|^{-d-2t} \le C(R, d, a, b)(1 + |y|)^{-d-2t}$. Consequently

$$\|\overline{V}\|_\infty \le C, \qquad \|\overline{f}\|_\infty \le C\|w\|_{L^t} \le CD.$$

These bounds require no derivative of the rough exterior coefficient. The lower-order terms need only be bounded at this stage.

We next establish the local boundedness needed before including $\overline{V}w$ in the forcing. Put $\tau = \min\{a, d/4\} > 0$, $\chi = d/(d - 2\tau) > 1$, $P = 4d$, and $\theta = d/(2\tau P) < 1$. For functions supported in a fixed ball, splitting $|x - y|$ at one and comparing the kernel powers bounds the H^τ seminorm by the H^t seminorm plus the L^2 norm, uniformly on I. The fixed-order fractional Sobolev inequality therefore gives

$$\|z\|^2_{2\chi} \le C([z]^2_{H^t} + \|z\|^2_2).$$

Positivity of $\overline{K}$ and smooth convex approximations to the positive and negative parts give the weak inequalities $L_{\overline{K}}w_\pm \le |g + \overline{f}| + \|\overline{V}\|_\infty w_\pm$. Use tests $\eta^2(w_\pm + D)^{m-1}$, initially with bounded truncations, for $m \ge 2$. The power estimate is

$$(A-B)(A^{m-1}-B^{m-1}) \geq \frac{4(m-1)}{m^2}(A^{m/2}-B^{m/2})^2 \quad (A,B\geq 0).$$

The quadratic cutoff integral is at most $C\|\nabla\eta\|_\infty^2/(1-b)+C/(2a)$ by the radial bounds. For a cutoff support at distance $\delta \leq 1$ from the exterior of the current interior ball, the exterior contribution uses only the linear tail:

$$\sup_x \int_{\text{exterior}} |w(y)|\overline{K}(x,y)\,dy \leq C\delta^{-d-2b}(\|w\|_{L^1(B_2^d)}+\|w\|_{L^t}) \leq CD\delta^{-d-2b}.$$

Since $w_\pm + D \geq D$, the resulting linear exterior terms are bounded by the corresponding interior mth-power integral. Interior cutoff products are bounded by the sum of the two mth powers; no higher exterior moment is needed. The bounded potential has the same bound. For $z = \eta(w_\pm + D)^{m/2}$, Hölder's inequality and interpolation give

$$\int |g+\overline{f}|D^{-1}z^2 \leq C\|z\|_{2P'}^2 \leq \epsilon\|z\|_{2\chi}^2 + C\epsilon^{-\theta/(1-\theta)}\|z\|_2^2, \qquad P' = P/(P-1).$$

Here $1/P' = 1-1/P$ yields precisely the stated interpolation exponent θ. Absorbing ϵ of size c/m^2 in the Sobolev energy inequality gives, for nested interior balls $B_s^d \subset B_r^d \subset B_{R/4}^d$,

$$\|w_\pm + D\|_{L^{m\chi}(B_s^d)} \leq \left[Cm^{2/(1-\theta)}(r-s)^{-N}\right]^{1/m}\|w_\pm + D\|_{L^m(B_r^d)},$$

with a fixed finite $N = N(d,a,b)$. Truncation and approximation in H^t justify the tests and the limit. Iterate with $m_j = 2\chi^j$ and radii decreasing geometrically from $R/4$ to $R/8$. The product of iteration constants is finite because $\sum_j (j+\log m_j)/m_j < \infty$. The initial L^2 norm is at most CD, so

$$\|w\|_{L^\infty(B_{R/8}^d)} \leq CD.$$

Choose a fixed smooth η equal to one on $B_{R/16}^d$ and supported in $B_{R/8}^d$, and put $u = \eta w$. Then $u \in H^t(\mathbb{R}^d)$ and $\|u\|_\infty + \|u\|_2 \leq CD$. In $B_{R/32}^d$, the exact cutoff identity of Fall's Lemma 2.3 is

$$L_{\overline{K}}u = g + \overline{f} - \overline{V}w + G_{w,\overline{K},\eta},$$
$$G_{w,\overline{K},\eta}(x) = \int w(y)(1-\eta(y))\overline{K}(x,y)\,dy.$$

The support gap and weighted-tail comparison give $\|G_{w,\overline{K},\eta}\|_\infty \leq CD$. Thus the reduced forcing has L^{4d} norm at most CD. This is the globally localized bounded solution, with bounded forcing norm, used below. Changes of scale are fixed and alter constants only by factors depending on R, a, b. Use a fixed finite covering of B_1^d by such interior balls.

For the approximation estimates, repeat the contradiction arguments in Fall's Propositions 3.3 and 4.1 with orders $t_n \in I$, passing to a subsequence $t_n \to t_* \in I$. Take $\beta = d/P = 1/4$ and the fixed growth exponents

$$\gamma_0 = \frac{3}{4}, \qquad \gamma_1 = \frac{5}{4}, \qquad \gamma_2 = \frac{3}{2}.$$

We record explicitly the affine-source estimates needed in both approximation arguments. For a symmetric kernel J satisfying

$$|J(x,y)| \leq C(1+|x|+|y|)^\alpha |x-y|^{-d-2t}, \qquad t \in I,$$

and a linear function $U(x) = c\cdot x$, split its weak pairing against a test ψ supported in B_R, $R \geq 1$, into $B_{2R}\times B_{2R}$ and the two exterior pieces. Cauchy–Schwarz and $|U(x)-U(y)| \leq |c||x-y|$ bound

the first piece by $C_R|c|(1-b)^{-1/2}[\psi]_{H^t(B_{2R})}$. On an exterior piece, $|x-y|$ is comparable to $|y|$ and its radial integral is $\int_{2R}^{\infty} r^{\alpha-2t}dr$. Consequently

$$|\langle L_J U, \psi\rangle| \le C_R|c|\big([\psi]_{H^t(\mathbb{R}^d)} + \|\psi\|_2\big).$$

Here and below C_R may depend on d, κ, a, b, R, but is independent of t. This defines an absolutely convergent weak pairing; no pointwise principal value of $L_J U$ is required. It also gives

$$[U]^2_{H^t_{|J|}(B_{2R})} \le C_R|c|^2/(1-b),$$

$$\sup_{x\in B_R}\int_{|y|>2R}|U(x)-U(y)||J(x,y)|dy \le C_R|c|/(2a-1-\alpha).$$

The weighted energy in this display denotes $\iint_{B^2_{2R}}|U(x)-U(y)|^2|J(x,y)|dxdy$. For an energy test $\psi=\eta^2 v$, with a fixed smooth cutoff η, multiplication by η is bounded on H^t uniformly on I, by the first radial bound above. The weak-source estimate and Young's inequality therefore give, for every $\epsilon>0$,

$$|\langle L_J U, \eta^2 v\rangle| \le \epsilon[\eta v]^2_{H^t} + C_{R,\epsilon}|c|^2 + C_R|c|\|v\|_{L^2(B_{2R})}.$$

Approximation in H^t justifies these energy tests. If v has the approximation arguments' uniformly bounded local L^2 norms and growth exponent $\gamma<2a$, its distant L^1 tail is uniformly bounded: dyadic annuli and Cauchy–Schwarz give $\int_{|y|>R}|v(y)||y|^{-d-2t}dy \le CR^{\gamma-2a}$. First cut off v on a larger fixed ball as in Lemma 2.3; the resulting distant forcing is bounded by this tail. Testing the resulting equation with $\eta^2 v$, the ordinary L^{4d} forcing contributes at most $C_R\|f\|_{4d}\|v\|_2$, because $(4d)'\le 2$. Absorb ϵ below the fixed ellipticity constant. The cutoff identity and these bounds yield $[v]_{H^t(B_R)}\le C_R$ when the forcing norms and $|c|$ are bounded. This supplies the uniform Caccioppoli estimate for the affine-perturbed equations, including those in the constant-approximation argument.

For an affine weak source the cutoff changes are also distant. If $U(x)=c\cdot x$ and a cutoff replaces it by ηU, then on the smaller region

$$L_J(\eta U) - L_J U = \int U(y)(1-\eta(y))J(x,y)\,dy.$$

For the polynomial kernel bound just used, the absolute value is at most $C_R|c|[1+(2a-1-\alpha)^{-1}]$. Extending a source kernel changes its affine pairing by the analogous distant integral of $U(x)-U(y)$, with the same bound. The near part remains the weak source and energy estimate established above; no pointwise diagonal commutator is asserted.

We give the simultaneous scale, moment, and growth properties in the contradiction constructions. Normalize the localized input data by one scalar factor so that the constant-stage sequences have $\|u_n\|_{L^2(\mathbb{R}^d)}\le 1$ and the affine-stage sequences have $\|u_n\|_{L^\infty(\mathbb{R}^d)}\le 1$. The forcing norms and any linear-source coefficients are also bounded independently of n. These normalizations use the cutoff bounds, rather than the regularity estimate being proved. Failure of a uniform approximation bound means that the scale-invariant errors below have unbounded suprema.

At the constant stage, with $\gamma=\gamma_0$, set $m_{n,r}=|B^d_r|^{-1}\int_{B^d_r}u_n$ and

$$D_n(r)=r^{-\gamma-d/2}\|u_n-m_{n,r}\|_{L^2(B^d_r)},\qquad \Theta_n(r)=\sup_{\rho\ge r}D_n(\rho).$$

Since the mean is the L^2 projection onto constants, $D_n(r)\le r^{-\gamma-d/2}$. Thus every $\Theta_n(r)$ is finite for $r>0$, even if the supremum over all positive radii is infinite for an individual function. At an affine stage, with $\gamma=\gamma_1$ or γ_2, let $P_{n,r}(x)=m_{n,r}+a_{n,r}\cdot x$ be the $L^2(B^d_r)$ affine projection and set

$$D_n(r)=r^{-\gamma}\|u_n-P_{n,r}\|_{L^\infty(B^d_r)},\qquad \Theta_n(r)=\sup_{\rho\ge r}D_n(\rho).$$

The projection formula gives $|m_{n,r}| \le 1$ and $|a_{n,r}| \le C_d r^{-1}$, so $D_n(r) \le C_d r^{-\gamma}$ and again $\Theta_n(r) < \infty$. Both envelopes range over all larger radii, including those beyond the localization radius.

Pass to a subsequence whose unbounded errors exceed n. Choose $\varepsilon_n > 0$ with $\Theta_n(\varepsilon_n) > n$ and then $r_n \ge \varepsilon_n$ with $D_n(r_n) \ge \Theta_n(\varepsilon_n)/2$. Monotonicity gives

$$\tfrac{1}{2}\Theta_n(r_n) \le D_n(r_n) \le \Theta_n(r_n), \qquad \Theta_n(r_n) > n/2.$$

Write $\Theta_n = \Theta_n(r_n)$. The global norm bounds force $r_n \to 0$, since otherwise $D_n(r_n)$ would remain bounded. Hence $\Theta_n \to \infty$ simultaneously, and $D_n(Rr_n) \le \Theta_n$ for every $R \ge 1$.

At the constant stage put

$$v_n(x) = \frac{u_n(r_n x) - m_{n,r_n}}{\Theta_n r_n^{\gamma_0}}.$$

Change of variables and the projection identity give

$$\int_{B_1^d} v_n = 0, \qquad \tfrac{1}{2} \le \|v_n\|_{L^2(B_1^d)} \le 1,$$
$$\|v_n - (v_n)_{B_R^d}\|_{L^2(B_R^d)} \le R^{d/2+\gamma_0} \quad (R \ge 1).$$

Comparing means on successive dyadic balls by Cauchy–Schwarz gives $|(v_n)_{B^d_{2^{k+1}}} - (v_n)_{B^d_{2^k}}| \le C_d 2^{k\gamma_0}$. The initial mean is zero and $\gamma_0 > 0$, so summing gives $|(v_n)_{B^d_{2^k}}| \le C 2^{k\gamma_0}$. Add the oscillation bound and enclose any B_R^d in the next dyadic ball to obtain

$$\|v_n\|_{L^2(B_R^d)} \le C R^{d/2+\gamma_0} \quad (R \ge 1).$$

At an affine stage use the v_n below, obtained by subtracting $P_{n,r_n}(r_n x)$. Its moments and normalization satisfy

$$\int_{B_1^d} v_n = \int_{B_1^d} v_n x_j = 0 \quad (1 \le j \le d), \qquad \tfrac{1}{2} \le \|v_n\|_{L^\infty(B_1^d)} \le 1.$$

The affine projection commutes with dilation and subtraction of an affine function. Thus

$$P_R v_n(x) = \frac{P_{n,Rr_n}(r_n x) - P_{n,r_n}(r_n x)}{\Theta_n r_n^{\gamma}},$$
$$\|v_n - P_R v_n\|_{L^\infty(B_R^d)} \le R^{\gamma} \quad (R \ge 1).$$

Write $P_{2^k} v_n = b_{n,k} + q_{n,k} \cdot x$. Comparison of the two residuals on $B^d_{2^k}$ gives $\|P_{2^{k+1}} v_n - P_{2^k} v_n\|_{L^\infty(B^d_{2^k})} \le C 2^{k\gamma}$. Consequently $|b_{n,k+1} - b_{n,k}| \le C 2^{k\gamma}$ and $|q_{n,k+1} - q_{n,k}| \le C 2^{k(\gamma-1)}$. The unit-ball moments give $b_{n,0} = q_{n,0} = 0$. Since $\gamma > 1$, summation yields $|b_{n,k}| \le C 2^{k\gamma}$ and $|q_{n,k}| \le C 2^{k(\gamma-1)}$. Add the residual bound and use dyadic enclosure to conclude

$$\|v_n\|_{L^\infty(B_R^d)} \le C R^{\gamma} \quad (R \ge 1).$$

The growth estimates hold also when Rr_n exceeds the localization radius. Their constants depend only on d and the fixed exponents, not on t_n or the equation's sources. Additional linear weak sources do not change this algebraic extraction. At the constant stage, subtracting a constant changes no weak operator. A term $L_{J_{n,j}}(c_{n,j} \cdot x)$ rescales to

$$L_{\overline{J}_{n,j}}\left(\frac{r_n^{1-\gamma_0}}{\Theta_n} c_{n,j} \cdot x\right), \qquad \overline{J}_{n,j}(x,y) = r_n^{d+2t_n} J_{n,j}(r_n x, r_n y).$$

For $r_n \le 1$ the polynomial kernel bound is preserved. The new coefficient tends to zero because $1-\gamma_0 = 1/4$ and the original coefficients are bounded. At an affine stage, the source is produced by subtracting the affine projection; the coefficient in the exact equation below is bounded by C/Θ_n using the previously established projection-slope bound. The weak-source and energy estimates therefore apply to precisely these rescaled sequences.

The constant-approximation argument at γ_0 gives a uniform preparatory $C^{3/4}$ estimate also for equations with finitely many such linear source functions having bounded coefficients. Indeed, in the constant-stage rescaling just constructed, a linear source coefficient becomes $cr_n^{1-\gamma_0}/\Theta_n$. Its rescaled kernel still has the displayed polynomial growth bound. Thus this coefficient tends to zero; the energy and weak-source bounds just proved are uniform. The same variable-order limit passage given below yields a homogeneous translation-invariant limit, and its sublinear growth contradicts the mean-zero nonzero L^2 normalization. This is the varying-order version of Proposition 3.3 and Corollary 3.6, now including their additional source terms. For the latter use the source parameters $\sigma_j = 1 > b$, $\alpha_j' = \alpha = 1/2$, $\beta = 1/4$, and $\gamma_0 = 3/4$. The requirements $\alpha_j' + \sigma_j = 3/2 < 2a$ and $\gamma_0 < 2a - \beta$ hold. Each $U_j(x) = c_j x_j$ has global Lipschitz seminorm at most $|c_j|$, and the source's required weighted norm is at most $C|c_j|/(\alpha + 2a - 1)$. All input bounds are therefore uniform. The Morrey forcing condition in that source follows directly from Hölder's inequality:

$$\sup_{z,\,0<r<1} r^{\beta-d} \int_{B_r(z)} |f| \le C\|f\|_{4d}, \qquad \beta = \frac{1}{4},$$

with forcing localized by the fixed cutoffs. No gradient estimate has been used in this preparatory argument.

Before constructing the normalized sequence at each affine stage, take a further fixed smooth cutoff supported strictly inside the region where the preceding stage's estimate holds, and equal to one on the smaller ball where the next approximation estimate is tested. The distant cutoff identity preserves bounded forcing and tail norms. The global $C^{1-\varrho}$ seminorm of this newly localized function is bounded by its established local seminorm and supremum norm. Recenter and rescale the smaller ball by fixed factors, then apply the envelope construction to these newly localized candidates. Any failure of the approximation bound on the inner ball persists, since this cutoff equals one there.

For the affine stages apply Proposition 4.1 first with $\varrho = 1/4$, $\gamma = \gamma_1$, and then with $\varrho = 0$, $\gamma = \gamma_2$. The first stage gives $C^{1,1/4}$, hence Lipschitz regularity needed for the second. Use the globally localized, normalized sequences just constructed; their forcing norms are bounded, and the cutoff-induced bounded potentials and distant forcing are included in f_n. Their affine projections are $m_{n,r} + a_{n,r} \cdot x$. The preceding preparatory bound at the first stage, and the first-stage Lipschitz bound at the second, give $[u_n]_{C^{1-\varrho}(\mathbb{R}^d)} \le C$. Since

$$a_{n,r,j} = \frac{\int_{B_r} (u_n(x) - u_n(0)) x_j dx}{\int_{B_r} x_j^2 dx}, \qquad \int_{B_r} x_j^2 dx = \frac{|B_r| r^2}{d+2},$$

we have $|a_{n,r}| \le C r^{-\varrho}$ uniformly in n. Let $r_n \downarrow 0$ and $\Theta_n \to \infty$ be the contradiction scale and amplitude divisor, and set

$$v_n(x) = \frac{u_n(r_n x) - m_{n,r_n} - r_n a_{n,r_n} \cdot x}{\Theta_n r_n^{\gamma}}.$$

The factor Θ_n is the envelope selected above; at the constant stage it is the square root of the squared envelope used in Fall's Proposition 3.3. At an affine stage put

$$\overline{K}_n(x,y) = r_n^{d+2t_n} K_n(r_n x, r_n y),$$
$$\mu_n(x,y) = A_{K_n}(0,0,(y-x)/|y-x|)|x-y|^{-d-2t_n},$$
$$J_n = r_n^{-\alpha}(\overline{K}_n - \mu_n), \qquad c_n = \Theta_n^{-1} r_n^{1+\alpha-\gamma} a_{n,r_n}.$$

The angular density of μ_n is even, so its weak operator annihilates affine functions. Coefficient freezing gives

$$|\overline{K}_n - \mu_n| \le C \min\{r_n^{\alpha}(|x|+|y|)^{\alpha}, 1\}|x-y|^{-d-2t_n},$$
$$|J_n(x,y)| \le C(|x|+|y|)^{\alpha}|x-y|^{-d-2t_n}.$$

Thus the exact rescaled weak equation is

$$L_{\overline{K}_n} v_n + L_{J_n}(c_n \cdot x) = F_n, \qquad F_n(x) = \Theta_n^{-1} r_n^{2t_n-\gamma} f_n(r_n x).$$

The projection bound and $1+\alpha-\gamma-\varrho = 0$ give $|c_n| \le C/\Theta_n \to 0$. On every fixed ball, Hölder's inequality and scaling give $\|F_n\|_{4d} \le C\Theta_n^{-1} r_n^{2t_n-\gamma-\beta} \to 0$, since the exponent is at least $1/8$. The just-proved weak-source estimate makes the affine source tend to zero on every smooth compact test and supplies its uniform energy control. Moreover, the preparatory $C^{3/4}$ estimate applies to v_n: its local L^2 norm and weighted tail are bounded by the growth exponent $\gamma < 2a$, and all its affine-source and forcing bounds have just been checked. This gives local uniform convergence at the affine stages without assuming the estimate being proved.

The established local energy bounds give local L^2 compactness of the rescaled functions through H^a. Fatou's lemma gives $H^{t_*}_{\mathrm{loc}}$ membership of the limit. Against a smooth compact test, Cauchy–Schwarz bounds the omitted diagonal weak pairing by $C\delta^{1-b}$, since

$$\int_0^{\delta} r^{1-2t_n} dr \le \frac{\delta^{2-2b}}{2(1-b)} \qquad (0<\delta<1).$$

The growth bounds give exterior tails at most CR^{γ_i-2a}, tending uniformly to zero as $R \to \infty$. On each intervening fixed annulus, the kernel powers converge and the frozen even angular coefficients have a weak-star convergent subsequence. The limiting angular density is even and lies between the common positive ellipticity bounds. The limiting weak equation is therefore the translation-invariant equation of order t_* with precisely the hypotheses required in the cited Liouville lemma. Fall's Lemma 3.2 makes its solution constant for the sublinear exponent γ_0, and affine for $\gamma_1, \gamma_2 < 2t_*$. The explicit zero moments and nonzero lower norms above contradict this conclusion. Strong local L^2 convergence preserves the constant-stage lower norm and zero mean. At the affine stages the preparatory local $C^{3/4}$ bound gives uniform convergence on the closed unit ball, preserving both the lower sup norm and all zero affine moments. Thus the approximation constants are uniform on I.

Corollary 4.2 and the two-stage bootstrap, fixed cutoffs and covering in Fall's Theorem 4.3 now give the required $C^{1,1/2}$ estimate with constant $C(d,\kappa,a,b)$. Choose $a = 15/16$ and $b = b(d,\kappa) < 1$ from the normalized near-one assertion, enlarging b if necessary. Combine the compact-interval bound on $[a,b]$ with the normalized bound on $[b,1)$. All remaining parameters are fixed, so the resulting constant depends only on d, κ. The forcing and membership hypotheses are exactly those in the statement. □

Proposition 3.8 (A normalized graph equation with Lipschitz forcing)**.** *For F_h of 2.1, let $d \ge 1$, $7/8 \le p < 1$, and suppose h is globally $C^{1,1}$ with*

$$\|\nabla h\|_{L^{\infty}(\mathbb{R}^d)} + [\nabla h]_{C^{0,1}(\mathbb{R}^d)} \le M_0.$$

If $F_h \in C^{0,1}(B_3^d)$, then

$$\|h - h(0)\|_{C^{2,1/2}(B_1^d)} \le C(d, M_0)\big(M_0 + [F_h]_{C^{0,1}(B_3^d)}\big),$$

where the constant is independent of p.

Proof. Use the graph constructions of 2.1 and the exact equation of 3.2, with G_p defined in 3.1:

$$F_h(x) = 2\sigma\,\mathrm{PV}\int_{\mathbb{R}^d} G_p\Big(\frac{h(x)-h(y)}{|x-y|}\Big)\,|x-y|^{-d-p}\,dy.$$

For $0 < |a| < 1/4$, set

$$w_a(x) = \frac{h(x+a)-h(x)}{|a|}, \qquad q_0(x,y) = \frac{h(x)-h(y)}{|x-y|}, \qquad q_a(x,y) = \frac{h(x+a)-h(y+a)}{|x-y|}.$$

We first justify the difference quotient in weak form. For $\psi \in C_c^\infty(B_2^d)$, extended by zero, symmetry and oddness give

$$\int_{\mathbb{R}^d} F_h(x)\psi(x)\,dx = \sigma \iint_{\mathbb{R}^d\times\mathbb{R}^d} G_p(q_0(x,y))(\psi(x)-\psi(y))|x-y|^{-d-p}\,dx\,dy.$$

The double integral is absolutely convergent. For $|x-y| < 1$, the secant is bounded by M_0 and $|\psi(x)-\psi(y)| \le C|x-y|$, so its radial majorant is $Cr^{-p}dr$; one variable is in a fixed bounded neighborhood of the test support. For $|x-y| \ge 1$, one variable is in the support and the majorant is $Cr^{-1-p}dr$. For evaluation points in a fixed compact neighborhood of the test support, tangent subtraction leaves a height remainder at most $(M_0/2)|x-y|^2$. Since $|G_p'(z)| \le 1$, the omitted normalized graph integral over $|x-y| < \epsilon$ is bounded by $CM_0\epsilon^{1-p}$ for each fixed $p < 1$. The tangent term cancels on each radial annulus. Thus the truncated principal values converge uniformly there; symmetrizing the truncated graph equation and removing the cutoff proves this identity. The same identity for the translated graph is valid on the test support, since $x + a \in B_3^d$ there.

Translate the integration variables in that identity and subtract. The fundamental theorem of calculus gives

$$G_p(q_a) - G_p(q_0) = b_a(x,y)\frac{|a|(w_a(x)-w_a(y))}{|x-y|},$$
$$b_a(x,y) = \int_0^1 G_p'((1-\theta)q_0 + \theta q_a)\,d\theta.$$

Thus, with $K_a(x,y) = b_a(x,y)|x-y|^{-d-1-p}$ and the weak operator of 3.6,

$$\int \frac{F_h(x+a)-F_h(x)}{|a|}\psi(x)\,dx = \sigma\iint (w_a(x)-w_a(y))(\psi(x)-\psi(y))K_a(x,y)\,dx\,dy$$
$$= 2\sigma\langle L_{K_a} w_a, \psi\rangle.$$

Here $\|w_a\|_\infty \le M_0$ and $\|\nabla w_a\|_\infty \le [\nabla h]_{C^{0,1}} \le M_0$. The last double integral is therefore also absolutely convergent, with near-diagonal radial majorant $Cr^{-p}dr$ and an integrable far tail. These bounds justify subtraction and removal of the truncations without differentiating a pointwise singular integral.

Put $t = (1+p)/2 \in [15/16, 1)$. The kernel is measurable, and it is symmetric because G_p' is even while both secants change sign under exchange of x, y. Since $|q_0|, |q_a| \le M_0$,

$$(1+M_0^2)^{-(d+2)/2} \le b_a(x,y) \le 1.$$

For its coefficient regularity, write

$$q_0(x, x+r\omega) = -\int_0^1 \nabla h(x+\lambda r\omega)\cdot\omega\, d\lambda,$$

and use the corresponding translated expression for q_a. On $0 \le r, r' \le 2$, each secant extension obeys

$$|q(x,r,\omega) - q(x',r',\omega')| \le CM_0(|x-x'| + |r-r'| + |\omega-\omega'|),$$

uniformly in a. The uniform derivatives of G_p' on $[-M_0, M_0]$ give a uniform Lipschitz bound for $A_{K_a}(x,r,\omega) = b_a(x, x+r\omega)$. At $r=0$ its value is

$$\int_0^1 G_p'(-((1-\theta)\nabla h(x) + \theta\nabla h(x+a))\cdot\omega)\, d\theta,$$

which is even in ω. Consequently K_a belongs to the class $\mathcal{K}^t(\kappa, 1/2, Q_2^{\mathrm{pol}})$ of 3.6 for a fixed $\kappa = \kappa(d, M_0) > 0$.

The normalization is exactly

$$1 - t = \frac{\sigma}{2}, \qquad (1-t)L_{K_a} w_a = \frac{1}{4}\frac{F_h(x+a) - F_h(x)}{|a|}.$$

The forcing has $L^{4d}(B_2^d)$ norm at most $C(d)[F_h]_{C^{0,1}(B_3^d)}$. The boundedness of w_a controls its local L^2 norm and, uniformly for $t \ge 15/16$, its tail:

$$\|w_a\|_{L^t(\mathbb{R}^d)} \le M_0 \int_{\mathbb{R}^d} (1+|y|)^{-d-15/8}\, dy \le C(d)M_0.$$

Its Lipschitz bound gives $w_a \in H^t(B_2^d)$ for every $t < 1$; the estimate requires this membership, not an order-uniform H^t seminorm bound. Proposition 3.7 now yields

$$\|w_a\|_{C^{1,1/2}(B_1^d)} \le C(d, M_0)(M_0 + [F_h]_{C^{0,1}(B_3^d)}).$$

For $a = \varepsilon e_j$ with $\varepsilon \downarrow 0$, $w_a(x) = \int_0^1 \partial_j h(x+\theta\varepsilon e_j)d\theta$ converges uniformly to $\partial_j h$. Arzelà–Ascoli and a diagonal argument on compact subsets of B_1^d give a subsequence on which each $\partial_i w_a$ converges to a continuous function v_{ij}. For every compact smooth test ψ,

$$\int v_{ij}\psi = -\int \partial_j h\, \partial_i \psi.$$

Thus v_{ij} is the distributional derivative $\partial_i\partial_j h$. A continuous function whose distributional first derivatives are continuous is C^1, as follows by mollification and the fundamental theorem of calculus on interior segments. Hence $h \in C^2(B_1^d)$ and $v_{ij} = \partial_i\partial_j h$. Passing to the limit in $|\partial_i w_a(x) - \partial_i w_a(y)| \le C|x-y|^{1/2}$ preserves the endpoint Hölder bound for every $x, y \in B_1^d$. Finally $|h(x) - h(0)| \le M_0|x|$, proving the asserted $C^{2,1/2}$ estimate. □

Corollary 3.9 (Uniform regularity from a critical tube)**.** *Use the graph and regular-boundary conventions of 2.1 and stationarity of 2.2. Let $7/8 \le p < 1$, let $h \in C^{1,1}(B_4^d)$, and suppose*

$$\|\nabla h\|_{L^\infty(B_4^d)} + [\nabla h]_{C^{0,1}(B_4^d)} \le M_0.$$

Let $0 < c \le 1/4$ be fixed. Suppose the graph is part of the regular boundary of a stationary set E and, after one fixed choice of its phase, E agrees with the local subgraph in $B_{c\sqrt{\sigma}}(X_h(x))$ for every $x \in B_3^d$. Then

$$\|h - h(0)\|_{C^{2,1/2}(B_1^d)} \le C(d, M_0, c).$$

This estimate is uniform in p and in the number of other interfaces of E. More precisely, fix a standard smooth cutoff $0 \le \chi \le 1$ supported in B_4^d, equal to one on $B_{7/2}^d$, and chosen once with $\|\nabla\chi\|_\infty + \|D^2\chi\|_\infty \le C(d)$. Extend by

$$\widetilde{h}(x) = h(0) + \chi(x)(h(x) - h(0)),$$

with value $h(0)$ outside B_4^d. This extension agrees with h on $B_{7/2}^d$, satisfies $\|\nabla\widetilde{h}\|_\infty + [\nabla\widetilde{h}]_{C^{0,1}} \le C(d)M_0$, and retains the same phase-agreement balls over B_3^d. Its normalized curvature satisfies

$$\|F_{\widetilde{h}}\|_{L^\infty(B_3^d)} \le C\sqrt{\sigma}, \qquad [F_{\widetilde{h}}]_{C^{0,1}(B_3^d)} \le C,$$

where the cutoff is fixed once and the constants depend only on d, M_0, c.

Proof. Use the graph curvature of 2.1 and the assumed stationarity in the sense of 2.2. Choose the fixed smooth cutoff in the statement, equal to one on $B_{7/2}^d$ and supported in B_4^d, with its stated derivative bounds. The formula

$$\widetilde{h}(x) = h(0) + \chi(x)(h(x) - h(0))$$

defines a global $C^{1,1}$ extension, with the expression equal to $h(0)$ outside B_4^d. Since $|h(x)-h(0)| \le 4M_0$ on B_4^d, its global gradient and gradient Lipschitz norms are bounded by $C(d)M_0$. The extension agrees with h on $B_{7/2}^d$. The radius is $c\sqrt{\sigma} \le 1/4$, smaller than the horizontal margin $7/2 - 3 = 1/2$, so every horizontal projection in an assumed ball stays in $B_{7/2}^d$. Thus those balls retain their phase agreement with the extended subgraph. Apply Lemma 3.4 with $\rho = c\sqrt{\sigma}$. By 3.5, its pulled-back forcing has both the asserted $C\sqrt{\sigma}$ size bound and a uniform Lipschitz bound. Proposition 3.8 applied to $\widetilde{h}$ proves the asserted uniform $C^{2,1/2}$ bound for $\widetilde{h} - \widetilde{h}(0) = h - h(0)$ on B_1^d. □

3.4. **The classical curvature comparison.**

Lemma 3.10 (The normalized curvature comparison on a fixed scale)**.** *For $F_h, \mathcal{M}, c_d, \sigma$ of 2.1, let $p \in [p_0, 1)$ with $p_0 > 0$, let $0 < \alpha \le 1$, and let h be a globally Lipschitz graph with $h \in C^{2,\alpha}(B_3^d)$. Suppose*

$$\|\nabla h\|_{L^\infty(\mathbb{R}^d)} \le M_0, \qquad \|D^2 h\|_{L^\infty(B_3^d)} + [D^2 h]_{C^{0,\alpha}(B_3^d)} \le M_1.$$

Then

$$\sup_{B_1^d} |F_h + c_d\mathcal{M}(h)| \le C(d, p_0, \alpha, M_0, M_1)\sigma.$$

In particular the comparison error is $o(\sqrt{\sigma})$ as $p \uparrow 1$. The estimate involves this one fixed extended graph and contains no sum over neighboring sheets.

Proof. Fix $x \in B_1^d$ and use 3.2 with 3.1 and the curvature conventions of 2.1. With $q = \nabla h(x)$, $A = D^2 h(x)$, and $y = x + r\omega$, Taylor's theorem along the segment from x to $x + r\omega$ gives, uniformly in $0 < r < 1$,

$$\frac{h(x) - h(x + r\omega)}{r} = -q \cdot \omega - \frac{r}{2} A[\omega, \omega] + O(r^{1+\alpha}).$$

The remainder in this secant expansion has absolute value at most $CM_1 r^{1+\alpha}$. The first two derivatives of G_p are bounded on the relevant bounded interval independently of $p \in [p_0, 1)$. Taylor expansion of G_p consequently leaves a remainder bounded by $C(M_1 r^{1+\alpha} + M_1^2 r^2)$, which is $O(r^{1+\alpha})$ because $0 < \alpha \le 1$ and $r < 1$. Therefore

$$G_p\left(\frac{h(x)-h(x+r\omega)}{r}\right)$$
$$= G_p(-q\cdot\omega) - \frac{r}{2}A[\omega,\omega](1+(q\cdot\omega)^2)^{-(d+1+p)/2} + O(r^{1+\alpha}).$$

The first term integrates to zero on the sphere because it is odd. Since $\sigma\int_0^1 r^{-p}\,dr = 1$, and

$$\sigma\int_0^1 r^{\alpha-p}\,dr = \frac{\sigma}{\alpha+1-p} \le \frac{\sigma}{\alpha},$$

the near part of the normalized graph equation is

$$-\int_{S^{d-1}} A[\omega,\omega](1+(q\cdot\omega)^2)^{-(d+1+p)/2}\,d\omega + O(\sigma).$$

The tail $r \ge 1$ is $O(\sigma)$: the argument of G_p is bounded by the global Lipschitz constant and $\int_1^\infty r^{-1-p}\,dr = 1/p \le 1/p_0$.

Replacing p by 1 in the angular weight changes the displayed leading angular integral by $O(\sigma)$, by the mean-value theorem in the exponent. The required angular identity is

$$\int_{S^{d-1}} \frac{\omega\otimes\omega}{(1+(q\cdot\omega)^2)^{(d+2)/2}}\,d\omega = \frac{c_d}{\sqrt{1+|q|^2}}\left(I - \frac{q\otimes q}{1+|q|^2}\right).$$

Here is a direct derivation. For a positive definite matrix Q, evaluation of the Gaussian integral in polar coordinates and after the linear change $z = Q^{1/2}x$ gives

$$\int_{S^{d-1}} (\omega\cdot Q\omega)^{-d/2}\,d\omega = |S^{d-1}|(\det Q)^{-1/2}.$$

In a sufficiently small matrix neighborhood of any positive definite Q, the quadratic form $\omega\cdot Q\omega$ has a common positive lower bound on the compact sphere. Its matrix derivatives are therefore uniformly bounded there, justifying differentiation under the angular integral. Differentiating in every symmetric matrix direction gives

$$\int_{S^{d-1}} (\omega\otimes\omega)(\omega\cdot Q\omega)^{-(d+2)/2}\,d\omega = c_d(\det Q)^{-1/2}Q^{-1}.$$

Taking $Q = I + q\otimes q$ proves the displayed angular matrix identity. Contracting that identity with A gives

$$\int_{S^{d-1}} A[\omega,\omega](1+(q\cdot\omega)^2)^{-(d+2)/2}\,d\omega = c_d\mathcal{M}(h)(x).$$

Substitution into the displayed leading angular integral proves the asserted $C\sigma$ comparison error, uniformly for $x \in B_1^d$. □

Corollary 3.11 (The classical curvature of a separated sheet is small)**.** *For the classical graph operator of 2.1, in a fixed smaller chart, a graph satisfying the hypotheses of Corollary 3.9 has*

$$|\mathcal{M}(h)| \le C\sqrt{\sigma}.$$

More precisely, the conclusion holds on $B_{1/2}^d$ under the hypotheses of that corollary, with a constant depending only on d, M_0, c.

Proof. Use the global cutoff extension $\widetilde{h}$ supplied by 3.9, with the graph conventions of 2.1. That corollary gives its uniform global slope bound, $|F_{\widetilde{h}}| \le C\sqrt{\sigma}$ on B_3^d, and uniform $C^{2,1/2}$ control of $h - h(0)$ on B_1^d.

For $z \in B_{1/2}^d$, set $r = 1/8$ and

$$k(y) = \frac{\widetilde{h}(z + ry) - \widetilde{h}(z)}{r}.$$

Its radius-three local base lies in B_1^d, since $|z| + 3r < 1$, and its global slope bound is uniform. Moreover $D^2k = rD^2h(z + r\,\cdot)$ and $[D^2k]_{C^{0,1/2}(B_3^d)} \le r^{3/2}[D^2h]_{C^{0,1/2}(B_1^d)}$. Thus 3.10 applies at zero. The similarity rule of 2.11 gives

$$F_k(0) = r^p F_{\widetilde{h}}(z), \qquad \mathcal{M}(k)(0) = r\mathcal{M}(h)(z).$$

It follows that $c_d r|\mathcal{M}(h)(z)| \le Cr^p\sqrt{\sigma} + C\sigma$. Since r is fixed, $c_d > 0$, and $\sigma \le \sqrt{\sigma}$, this proves the conclusion uniformly over z. □

4. Adjacent-gap inequalities and sharp sheet separation

Stationarity compares the two sides of an adjacent gap, while stability controls its inverse square. Their matching coefficient permits an inverse-power estimate strong enough in horizontal dimension six. The last argument creates the needed critical tubes by point selection and proves separation for the whole clipped chart.

4.1. **Adjacent interfaces and phase comparison.**

Lemma 4.1 (The combined exterior term is a volume-controlled flux)**.** *Use $m = d + 1$, $p \in (0, 1)$, $\sigma = 1 - p$, outward normals, and curvature from 2.1. Let E be stationary in the sense of 2.2, with boundary Γ regular in an open neighborhood of the compact support S of a normal test ϕ. Let $\chi \in C_c^\infty(\mathbb{R}^{d+1})$ satisfy $0 \le \chi \le 1$ and equal one in a neighborhood of S, and assume that the distance from S to $\{\chi \ne 1\}$ is positive. Split the quadratic form 2.10 using the factors $\chi(X)\chi(Y)$ and $1 - \chi(X)\chi(Y)$. The latter part is exactly*

$$\int_\Gamma \phi(X)^2\nu_E(X) \cdot \int_\Gamma (1 - \chi(Y))\nu_E(Y)|X - Y|^{-d-1-p}\, d\mathcal{H}_Y^d\, d\mathcal{H}_X^d.$$

For $p \ge p_0 > 0$, its absolute value is bounded by

$$C(d, p_0, \chi, S) \int_\Gamma \phi^2.$$

In particular its contribution to $2\sigma Q_p[\phi]$ is $O(\sigma)\int_\Gamma \phi^2$, independently of the area or the number of interfaces outside the cutoff.

If the remote boundary does not carry a classical normal measure, the inner integral in the displayed flux means the distributional quantity

$$\int_E \nabla_Y\big((1 - \chi(Y))|X - Y|^{-m-p}\big)\, dY;$$

this is absolutely convergent and gives the same conclusion for a phase that is only locally regular.

Proof. Use the dimension, normalization, and outward-normal conventions of 2.1. All remote surface-flux notation in this proof has the paired meaning of 2.1. When a weighted surface integral does not converge separately, it denotes the outer-cutoff limit defined by the absolutely convergent volume-gradient integral. This convention also applies when classical normals exist remotely but their unsigned weighted surface measure is infinite. If X belongs to the support of ϕ and $1 - \chi(Y) \ne 0$, then $\phi(Y) = 0$ and $\chi(X) = 1$. The two ordered Dirichlet terms, including their factor $1/2$, contribute $\phi(X)^2|X - Y|^{-d-1-p}$. The normal-potential term subtracts $\phi(X)^2(1 - \nu_E(X) \cdot \nu_E(Y))|X - Y|^{-d-1-p}$. By 2.10, their sum is exactly the combined exterior integrand stated above.

For the outward normal, the distributional Gauss–Green formula is

$$\int_{\partial E} f(Y)\nu_E(Y)\,d\mathcal{H}^d_Y = \int_E \nabla_Y f(Y)\,dY.$$

Apply it with $f(Y) = (1-\chi(Y))|X-Y|^{-d-1-p}$. This function has no singularity near X, and its gradient is integrable. In fact, with $m = d+1$, the two terms to be estimated are

$$|\nabla\chi(Y)|\,|X-Y|^{-m-p}, \qquad (m+p)(1-\chi(Y))|X-Y|^{-m-p-1}.$$

Put $\rho = \operatorname{dist}(S, \{\chi \neq 1\}) > 0$. The supports of $1-\chi$ and $\nabla\chi$ have distance at least ρ from each $X \in S$; for the latter use $0 \le \chi \le 1$, so its derivative vanishes at points where $\chi = 1$. Compactness of $\operatorname{supp}\nabla\chi$ therefore gives

$$\int |\nabla\chi(Y)|\,|X-Y|^{-m-p}\,dY \le \|\nabla\chi\|_\infty |\operatorname{supp}\nabla\chi| \max\{\rho^{-m-p_0}, \rho^{-m-1}\}.$$

For the second term, radial integration yields

$$(m+p)\int_{|Y-X|\ge\rho} |X-Y|^{-m-p-1}\,dY = \frac{(m+p)|S^{m-1}|}{p+1}\rho^{-p-1} \le C(m)\max\{\rho^{-p_0-1}, \rho^{-2}\}.$$

Both bounds are uniform for $X \in S$ and $p \in [p_0, 1)$. This proves the asserted uniform bound. If the remote boundary is not regular, the same expression is defined by the distributional derivative of χ_E and gives the identical bound. For the noncompact pairing, choose $\eta \in C^\infty_c(U)$ equal to one near S, where U is an open regular-boundary neighborhood of S. The kernel $(1-\eta(Y))|X-Y|^{-m-p}$ has the exhaustion-independent pairing stated in 2.7. Its difference from the kernel with χ is

$$(1-\chi(Y))|X-Y|^{-m-p} - (1-\eta(Y))|X-Y|^{-m-p} = (\eta(Y)-\chi(Y))|X-Y|^{-m-p}.$$

This is a compact smooth test vanishing near every evaluation point in S. Its distributional normal pairing is the volume-gradient integral of 2.1, even if its support meets an irregular part of the boundary. Adding this compact pairing to the noncompact pairing from 2.7 justifies the split for the original χ and every outer exhaustion. The uniform-in-p bound is the volume-gradient estimate proved above. No estimate of remote surface measure is involved. □

Proposition 4.2 (An adjacent-pair curvature bracket)**.** *Use the curvature and normalized graph curvature $F_h = \sigma H_p[B_h]$ of 2.1, and stationarity of 2.2. Let $h_- < h_+$ be globally defined C^2 graphs in $\mathbb{R}^{d+1}$, and write*

$$B_\pm = \{t < h_\pm(y')\}, \qquad S = \{h_-(y') < t < h_+(y')\}, \qquad U = \{t > h_+(y')\}, \qquad L = \{t < h_-(y')\}.$$

Suppose the two graphs are adjacent regular boundary sheets of a stationary phase E in an open cylinder Q: the open intervening slab contains no boundary, and E occupies exactly one local side at every graph point in Q. Fix $z_\pm = (x', h_\pm(x')) \in Q$ and assume that E fills S in Q. Put $K_\pm(Y) = |z_\pm - Y|^{-d-1-p}$. Then

$$F_{h_-}(x') - F_{h_+}(x') = 2\sigma\int_U K_- + 2\sigma\int_L K_+ - 2\sigma\int_{(U\cup L)\cap E}(K_- + K_+) + 2\sigma\int_{S\setminus E}(K_- + K_+).$$

Consequently, if both $z_\pm$ have distance at least r from Q^c, then, uniformly for $p \ge p_0 > 0$,

$$F_{h_-}(x') - F_{h_+}(x') \le 2\sigma\int_U K_- + 2\sigma\int_L K_+ + C(d,p_0)\sigma r^{-p}.$$

If E is empty in the intervening slab, apply the assertion to E^c instead. No condition on the number of other sheets is needed.

Proof. Use H_p, F_h from 2.1 and the flow notion of stationarity from 2.2. Around each regular sheet point $z_\pm$, choose a sufficiently small open neighborhood contained in Q in which the actual boundary is the stated C^2 graph and E occupies exactly one side. It is therefore a genuine $C^{1,1}$ phase boundary there in the sense of 2.3. Every compact smooth flow supported in that neighborhood is admissible in Q, so the assumed stationarity restricts to it. Apply 2.9 in these neighborhoods, with the locally finite perimeter-difference convention of 2.1; that proposition allows arbitrary exterior data and yields $H_p[E](z_-) = H_p[E](z_+) = 0$. The following pointwise identities hold in the three regions L, S, U:

$$1 - 2\chi_E = -(1 - 2\chi_{B_-}) + 2(1 - \chi_E)\chi_U - 2\chi_E\chi_L + 2(1 - \chi_E)\chi_S,$$
$$1 - 2\chi_E = (1 - 2\chi_{B_+}) + 2(1 - \chi_E)\chi_L - 2\chi_E\chi_U + 2(1 - \chi_E)\chi_S.$$

Integrate the first identity on $\mathbb{R}^{d+1} \setminus B_\delta(z_-)$ against σK_-, and the second on $\mathbb{R}^{d+1} \setminus B_\delta(z_+)$ against σK_+. The graph and phase curvatures have principal values by 2.6. Each correction term vanishes near its relevant evaluation point and has an integrable far tail, so letting $\delta \downarrow 0$ leaves the ordinary correction integrals and the stationary curvatures zero. Solving these two identities for F_{h_-} and F_{h_+} and subtracting gives the exact four-term identity in the statement. The displayed correction integrals are ordinary absolutely convergent integrals: the opposite graphical halfspace stays away from the evaluation point, and the correction phases vanish near each relevant point. The third term on the right of that identity is nonpositive. The last is supported in Q^c, since E fills S in Q. The identity

$$\int_{|Y-z|\geq r} |z - Y|^{-d-1-p}\, dY = \frac{|S^d|}{p} r^{-p}$$

proves the upper bound, with the constant uniform for $p \geq p_0$.

For an empty intervening slab, apply 2.11: E^c is stationary in the same cylinder and has the same regular graph interfaces. Complementation reverses their one-sided occupancy and fills the slab. Thus all hypotheses of the filled-slab assertion hold for this single chosen phase, giving the complementary case. □

Corollary 4.3 (Ordered local extensions and the local phase bracket)**.** *Use H_p, F_h, B_h, $m = d+1$, and $\sigma = 1 - p$ from 2.1. Let $D_0 \Subset D_1 \Subset D_2 \subset \mathbb{R}^d$ be horizontal balls, and let $h_- < h_+$ be C^2 on a neighborhood of $\overline{D}_2$. Suppose E is stationary in $Q = D_1 \times (-T, T)$ in the sense of 2.2. The two graphs are actual C^2 boundary sheets there, the open intervening slab contains no boundary, and E occupies exactly one local side at each of their points in Q. Suppose their complete patches over D_1 lie between the caps and E fills their intervening slab in Q; alternatively replace E by E^c. Choose a common smooth cutoff $0 \leq \theta \leq 1$ that equals one near $\overline{D}_1$ and has support in D_2, and any constants $C_- < C_+$. The global ordered extensions*

$$\widetilde{h}_\pm = \theta h_\pm + (1 - \theta) C_\pm$$

are C^2 and equal to the original graphs near $\overline{D}_1$. At $z_\pm = (x', h_\pm(x'))$ for $x' \in D_0$, set $a = \min_\pm \operatorname{dist}(z_\pm, Q^c) > 0$, $\widetilde{U} = \{t > \widetilde{h}_+(y')\}$, $\widetilde{L} = \{t < \widetilde{h}_-(y')\}$, and $K_\pm(Y) = |z_\pm - Y|^{-d-1-p}$. For $p \geq p_0 > 0$,

$$F^{\widetilde{}}_{\widetilde{h}_-}(x') - F^{\widetilde{}}_{\widetilde{h}_+}(x') \leq 2\sigma \int_{\widetilde{U}} K_- + 2\sigma \int_{\widetilde{L}} K_+ + C(d, p_0)\sigma a^{-p}.$$

If another global extension $\widehat{h}$ agrees with $\widetilde{h}$ on D_1, then at $x' \in D_0$, writing $a_0 = \operatorname{dist}(D_0, D_1^c) > 0$,

$$|F^{\widehat{}}_{\widehat{h}}(x') - F^{\widetilde{}}_{\widetilde{h}}(x')| \leq C(d, p_0)\sigma a_0^{-p}.$$

Both constants are independent of extension heights and the number of other interfaces.

Proof. The common cutoff preserves ordering:

$$\widetilde{h}_+ - \widetilde{h}_- = \theta(h_+ - h_-) + (1-\theta)(C_+ - C_-) > 0.$$

The cutoff θ vanishes near D_2^c, so both formulas define global C^2 graphs; they coincide with the original boundary patches near $\overline{D}_1$. The stationary phase and its genuine intervening slab are therefore unchanged in Q. Applying Proposition 4.2 at $z_\pm$ gives the first assertion, using the subgraph and curvature conventions of Definition 2.1 and stationarity as in Definition 2.2.

For the extension comparison, equality of the graphs on D_1 implies equality of their subgraph characteristic functions there. Their difference is supported over D_1^c, hence at distance at least a_0 from $z = (x', h(x'))$. The local singularity disappears when the two curvatures are subtracted, and direct volume integration yields

$$|F_{\widehat{h}}(x') - F_{\widetilde{h}}(x')| \le 2\sigma \int_{|Y-z|\ge a_0} |Y-z|^{-m-p}\, dY = \frac{2|S^{m-1}|}{p}\sigma a_0^{-p}.$$

This estimate uses only the distance of the horizontal bases; extension heights and remote interface areas do not enter it. □

4.2. **Inverse powers and alternating interactions.**

Lemma 4.4 (Weak regularity suffices for the inverse-power estimate)**.** *Let $V \subset \mathbb{R}^d$ be open, $\kappa > 0$, and $v \in C^1(V)$ be positive. Assume $\Delta v \le \kappa/v$ in distributions and*

$$\kappa \int_V v^{-2}\psi^2 \le \int_V |\nabla\psi|^2 \qquad (\psi \in C_c^1(V)).$$

Then for every $2 < q < 4 + 2\sqrt{2}$ and every compactly supported Lipschitz $0 \le \zeta \le 1$,

$$\int_V v^{-q}\zeta^q \le C(q,\kappa)\int_V |\nabla\zeta|^q.$$

If $V = \mathbb{R}^d$ and $d \le 6$, no such positive function exists.

Proof. The distributional inequality means $-\int \nabla v\cdot\nabla f \le \kappa \int f/v$ for every nonnegative $f \in C_c^\infty(V)$. A compactly supported nonnegative C^1 function f can be extended by zero and mollified by nonnegative kernels in a fixed compact subset of V. The mollifications converge in C^1. Since v is positive and has a positive minimum on that set, both integrals pass to the limit. Thus the inequality also holds for nonnegative C_c^1 tests.

Put $t = (q-2)/2 > 0$, choose $\eta \in C_c^\infty(V)$, and define

$$I = \int v^{-2t-2}\eta^2, \quad J = \int v^{-2t-2}|\nabla v|^2\eta^2, \quad K_0 = \int v^{-2t-1}\eta\nabla v\cdot\nabla\eta, \quad H_0 = \int v^{-2t}|\nabla\eta|^2.$$

The admissible nonnegative distributional test $f = v^{-2t-1}\eta^2$ and the stability test $\psi = v^{-t}\eta$ give

$$(2t+1)J \le \kappa I + 2K_0, \qquad \kappa I \le t^2 J - 2tK_0 + H_0.$$

For $a = 2t+1$, $b = t^2$, and $0 < \varepsilon \le \min\{a/4, (a-b)/4\}$, which is possible because $q < 4 + 2\sqrt{2}$, Young's inequality gives

$$(a-\varepsilon)J \le \kappa I + \varepsilon^{-1}H_0, \qquad \kappa I \le (b+\varepsilon)J + (1 + b/\varepsilon)H_0.$$

Eliminating J gives

$$\kappa\frac{a-b-2\varepsilon}{a-\varepsilon}I \le \left[\frac{b+\varepsilon}{\varepsilon(a-\varepsilon)} + 1 + \frac{b}{\varepsilon}\right]H_0.$$

Both the denominator and the coefficient on the left are positive, so $I \le C(t,\kappa)H_0$. By C^1 approximation this holds also for compactly supported C^1 η. Take $\eta = \zeta^{t+1}$ first for a smooth nonnegative cutoff ζ; this power is C^1 even at its zeros. Since $q = 2t+2$, Hölder gives

$$H_0 \le (t+1)^2 I^{(q-2)/q} \left(\int |\nabla \zeta|^q \right)^{2/q}.$$

If $I > 0$, division by $I^{(q-2)/q}$ and raising to $q/2$ gives the asserted estimate; for $I = 0$ it is immediate. A compactly supported Lipschitz ζ extended by zero admits smooth nonnegative mollifications in $[0,1]$, with support in one compact subset of V, uniform convergence and gradient convergence in L^q. Positivity of v on that compact set lets the inverse-power integral pass by dominated convergence, proving the same estimate for ζ.

Finally choose $\max\{2,d\} < q < 4+2\sqrt{2}$ when $d \le 6$, and a cutoff equal to one on B_R, supported in B_{2R}, with gradient at most C/R. Then $\int_{B_R} v^{-q} \le CR^{d-q} \to 0$. The integral on any fixed ball must be zero, contrary to the positive continuous integrand. □

Lemma 4.5 (Inverse moments from a matching equation and stability)**.** *Let $v > 0$ on an open set $V \subset \mathbb{R}^d$. Suppose that $v \in C^1(V) \cap W^{2,r}_{\mathrm{loc}}(V)$ for some $r > d$, and that, for a constant $\kappa > 0$,*

$$\Delta v \le \frac{\kappa}{v}, \qquad \kappa \int_V v^{-2}\psi^2 \le \int_V |\nabla \psi|^2 \quad (\psi \in C^1_c(V)).$$

For every

$$2 < q < 4 + 2\sqrt{2}$$

and every compactly supported Lipschitz function $0 \le \zeta \le 1$,

$$\int_V v^{-q}\zeta^q \le C(q,\kappa) \int_V |\nabla \zeta|^q.$$

In particular, if $d \le 6$, no positive function of the stated regularity on all of $\mathbb{R}^d$ satisfies both displayed hypotheses.

Proof. For a nonnegative $f \in C^\infty_c(V)$, the assumed local Sobolev regularity permits integration by parts and gives

$$-\int_V \nabla v \cdot \nabla f = \int_V (\Delta v) f \le \kappa \int_V f/v.$$

Thus the differential inequality holds in distributions. The positivity, C^1 regularity and stability inequality are exactly the hypotheses of 4.4. Applying that lemma proves both the inverse-moment bound for compactly supported Lipschitz cutoffs and the entire-space contradiction when $d \le 6$. □

Lemma 4.6 (Every adjacent gap retains coefficient four)**.** *Let $V \subset \mathbb{R}^d$ be open, let the indices form an interval $I \subset \mathbb{Z}$, finite or infinite, and let w_i be strictly ordered functions on V. Interpret the displayed form as a finite, well-defined integral for each compactly supported C^1 test with finitely many nonzero components. In the infinite case suppose the consecutive gaps have a uniform positive lower bound on every compact subset of V. Let A be a constant positive definite matrix and $L > 0$. Assume the form*

$$\int_V \sum_i \nabla \phi_i \cdot A \nabla \phi_i + 2L \int_V \sum_{i<k} (-1)^{i-k} \frac{(\phi_i - \phi_k)^2}{(w_k - w_i)^2} \ge 0$$

holds for compactly supported tests with finitely many nonzero components. For an adjacent gap $v = w_{i+1} - w_i$, it follows that

$$4L\int_V \frac{\eta^2}{v^2} \le \int_V \nabla\eta\cdot A\nabla\eta.$$

Here $\eta \in C_c^1(V)$. It is enough to assume the displayed form only for the test $\phi_i = -\eta$, $\phi_{i+1} = \eta$, all other components zero.

Proof. That test has gradient term $2\int \nabla\eta\cdot A\nabla\eta$ and adjacent-pair potential $-8L\int v^{-2}\eta^2$. For an index $k<i$, put $a_k = w_i - w_k > 0$. Its remaining contribution is

$$2L(-1)^{i-k}\eta^2\left(\frac{1}{a_k^2} - \frac{1}{(a_k+v)^2}\right).$$

For each fixed $v>0$, the function $b_v(a) = a^{-2} - (a+v)^{-2}$ is positive for $a>0$, since $a < a+v$. Its derivative is

$$b_v'(a) = -2a^{-3} + 2(a+v)^{-3} < 0 \qquad (a>0),$$

so it is strictly decreasing. Reading the left indices from $i-1$ outwards gives an alternating sum starting with a negative term: pair successive terms as $-b_1 + b_2 \le 0$, $-b_3 + b_4 \le 0$, and so on, with a possible final negative term. This proves that contribution is nonpositive. On the right put $a_k = w_k - w_{i+1} > 0$ for $k > i+1$. The combined contribution of the two tested endpoints is

$$2L(-1)^{i+1-k}\eta^2\left(\frac{1}{a_k^2} - \frac{1}{(a_k+v)^2}\right),$$

and the right indices read from $i+2$ outwards again start with a negative term and have decreasing magnitudes. The same grouping proves their sum nonpositive. In the infinite case, fix the compact support K of η and let $\delta_K > 0$ be its lower bound for consecutive gaps. Then $|w_k - w_i| \ge \delta_K|k-i|$ and $|w_k - w_{i+1}| \ge \delta_K|k-i-1|$. Every remaining endpoint term is dominated on K by $C_K\eta^2|k-i|^{-2}$, after changing the constant for the finitely many nearest indices. This majorant is summable in k and integrable in the spatial variable. Thus the sums converge absolutely under the integral and the negative-first grouping is legitimate. Thus the left side of the assumed form is bounded above by

$$2\int \nabla\eta\cdot A\nabla\eta - 8L\int v^{-2}\eta^2.$$

Its nonnegativity proves the stated coefficient-four bound after division by two. □

4.3. **The finite-packet limit.**

Proposition 4.7 (The sharp equation and stability of a limit gap)**.** *Use normalized curvature and graph notation from 2.1 and stationarity and stability from 2.2. Let $V \subset \mathbb{R}^d$ be open, let V_j exhaust V, and let $T_j \to \infty$. Suppose $p_j \uparrow 1$, $\sigma_j = 1-p_j$, and E_j is stationary and stable in $V_j \times (-T_j, T_j)$. Its entire boundary there is the clipped union of a fixed finite ordered family $h_{j,1} < \cdots < h_{j,K}$, and the phase changes side at each graph. Suppose, on every compact subset of V, that the graphs have a common $C^{2,\alpha}$ bound for some $\alpha>0$, and*

$$\frac{h_{j,i}}{\sqrt{\sigma_j}} \longrightarrow g_i \quad \text{in } C^1_{\mathrm{loc}}(V), \qquad g_1 < \cdots < g_K.$$

Then every adjacent gap $v = g_{i+1} - g_i$ satisfies

$$\Delta v \le \frac{4}{v}, \qquad 4\int_V v^{-2}\eta^2 \le \int_V |\nabla\eta|^2 \quad (\eta \in C_c^1(V)).$$

The differential inequality is distributional. In particular the proposition applies when $V = \mathbb{R}^d \setminus \{0\}$.

Proof. Use the normalized curvature and graph operator of 2.1 and the variational conventions of 2.2. Write $X_{j,i}(x') = (x', h_{j,i}(x'))$ and $n_{j,i} = (-\nabla h_{j,i}, 1)/\sqrt{1+|\nabla h_{j,i}|^2}$ for the upward unit normal. Fix compact horizontal sets $D_0 \Subset D_1 \Subset V$, with D_1 containing a neighborhood of D_0; all local estimates below are taken on a still larger compact subset of V. After discarding finitely many indices, these sets lie in V_j, all the relevant heights tend uniformly to zero, and the vertical caps of the stationary cylinder are a fixed positive distance away. Local graph extensions can be chosen in an ordered manner using one common smooth cutoff and constants $i\sqrt{\sigma_j}$:

$$\widetilde{h}_{j,i} = \theta h_{j,i} + (1-\theta) i\sqrt{\sigma_j}.$$

Their ordering follows because each consecutive difference is a convex combination of the positive original gap and $\sqrt{\sigma_j}$. Convergence of $h_{j,i}/\sqrt{\sigma_j}$ in local C^1 gives global slopes $O(\sqrt{\sigma_j})$ for these extensions; the local $C^{2,\alpha}$ bounds remain uniform. They coincide with the original graphs near D_1. Every resulting curvature-extension error is $O(\sigma_j)$ by 4.3. Cover D_0 by finitely many smaller balls in V when it is not contained in one horizontal ball; the local bracket is applied on each ball. For the calculation below continue to denote an extension by $h_{j,i}$.

The differential inequality. For the selected adjacent pair let $v_j = (h_{j,i+1} - h_{j,i})/\sqrt{\sigma_j}$. The single-sheet comparison of Lemma 3.10, and the exact identity for the classical graph operator, give

$$\frac{F_{h_{j,i}} - F_{h_{j,i+1}}}{c_d\sqrt{\sigma_j}} = \operatorname{div}(a_j \nabla v_j) + O(\sqrt{\sigma_j}), \qquad a_j = \int_0^1 D\left(\frac{z}{\sqrt{1+|z|^2}}\right)\Big|_{z=(1-t)\nabla h_{j,i}+t\nabla h_{j,i+1}} dt.$$

The single-sheet comparison is applied on a finite covering by translated and fixed-scale horizontal balls; these fixed scales change its constants uniformly for $p_j \geq 1/2$. The error is uniform on D_0. Since the rescaled heights converge in C^1, $a_j \to I_d$ uniformly and $v_j \to v$ in C^1 there. Consequently the displayed left side converges to Δv in distributions.

On a small cylinder about any point of D_0, the phase either fills the selected slab or its complement does. Apply the local form 4.3 to the appropriate phase; its conclusion for the ordered subgraph curvatures is the same in either case. The distance to the cylinder boundary is uniformly positive on a smaller cylinder. Thus its error, after division by $\sqrt{\sigma_j}$, tends to zero.

In the upper-halfspace term use $Y' = x' + \sqrt{\sigma_j}\zeta$ and $Y_{d+1} = h_{j,i}(x') + \sqrt{\sigma_j}t$. The lower threshold is

$$\frac{h_{j,i+1}(x'+\sqrt{\sigma_j}\zeta) - h_{j,i}(x')}{\sqrt{\sigma_j}} \longrightarrow v(x')$$

uniformly in $x' \in D_0$ and in bounded ζ sets. The scaling factor in the normalized volume integral is $\sigma_j^{(1-p_j)/2} \to 1$, because $\sigma_j \log \sigma_j \to 0$. On bounded ζ sets the positive minimum of v excludes the singularity, and the integrable vertical tails permit dominated convergence. On $|\zeta| > R \geq 1$, integrating over all $t \in \mathbb{R}$ gives the uniform bound $CR^{-p_j} \leq CR^{-1/2}$. Hence the volume integral converges locally uniformly to

$$\int_{\mathbb{R}^d}\int_{v(x')}^{\infty} (|\zeta|^2 + t^2)^{-(d+2)/2}\, dt\, d\zeta = \frac{c_d}{v(x')}.$$

Here the horizontal integral is $c_d t^{-2}$, as follows by radial integration (or the beta integral); the same constant is $c_d = \pi^{d/2}/\Gamma(1+d/2)$. The lower-halfspace term has the identical limit. Testing the bracket with a nonnegative compactly supported smooth function and passing to the limit therefore gives $\Delta v \leq 4/v$ in distributions.

The stability inequality. Fix $\phi_1, \ldots, \phi_K \in C_c^\infty(V)$ and choose D_0 to contain all their supports. Let $\alpha \in C_c^\infty(V)$ satisfy $0 \leq \alpha \leq 1$, $\alpha = 1$ on a neighborhood of those supports, and $\operatorname{supp}\alpha \Subset D_1$.

Choose a vertical cutoff β that is one on a fixed neighborhood of zero and is supported in a fixed bounded interval. For large j, the ambient cutoff $\chi(x',t) = \alpha(x')\beta(t)$ lies inside the stationary cylinder and restricts to $\alpha(x')$ on every graph over $\operatorname{supp}\alpha$. If the test supports meet several components of V, only finitely many components are involved: their supports are compact and have a positive distance from V^c. Choose separated compact cutoff neighborhoods in those components. The cross contributions between the neighborhoods are $O(\sigma_j)$, by their fixed positive horizontal distance and bounded local areas. Hence it suffices to sum the following calculation over connected cutoff neighborhoods.

Write $s_i = (-1)^i$. The normals have the form $\nu_{j,i} = \varepsilon_j s_i n_{j,i}$ on each connected chart, where $n_{j,i}$ is the upward graph normal and $n_{j,i} \to e_{d+1}$ uniformly. Prescribe scalar physical normal speed $s_i\phi_i$ on sheet i. For each fixed j, these speeds define a compactly supported C^1 scalar normal test on the regular graph boundary: the finitely many disjoint sheets have positive separation over the compact test base. Apply 2.8 to approximate it by traces of compact smooth ambient fields in the stationary cylinder, with convergence of the quadratic form for this fixed $p_j < 1$. Hence the prescribed traces are admissible in the stability inequality of 2.10. No approximation bound uniform in j is required.

Localize the entire quadratic form with the nonnegative $\chi(X)\chi(Y)$ before removing any potential. Lemma 4.1 bounds the exterior contribution to $2\sigma_j Q_{p_j}$ by $C\sigma_j$; the fixed finite packet has uniformly bounded local area. Remove only the nonpositive same-sheet normal potentials from the localized form. Call the resulting retained form $\mathcal{R}_j$. Stability implies $\mathcal{R}_j \geq -C\sigma_j$.

The same-sheet Dirichlet part of $\mathcal{R}_j$ converges to $c_d \int |\nabla\phi_i|^2$. Here is a direct localization argument. On the compact support of α, the slopes are $O(\sqrt{\sigma_j})$. Thus the surface Jacobians and the ratio of the graph kernel to the horizontal kernel tend to one uniformly; their error multiplies a uniformly bounded normalized horizontal energy. Choose a fixed radius ρ smaller than the distance from the test supports to $\{\alpha \neq 1\}$. In that radius the cutoff weights are one whenever the test difference is nonzero. Taylor expansion of the fixed smooth test gives the angular factor $c_d|\nabla\phi_i|^2$, and

$$\sigma_j \int_0^\rho r^{-p_j}\, dr = \rho^{\sigma_j} \to 1.$$

The Taylor remainder is bounded by $C\sigma_j \int_0^\rho r^{1-p_j}\, dr \to 0$. The part with horizontal distance at least ρ is $O(\sigma_j)$, because the horizontal integration region is fixed and bounded. This proves the asserted limit.

For $i \neq k$ use the *localized* cross row

$$R_{j,ik}(x') := \sigma_j \int_{D_1} \alpha(y')\, |X_{j,i}(x') - X_{j,k}(y')|^{-d-1-p_j} \sqrt{1 + |\nabla h_{j,k}(y')|^2}\, dy' \longrightarrow \frac{c_d}{(g_k(x') - g_i(x'))^2}$$

uniformly on the test supports. To prove this, set $y' - x' = \sqrt{\sigma_j}\zeta$. On bounded ζ sets the cutoff is one, the scaled vertical difference tends uniformly to $g_k(x') - g_i(x')$, and the surface Jacobian tends to one. Strict limiting separation on the compact set excludes the singularity. The scaling factor is again $\sigma_j^{(1-p_j)/2} \to 1$. For $|\zeta| > R \geq 1$, the horizontal distance alone and the uniform slope bound give a row-tail bound $CR^{-1-p_j} \leq CR^{-3/2}$. The limiting horizontal integral is

$$\int_{\mathbb{R}^d} (|\zeta|^2 + (g_k - g_i)^2)^{-(d+2)/2}\, d\zeta = \frac{c_d}{(g_k - g_i)^2}.$$

The same tail splitting shows that a fixed continuous test at y' can be replaced by its value at x' in the cross integral. All rows with a nonzero test at one endpoint are therefore bounded uniformly; the reverse endpoint is treated by interchanging x' and y'.

For an unordered pair $i < k$, the exact combined local cross integrand in $2\sigma_j Q_{p_j}$ is

$$2\sigma_j\chi(X)\chi(Y)\left[\nu_{j,i}(X)\cdot\nu_{j,k}(Y)(\phi_i(x')^2+\phi_k(y')^2)-2s_is_k\phi_i(x')\phi_k(y')\right]|X-Y|^{-d-1-p_j}.$$

The scalar product of the normals tends uniformly to $s_is_k=(-1)^{i-k}$, and the localized row limits just proved justify every term in its integral. Consequently this pair contributes in the limit

$$2c_d\int_V(-1)^{i-k}\frac{(\phi_i-\phi_k)^2}{(g_k-g_i)^2}.$$

There are only finitely many pairs. Thus $\mathcal{R}_j$ has a limit, and $\mathcal{R}_j\ge -C\sigma_j$ proves

$$\int_V\sum_i|\nabla\phi_i|^2+2\int_V\sum_{i<k}(-1)^{i-k}\frac{(\phi_i-\phi_k)^2}{(g_k-g_i)^2}\ge 0.$$

Apply Lemma 4.6 with $A=I_d$ and $L=1$, first for smooth η and then by compactly supported C^1 approximation. The gap is bounded away from zero on the test support, so this approximation also passes through the potential term. This gives the claimed coefficient-four stability inequality. □

4.4. **Separation without a sheet-number bound.**

Theorem 4.8 (Separation in a clipped graph chart)**.** *Use $Q_R=B_R^d\times(-R,R)$ and $\sigma=1-p$ from 2.1 and the phase notions of 2.2. Let $1\le d\le 6$ and $M<\infty$. There are $p_*(d,M)<1$ and $c(d,M)>0$ with the following property. Suppose $p\in(p_*,1)$, E is stationary and stable in Q_2, and its entire boundary there is the clipped union of a finite ordered family of C^2 graphs $h_1<\cdots<h_K$ on B_3^d, satisfying*

$$\sup_i\|\nabla h_i\|_{L^\infty(B_3)}\le\frac{1}{100},\qquad\sup_i\|D^2h_i\|_{L^\infty(B_3)}\le M.$$

The phase occupies one local side at every graph point in Q_2. If the two points of an adjacent pair above x' lie in Q_1, then

$$h_{i+1}(x')-h_i(x')\ge c(d,M)\sqrt{1-p}.$$

There is no bound on K, no initial Hölder modulus of the Hessians is assumed, and graphs may cross both caps of either cylinder.

Proof. Use 2.1 for normalized curvatures and graph notation, and 2.2 for stationary and stable phases. Translation and positive dilation preserve both by 2.11.

Step 1: point selection with horizontal and vertical margins. If no such constants existed, we could choose $p_j\uparrow 1$ and a sequence of charts for which the ratio of an interior adjacent gap to $\sqrt{\sigma_j}$ tends to zero. Here $\sigma_j=1-p_j$, K_j is the finite number of graphs in the jth chart, and the original labels are $1,\dots,K_j$. Such a monotone sequence is obtained inductively by choosing the threshold p_* above both $1-1/j$ and the preceding chosen p_j, and demanding a gap ratio below $1/j$. Set

$$g_{j,i}(x')=h_{j,i+1}(x')-h_{j,i}(x'),$$

and, on $\overline{B}_{3/2}$, let

$$b_{j,i}(x')=\min\left\{\frac{3}{2}-|x'|,\frac{3}{2}-|h_{j,i}(x')|,\frac{3}{2}-|h_{j,i+1}(x')|\right\}_+.$$

This is the smaller boundary margin of the two endpoints in $Q_{3/2}$, or zero when one endpoint is outside that cylinder. For each j, maximize

$$\frac{b_{j,i}(x')\sqrt{\sigma_j}}{g_{j,i}(x')}$$

over all adjacent indices and $\overline{B}_{3/2}$. The maximum exists: the family is finite and each graph gap has a positive minimum on this compact horizontal base. The assumed violation in Q_1 makes the maximum tend to infinity. At a maximizing pair and point, write

$$\rho_j = g_{j,i_j}(x'_j), \qquad b_j = b_{j,i_j}(x'_j), \qquad R_j = \frac{\rho_j}{\sqrt{\sigma_j}}, \qquad L_j = \frac{b_j}{R_j}.$$

Then $R_j \to 0$ and $L_j \to \infty$. Let Z_j be the midpoint of the selected vertical segment. Its distance to $\partial Q_{3/2}$ is at least b_j.

Whenever both endpoints of any adjacent pair lie in $B_{b_j/4}(Z_j)$, their margins are at least $b_j/2$. Maximality gives

$$g_{j,k}(y') \geq \rho_j/2.$$

If one endpoint lies in $B_{b_j/8}(Z_j)$ but its adjacent neighbor is outside $B_{b_j/4}(Z_j)$, their vertical distance is at least $b_j/8$. Thus no small gap next to an interior sheet is omitted by this selection.

Translate by Z_j and dilate every ambient coordinate by R_j^{-1}. Retain the original vertical direction. The stationary region contains balls of radius tending to infinity; the selected graph gap is $\sqrt{\sigma_j}$ at the origin of the horizontal coordinates, and all relevant neighboring gaps in $B_{L_j/8}$ are bounded below by a fixed multiple of $\sqrt{\sigma_j}$. The new Hessian bound is $MR_j \to 0$ and the slope bound is unchanged. We continue to write $h_{j,i}$ for these rescaled graphs and label the selected indices $0, 1$.

For clarity, the maximizing quotient is exactly L_j; since $b_j \leq 3/2$, we have $R_j = b_j/L_j \to 0$. In the rescaled coordinates the two selected central heights are $h_{j,0}(0) = -\sqrt{\sigma_j}/2$ and $h_{j,1}(0) = \sqrt{\sigma_j}/2$. Fix any compact ambient box. For all sufficiently large j it lies in $B_{L_j/8}(0)$. An immediate neighbor of any sheet point in this box either lies in $B_{L_j/4}(0)$, in which case its vertical gap is at least $\sqrt{\sigma_j}/2$, or lies outside that ball, in which case its vertical gap is at least $L_j/8$. The latter is also at least $\sqrt{\sigma_j}/2$ for large j. For the distance estimate, let X be a tested point and let another graph have vertical separation $a > 0$ from it over the same horizontal coordinate. Put $r = |X' - Y'|$ for a point Y on that graph and $\delta = 1/100$. The slope bound gives $|X - Y|^2 \geq r^2 + s^2$, where $s = (a - \delta r)_+$. Since $a \leq s + \delta r$, we have

$$r^2 + a^2 \leq (2 + 2\delta^2)(r^2 + s^2).$$

Thus a fixed ball of radius $c_0\sqrt{\sigma_j}$ about an interior sheet point meets no other graph, for example with $c_0 < 1/(4\sqrt{2 + 2\delta^2})$. Ordering controls every nonadjacent graph by an immediate neighbor. The ball is also inside the expanding stationary region. On its two sides of the tested graph the phase is constant; the assumed local one-side occupancy therefore gives agreement with that graph's subgraph or its complement. This proves the critical-tube hypothesis before invoking any graph regularity.

Step 2: uniform graph regularity and individual curvature bounds. Fix an arbitrary finite physical box in the rescaled coordinates and a slightly larger box. Every sheet that enters the first continues as one of the original graphs over the horizontal base of the second, for all sufficiently large j. Its small slope bounds its height variation there. The selected critical tube exists at every point of the resulting smaller graph patches. For each fixed sheet over this connected larger horizontal ball, let $\varepsilon(x') \in \{-1, 1\}$ record whether E_j locally occupies its lower or upper side. Genuine one-sided occupancy makes this sign locally constant: neighboring graph neighborhoods overlap on both sides of the interface. It is therefore constant on the connected patch. Choose E_j or E_j^c once according to that sign. In every selected tube ball the chosen phase agrees with this sheet's subgraph, because no other interface is present and the two sides of the graph in the ball are connected. Extend the patch once to a global graph, with uniform $C^{1,1}$ bounds, after subtracting a vertical constant; choose its cutoff equal to one on a neighborhood of the whole smaller patch and its tube balls. Lemma 3.4, its critical-scale forcing bound 3.5, and Corollary 3.9 give, uniformly in the sheet index,

$$|F_{h_{j,k}}| \le C\sqrt{\sigma_j}, \qquad \|h_{j,k} - h_{j,k}(0)\|_{C^{2,1/2}} \le C$$

on smaller patches; the constants may depend on the fixed box. Lemma 3.10 then gives

$$|\mathcal{M}(h_{j,k})| \le C\sqrt{\sigma_j}.$$

There is no sum over sheets in either estimate. The original rescaled Hessian bound $MR_j \to 0$ remains valid as well.

Step 3: relative heights and a possibly infinite interval of indices. After passing to a subsequence, the lower and upper endpoint indices relative to $0, 1$ have limits in the extended integers. Also let

$$q = \lim_j \nabla h_{j,0}(0), \qquad L = \sqrt{1+|q|^2}, \qquad G = I_d + q \otimes q, \qquad A_q = L^{-1}G^{-1}.$$

Because the rescaled Hessians tend to zero, the reference slopes tend locally uniformly to q.

For two ordered graphs that remain in bounded physical boxes, the difference has the exact equation

$$\operatorname{div}(a_{j,ik}\nabla(h_{j,k} - h_{j,i})) = \mathcal{M}(h_{j,k}) - \mathcal{M}(h_{j,i}),$$

where

$$a_{j,ik} = \int_0^1 D\left(\frac{z}{\sqrt{1+|z|^2}}\right)\Big|_{z=(1-t)\nabla h_{j,i}+t\nabla h_{j,k}} dt.$$

Explicitly,

$$D\left(\frac{z}{\sqrt{1+|z|^2}}\right) = \frac{1}{\sqrt{1+|z|^2}}\left(I_d - \frac{z\otimes z}{1+|z|^2}\right).$$

Its eigenvalues lie between $(1+|z|^2)^{-3/2}$ and $(1+|z|^2)^{-1/2}$. Thus these matrices are uniformly elliptic, with uniform local $C^{1,1/2}$ bounds from Step 2. By the individual classical-curvature bound of Step 2, every positive normalized difference $(h_{j,k}-h_{j,i})/\sqrt{\sigma_j}$ has a uniformly bounded right-hand side in this divergence equation.

On fixed nested balls $B_R \Subset B_{2R}$, the positive difference $u_j = (h_{j,k} - h_{j,i})/\sqrt{\sigma_j}$ satisfies

$$\operatorname{div}(a_{j,ik}\nabla u_j) = f_j, \qquad \|f_j\|_{L^\infty(B_{2R})} \le C_R.$$

Apply the Harnack and interior $W^{2,r}$ estimates of 2.13, with their stated Lipschitz coefficient and bounded-forcing hypotheses. Here the positive solution is u_j, the principal matrix is $a_{j,ik}$ with uniform ellipticity and $C^{1,1/2}$ norm, and f_j has the stated uniform L^∞ bound. The Harnack estimate gives

$$\sup_{B_R} u_j \le C_R\left(\inf_{B_R} u_j + 1\right).$$

Consequently a bounded $u_j(0)$ bounds u_j on B_R, whereas $u_j(0) \to \infty$ implies $\inf_{B_R} u_j \ge C_R^{-1}u_j(0) - 1 \to \infty$. The uniformly $C^{1,1/2}$ coefficients permit the interior estimate, for each $1 < r < \infty$,

$$\|u_j\|_{W^{2,r}(B_R)} \le C_{R,r}\left(\|u_j\|_{L^\infty(B_{2R})} + \|f_j\|_{L^r(B_{2R})}\right),$$

after increasing the outer ball if needed. One can write the equation as $a^{ab}_{j,ik}\partial_{ab}u_j + (\partial_a a^{ab}_{j,ik})\partial_b u_j = f_j$; the principal coefficients and first-order coefficients have the required uniform bounds. For $r > d$, compact Sobolev embedding now gives $C^{1,\beta}$ compactness for $\beta < 1 - d/r$. For negative relative indices, apply the same argument to $(h_{j,0} - h_{j,k})/\sqrt{\sigma_j}$.

Pass to a further diagonal subsequence over the countably many fixed labels that eventually exist. For each such k, the positive central relative height $a_{j,k} = |h_{j,k}(0) - h_{j,0}(0)|$ and its quotient $a_{j,k}/\sqrt{\sigma_j}$ have limits in $[0,\infty]$. A finite physical limit makes $a_{j,k}$ bounded, while an infinite physical limit makes it tend to infinity. Since $h_{j,0}(0) = -\sqrt{\sigma_j}/2 \to 0$, this gives precisely the bounded and physically escaping alternatives used below. For a fixed index k, there are no missing alternatives in this assertion. If its central physical height tends to infinity in absolute value, the uniform slope bound makes it escape every fixed physical box. If that height is bounded, its whole patch on a fixed horizontal ball lies in a fixed physical box, so the preceding curvature and Harnack estimates apply. In particular they apply whenever its central height relative to sheet 0, divided by $\sqrt{\sigma_j}$, is bounded. A diagonal subsequence over fixed indices and an exhaustion of the horizontal space therefore gives an interval $I \subset \mathbb{Z}$ containing $0, 1$ and limits. More precisely, first extract a limit in $[0,\infty]$ for each central positive normalized difference, then diagonalize over the fixed indices, integer-radius balls, and a countable unbounded sequence of Sobolev exponents $r > d$. The finite alternatives have the limits

$$w_k = \lim_j \frac{h_{j,k} - h_{j,0}}{\sqrt{\sigma_j}} \quad \text{in } C^{1,\beta}_{\mathrm{loc}}(\mathbb{R}^d), \qquad k \in I, \qquad w_0 = 0.$$

All fixed indices outside I either cease to exist or escape locally uniformly in normalized height. Ordering makes I contiguous. It may be finite, one-sided infinite, or two-sided infinite. The selection bound gives $w_{k+1} - w_k \geq c_0 > 0$ on every fixed compact set, with a uniform positive c_0 for this selected sequence. The selected gap

$$v = w_1 - w_0$$

is positive on $\mathbb{R}^d$ and satisfies $v(0) = 1$. No boundedness or convergence of $h_{j,0}/\sqrt{\sigma_j}$ is asserted or needed. For $k \in I$, the convergence of the relative gradients shows $\nabla h_{j,k} \to q$, and hence $a_{j,01} \to A_q$ locally uniformly. The local $W^{2,r}$ bounds also pass to v.

Step 4: the equation of the selected gap. Here the local cylinder used in the phase bracket can be chosen without imposing an empty band at either cap. Fix a compact horizontal ball and a slightly larger connected ball B_{2R}. The selected sheets tend in unnormalized physical height to $x' \mapsto q \cdot x'$ on this ball. Choose a fixed T so large that their complete patches over B_{2R} lie strictly between $-T$ and T, with a fixed margin. For large j, $B_{2R} \times (-T, T)$ is inside the stationary region. The intervening slab over B_{2R} is connected, contains no boundary sheet, and lies entirely in this cylinder. Its phase is therefore constant, and complementing E_j if needed makes it filled. Other sheets may cross both cylinder caps. For the ordered extension choose one fixed smooth cutoff θ, equal to one near $\overline{B}_{2R}$ and supported in B_{3R}, and use the central-height constants:

$$\widetilde{h}_{j,a} = \theta h_{j,a} + (1-\theta) h_{j,a}(0), \qquad a \in \{0, 1\}.$$

The difference of the two extensions is a convex combination of the positive local gap and $h_{j,1}(0) - h_{j,0}(0) = \sqrt{\sigma_j}$, so they remain ordered. Moreover $|h_{j,a}(x') - h_{j,a}(0)| \leq \delta |x'|$ with $\delta = 1/100$, which gives a uniform global Lipschitz bound depending on this fixed cutoff and R, independently of j. The bounds of Step 2 on the larger fixed patch give the required local $C^{2,1/2}$ bounds as well. Apply 4.3 on the smaller horizontal ball. The cylinder and extension errors are $O(\sigma_j)$ because their distances from the tested points have a fixed positive lower bound.

The error in replacing each $F_{h_{j,i}}$ by its classical curvature is $O(\sigma_j)$ by Step 2, hence is negligible after division by $\sqrt{\sigma_j}$. The exact graph difference equation and the relative-height convergence of Step 3 show that the left side, so divided, tends to $c_d \operatorname{div}(A_q \nabla v)$ in distributions.

In the upper-halfspace term, use $Y' = x' + \sqrt{\sigma_j}\,\zeta$ and $Y_{d+1} = h_{j,0}(x') + \sqrt{\sigma_j}\,t$. The threshold tends to $q \cdot \zeta + v(x')$. This is a halfspace at normal distance $v(x')/L$ from the origin. Rotational invariance gives

$$\int_{t>q\cdot\zeta+v(x')} (|\zeta|^2+t^2)^{-(d+2)/2}\,d\zeta\,dt = \frac{c_d L}{v(x')}.$$

The scale factor is $\sigma_j^{(1-p_j)/2} \to 1$ because $\sigma_j \log \sigma_j \to 0$. On $|\zeta| > R \geq 1$, extend the vertical integration to all of $\mathbb{R}$; integration in t and then in the horizontal radius gives

$$\int_{|\zeta|>R}\int_{\mathbb{R}} (|\zeta|^2+t^2)^{-(d+1+p_j)/2}\,dt\,d\zeta \leq CR^{-p_j} \leq CR^{-1/2}.$$

On bounded ζ sets the limiting plane has normal distance $v/L > 0$ from the origin, uniformly positive over the tested compact set; the convergence of thresholds then excludes a fixed neighborhood of the singularity. Uniformly integrable vertical tails and dominated convergence apply on that part. The other halfspace contributes the same limit. The cylinder and extension error is $O(\sigma_j) = o(\sqrt{\sigma_j})$. Testing against arbitrary nonnegative compactly supported smooth functions gives

$$\operatorname{div}(A_q \nabla v) \leq \frac{4L}{v} \quad \text{on } \mathbb{R}^d.$$

Step 5: stability with an unbounded number of sheets. Let $\eta \in C_c^\infty(\mathbb{R}^d)$. In the notation of the alternating form, set $\phi_0 = -\eta$, $\phi_1 = \eta$, and $\phi_k = 0$ for every other sheet. The physical normal speed on sheet k is $(-1)^k\phi_k$. As in the finite packet proof, these are legitimate compactly supported tests for each j. For any fixed finite collection of nonescaping indices, all of their complete patches and every intervening patch lie in a connected larger cylinder for large j. Each intervening slab is connected and contains no boundary. One-side occupancy therefore gives $\nu_{j,k} = \varepsilon_j(-1)^k n_{j,k}$ on this chart, with one common sign $\varepsilon_j \in \{-1,1\}$. The common sign cancels in all normal scalar products. Escaping rows below are estimated absolutely and need no limiting parity.

Here $\Gamma_{j,k}$ is the graph of $h_{j,k}$, $X_{j,k}(x') = (x', h_{j,k}(x'))$, and $n_{j,k} = (-\nabla h_{j,k}, 1)/\sqrt{1+|\nabla h_{j,k}|^2}$ is its upward normal.

For each fixed j, the two prescribed speeds define a compactly supported C^1 normal test, zero on all other sheets. Positive separation of the finite collection on the compact test base makes it C^1 on the regular boundary. By 2.8 this test is the C^1 trace limit of smooth compactly supported ambient fields inside the stationary region, and its quadratic form is their limit. Thus it is admissible for 2.10, without a uniform-in-j transverse derivative bound. Choose a single smooth cutoff χ, equal to one on a fixed open neighborhood N of the limiting plane over $\operatorname{supp}\eta$ and supported in a larger fixed box. Choose a compact set $S_* \Subset N$ whose interior contains that limiting planar test support. Uniform convergence of the two actual tested patches to this plane gives $S_j \subset \operatorname{int} S_*$ for all large j, where S_j is their combined test support. Thus $\operatorname{dist}(S_j, \{\chi \neq 1\}) \geq \operatorname{dist}(S_*, \{\chi \neq 1\}) > 0$. The volume-gradient estimate in 4.1 consequently has one constant uniform in j, determined by this fixed support envelope. Their areas are bounded by the slope estimate, independently of K_j, so the combined exterior contribution to $2\sigma_j Q_{p_j}$ is $O(\sigma_j)$.

Localize with this χ by the factors $\chi(X)\chi(Y)$. Inside the cutoff, discard the nonpositive same-sheet normal potentials. Their Dirichlet terms have the limits

$$c_d L \int \nabla\phi_i \cdot G^{-1}\nabla\phi_i, \qquad i = 0, 1.$$

We compute the tilted coefficient on the common compact support. On the tested compact base, $\nabla h_{j,i} \to q$, the graph area elements tend uniformly to $L\,dx'$, and Taylor expansion in $y' = x' + r\omega$ gives the leading horizontal kernel $r^{-d-1-p_j}(1+(q\cdot\omega)^2)^{-(d+1+p_j)/2}$. A radius $\rho > 0$ can be chosen such that $\chi = 1$ wherever the same-sheet test difference is nonzero and $r < \rho$. The expansion of that difference has square $r^2(\nabla\phi_i\cdot\omega)^2 + O(r^3)$. Thus $\sigma_j \int_0^\rho r^{-p_j}\,dr = \rho^{\sigma_j} \to 1$ supplies the radial

factor, while the remainder contributes $O(\sigma_j \int_0^\rho r^{1-p_j}\,dr) \to 0$. The contribution at $r \ge \rho$ is $O(\sigma_j)$ on the bounded cutoff base. The two area elements supply L^2, and the angular identity is

$$\int_{S^{d-1}} \frac{\omega \otimes \omega}{(1+(q\cdot\omega)^2)^{(d+2)/2}}\,d\omega = \frac{c_d}{L} G^{-1}.$$

Indeed $\omega \cdot G\omega = 1+(q\cdot\omega)^2$ and $\det G = L^2$. Differentiating the Gaussian identity $\int_{\mathbb{R}^d} e^{-z\cdot Gz}\,dz = \pi^{d/2}/L$ with respect to the entries of G, and expressing the resulting second-moment integral in polar coordinates, gives the displayed identity, since $\pi^{d/2}/\Gamma(1+d/2) = c_d$. Contraction against $\nabla\phi_i$ gives exactly $c_d L \int \nabla\phi_i \cdot G^{-1}\nabla\phi_i$. The uniform $C^{2,1/2}$ bounds from Step 2 justify every Taylor remainder independently of j.

For a fixed $k \in I$, the corresponding cross row tends to

$$\sigma_j \int_{\Gamma_{j,k}} |X_{j,i}(x') - Y|^{-d-1-p_j}\,d\mathcal{H}^d_Y \longrightarrow \frac{c_d L^2}{(w_k(x') - w_i(x'))^2}, \qquad i \in \{0,1\},\ k \ne i,$$

with the integral localized as just described. Write $w = w_k(x') - w_i(x') \ne 0$, and set $y' = x' + \sqrt{\sigma_j}\zeta$. The scaled graph separation converges on bounded ζ sets to $q\cdot\zeta + w$, the inner surface Jacobian tends to L, and the prefactor is $\sigma_j^{(1-p_j)/2} \to 1$. The limiting squared distance is

$$|\zeta|^2 + (q\cdot\zeta + w)^2 = \left(\zeta + \frac{wq}{L^2}\right)\cdot G\left(\zeta + \frac{wq}{L^2}\right) + \frac{w^2}{L^2}.$$

With $z = G^{1/2}(\zeta + wq/L^2)$ we have $dz = L\,d\zeta$. Therefore

$$L\int_{\mathbb{R}^d} (|\zeta|^2 + (q\cdot\zeta + w)^2)^{-(d+2)/2}\,d\zeta = \int_{\mathbb{R}^d} (|z|^2 + w^2/L^2)^{-(d+2)/2}\,dz = \frac{c_d L^2}{w^2}.$$

The positive lower gap on the tested compact set excludes the singularity on bounded ζ sets. On $|\zeta| > R \ge 1$, the horizontal distance alone bounds the scaled tail by $CR^{-1-p_j} \le CR^{-3/2}$. This proves uniform convergence of the localized row. The same splitting permits replacement of each fixed continuous test value at y' by its value at x'. The normals of fixed nonescaping sheets tend uniformly to the common upward normal, multiplied by $\varepsilon_j(-1)^k$; hence their scalar product tends to $(-1)^{i-k}$. For an unordered fixed pair, 2.10 then gives the limiting combined cross contribution

$$2c_d L^3 \int_{\mathbb{R}^d} (-1)^{i-k} \frac{(\phi_i - \phi_k)^2}{(w_k - w_i)^2}\,dx'.$$

We justify passage past the remaining indices. If an interior point of a sheet k lies in the fixed larger box, its value over the horizontal coordinate of a tested point lies in a slightly larger fixed box, by the slope bound. All intermediate heights then lie in that box as well. For large j they are inside the selection region, so the selected lower gap bound from Step 1, summed over intermediate indices, gives vertical separation at least $c_0|k-i|\sqrt{\sigma_j}$. The graph-distance estimate of Step 1 and elementary radial integration imply

$$\begin{aligned}\sigma_j \int_{\Gamma_{j,k}\cap\{\text{fixed larger box}\}} |X_{j,i}(x') - Y|^{-d-1-p_j}\,d\mathcal{H}^d_Y &\le \frac{C\sigma_j}{(c_0|k-i|\sqrt{\sigma_j})^{1+p_j}} \\ &\le C|k-i|^{-1-p_j}.\end{aligned}$$

The actual combined cross term between a tested sheet i and an untested sheet k is bounded in absolute value by $C\|\eta\|^2_\infty$ times the displayed row: the test on sheet k is zero and $|\nu_i \cdot \nu_k| \le 1$. The relevant radial estimate is, uniformly for $p \ge 1/2$,

$$\int_{\mathbb{R}^d} (r^2 + a^2)^{-(d+1+p)/2}\,dy' \le C_d a^{-1-p}.$$

Substitution of $a \geq c_0|k-i|\sqrt{\sigma_j}$ gives the factor $\sigma_j^{(1-p_j)/2} \leq 1$ in the index-row bound. For a fixed sheet escaping only in normalized height, its actual separation satisfies $a_j/\sqrt{\sigma_j} \to \infty$ uniformly on the tested base by Harnack. The same radial bound is at most

$$C\sigma_j^{(1-p_j)/2}(a_j/\sqrt{\sigma_j})^{-1-p_j} \longrightarrow 0.$$

These estimates have constants uniform on the tested compact set and independent of the total number of sheets. Since eventually $p_j \geq 1/2$, the sum for $|k-i| > J$ is at most $CJ^{-1/2}$. An index escaping in physical height eventually misses the box altogether. Interactions beyond that box were already bounded as the combined exterior flux, so no unestimated positive Dirichlet tail remains.

It is now legitimate first to pass $j \to \infty$ with a fixed finite index cutoff and then to let the cutoff tend to infinity. The retained localized form, after discarding only the same-sheet nonpositive potentials, is at least $-C\sigma_j$ by stability and 4.1. Combining the cross Dirichlet and normal terms as computed above and dividing by $c_d L^2 > 0$ therefore gives

$$\int_{\mathbb{R}^d} \sum_{i\in\{0,1\}} \nabla\phi_i \cdot A_q \nabla\phi_i + 2L \int_{\mathbb{R}^d} \sum_{i<k\,;\ i,k\in I} (-1)^{i-k} \frac{(\phi_i-\phi_k)^2}{(w_k-w_i)^2} \geq 0,$$

where all components except $0, 1$ are zero. The absolute convergence of the limiting tails follows from $|w_k - w_i| \geq c_0|k-i|$ on compact sets. This is precisely the two-component hypothesis sufficient in 4.6, with $A = A_q$ and the present L. Applying that lemma gives

$$4L \int_{\mathbb{R}^d} v^{-2}\eta^2 \leq \int_{\mathbb{R}^d} \nabla\eta \cdot A_q \nabla\eta.$$

Step 6: the entire-gap contradiction. Use the constant linear change $x' = A_q^{1/2} y$ and put $\widetilde{v}(y) = v(A_q^{1/2} y)$. The differential inequality of Step 4 and the stability inequality of Step 5 become

$$\Delta\widetilde{v} \leq \frac{4L}{\widetilde{v}}, \qquad 4L \int \widetilde{v}^{-2}\psi^2 \leq \int |\nabla\psi|^2 \quad (\psi \in C_c^1(\mathbb{R}^d)).$$

The relation $\nabla_y \psi = A_q^{1/2} \nabla_{x'} \eta$, for $\psi(y) = \eta(A_q^{1/2} y)$, transforms the energy exactly to $\int |\nabla_y \psi|^2$ after cancellation of the constant Jacobian $\det(A_q^{1/2})$. The same linear change transforms the distributional operator $\mathrm{div}_{x'}(A_q \nabla_{x'} v)$ to $\Delta_y \widetilde{v}$. The stability estimate first obtained for smooth tests extends to C_c^1 tests by C^1 approximation; the positive continuous gap has a positive minimum on their common compact support, so the potential integral also converges. The limit is positive, belongs locally to $W^{2,r}$ for every finite r, and has value one at the origin. Lemma 4.5, with $\kappa = 4L$ and $d \leq 6$, gives a contradiction. This contradiction proves the claimed uniform separation. □

5. Stable classical limits and uniform curvature

Separation makes the local graph estimates available under a curvature bound. To retain stability when the number of sheets diverges, prescribe the same scalar normal speed on the entire packet and average by its degree. The same-sheet terms converge to the classical Jacobi form, while separated cross-sheet Dirichlet terms vanish. This produces stable classical leaves and supplies the rigidity needed to rule out curvature blow-up.

5.1. **Stability under packet averaging.**

Lemma 5.1 (The combined exterior term)**.** *Let U be a bounded smooth domain, let $F \Subset U$, and suppose that a phase E has a smooth boundary $\mathcal{S}$ near $\overline{U}$. If ϕ is a smooth scalar normal speed supported in $\mathcal{S} \cap F$, the part of the quadratic form 2.10 with one point outside U is*

$$\int_{\mathcal{S}\cap F}\phi(X)^2\nu(X)\cdot\left(\int_{\mathcal{S}\setminus U}\nu(Y)K_p(X,Y)\,d\mathcal{H}^d_Y\right)\,d\mathcal{H}^d_X.$$

The inner signed integral, interpreted by the volume formula if necessary, is bounded by $C(F,U,m)$ for $1/2\le p<1$. Consequently the absolute value of the normalized exterior term is at most

$$C\sigma\int_{\mathcal{S}\cap F}\phi^2.$$

The constant is independent of the number and area of the interfaces outside U.

Proof. Use the conventions of 2.1: $m=d+1$, $0<p<1$, $\sigma=1-p$, $K_p(X,Y)=|X-Y|^{-m-p}$, and ν is the exterior unit normal. The quadratic form is the displayed form in 2.10; its algebraic expression is defined for every regular phase, whether or not it is stationary. First suppose the exterior surface integrals exist. The factor $1/2$ in the Dirichlet term cancels the two ordered cross regions, so its contribution with one point outside U is

$$\int_{\mathcal{S}\cap F}\phi(X)^2\int_{\mathcal{S}\setminus U}K_p(X,Y)\,d\mathcal{H}^d_Y\,d\mathcal{H}^d_X.$$

The corresponding potential contribution is

$$-\int_{\mathcal{S}\cap F}\phi(X)^2\int_{\mathcal{S}\setminus U}(1-\nu(X)\cdot\nu(Y))K_p(X,Y)\,d\mathcal{H}^d_Y\,d\mathcal{H}^d_X.$$

Their sum is the signed cross-region expression in the statement. In particular it is this combined signed expression that must be retained if the two unsigned exterior surface integrals are not separately finite.

Let $\tau_U\chi_E$ denote the one-sided trace of the characteristic function of E on ∂U, taken from the interior of U. Thus $0\le\tau_U\chi_E\le1$ almost everywhere on ∂U. The volume-flux formula is

$$\int_{\mathcal{S}\setminus U}\nu(Y)K_p(X,Y)\,d\mathcal{H}^d_Y=\int_{E\setminus U}\nabla_Y K_p(X,Y)\,dY+\int_{\partial U}(\tau_U\chi_E)(Y)n_U(Y)K_p(X,Y)\,d\mathcal{H}^d_Y.$$

The trace convention matters if an interface shares a patch with ∂U: the left side includes that patch because U is open. To justify the formula, choose $\eta\in C^\infty_c(U)$ equal to one on a neighborhood of F and set

$$g_X(Y)=(1-\eta(Y))K_p(X,Y).$$

This function vanishes near X. The exhaustion-independent remote pairing of 2.7 is

$$\int_{\mathcal{S}}\nu(Y)g_X(Y)\,d\mathcal{H}^d_Y=\int_E\nabla_Y g_X(Y)\,dY.$$

The left side uses the distributional convention of 2.1 when the remote boundary is not regular or its unsigned weighted area is infinite. On the bounded smooth domain U, the Gauss–Green formula with the interior trace gives

$$\int_{\mathcal{S}\cap U}\nu(Y)g_X(Y)\,d\mathcal{H}^d_Y=\int_{E\cap U}\nabla_Y g_X(Y)\,dY-\int_{\partial U}(\tau_U\chi_E)(Y)n_U(Y)g_X(Y)\,d\mathcal{H}^d_Y.$$

Smoothness of the phase near $\overline{U}$ ensures its local finite perimeter and the trace formula; tangency or coincidence with ∂U is handled by that interior trace. Subtract the local identity from the remote one. Since $\eta=0$ near ∂U and outside U, there $g_X=K_p$ and $\nabla g_X=\nabla K_p$. This gives precisely the displayed exterior volume-flux formula, including its plus sign and any interface patch on ∂U. The same subtraction defines the signed exterior term for an arbitrary remote phase. Thus no separate exterior area estimate or second outer-exhaustion argument is needed.

Let $a = \operatorname{dist}(F, \partial U) > 0$. Direct radial integration gives

$$\int_{|X-Y|\geq a} |\nabla_Y K_p(X,Y)|\, dY = \frac{|S^{m-1}|(m+p)}{p+1} a^{-1-p}.$$

This is uniformly bounded for $p \in [1/2, 1)$ and fixed a. Moreover

$$\int_{\partial U} K_p(X,Y)\, d\mathcal{H}^d_Y \leq \mathcal{H}^d(\partial U) \max\{a^{-m-1/2}, a^{-m-1}\}, \qquad X \in F.$$

The volume-flux identity therefore bounds the absolute value of its inner signed integral independently of the exterior interface area or number. Integrating against ϕ^2 and multiplying by $\sigma = 1-p$ proves the normalized bound in the statement. □

Lemma 5.2 (Stability survives arbitrary local multiplicity)**.** *Let $p_j \uparrow 1$ and let E_j be stationary stable phases in an open set $U \subset \mathbb{R}^m$. Suppose that in a tubular neighborhood $N \subset U$ of a compact connected domain L_1 of a smooth embedded minimal hypersurface L, their interfaces form a finite covering packet of K_j sheets. Here this means the following precise property. Let $\pi : N \to L$ be normal projection. Its restriction to $\partial E_j \cap N \cap \pi^{-1}(\operatorname{int} L_1)$ is a finite covering onto $\operatorname{int} L_1$ of constant positive degree $K_j \geq 1$. For every sufficiently small connected graph chart $V \subset \operatorname{int} L_1$ with a local unit normal ν_L, the entire set $\partial E_j \cap N \cap \pi^{-1}(V)$ consists exactly of K_j pairwise disjoint C^2 normal graphs $\{x + h_{j,a}(x)\nu_L(x) : x \in V\}$, ordered as $h_{j,1} < \cdots < h_{j,K_j}$. Thus the packet exhausts all interfaces in that part of the tube. A global normal direction or global sheet labeling is not assumed: chart transitions and monodromy may change the labels. Consecutive sheets and the graph hypotheses below refer to these local ordered normal-line trivializations. Assume that all these sheets converge to L_1 uniformly in C^2, have uniformly bounded $C^{2,\alpha}$ graph norms for some $\alpha > 0$, and that in every smaller graph chart consecutive sheets are separated by $c\sqrt{\sigma_j}$. Then, for every $\eta \in C_c^\infty(\operatorname{int} L_1)$,*

$$\int_L |A_L|^2 \eta^2 \leq \int_L |\nabla_L \eta|^2.$$

The conclusion is a scalar inequality and remains valid on every cover of L when the hypotheses hold on all compact domains.

Proof. Use 2.1 for $d = m-1$, $\sigma_j = 1-p_j$, $c_d = |S^{d-1}|/d$ and the quadratic form Q_{p_j}. Stationarity and stability mean the unrestricted phase variations of 2.2. Choose compact connected collared domains $D \Subset D_1 \Subset \operatorname{int} L_1$ with $\operatorname{supp}\eta \subset \operatorname{int} D$. Write $N \subset U$ for the packet neighborhood from the target. There is a fixed normal tube U_0 over D_1 whose closure is contained in N, and hence is a compact subset of the stationarity and stability domain U. All normal speeds constructed below therefore have compact support in U. Uniform convergence of all packet sheets to the zero section leaves a fixed empty collar at the normal boundary of this tube for all sufficiently large j. After smoothing its boundary away from the speed support, use this tube as a bounded localization domain. Write π for normal projection, which is defined also when the normal bundle of L is nontrivial. Choose a fixed ambient smooth cutoff function Φ equal to $\eta \circ \pi$ near the packet over D and supported over D_1 inside the tube. It can be chosen so that its trace on every packet sheet is exactly $\eta \circ \pi$. Its Lipschitz norm is independent of j.

Prescribe the scalar speed $\phi_j = \Phi$ on the interfaces. The scalar speed is measured with respect to the exterior normal of the phase; thus adjacent sheets have the same scalar speed, although their normal velocity vectors point in opposite directions. For each fixed j, 2.8 realizes this compact C^1 normal trace by an ambient field and admits it by approximation with smooth fields in 2.10. No uniform bound on the derivatives of those realizing vector fields is needed.

The finite covering has a constant fiber cardinality K_j on the connected collar. A normal fiber has a fixed bounded length. Its ordered interfaces satisfy the spacing hypothesis in a local trivialization, so

$$(K_j-1)c\sqrt{\sigma_j}\le C,\qquad K_j\le C\sigma_j^{-1/2}.$$

The constants are uniform over a finite collection of charts. Uniform graph Jacobian bounds also give $\int\phi_j^2\le CK_j$. We apply stability to ϕ_j, normalize by σ_j, and average by K_j. Choose a fixed smooth $0\le\chi\le1$ supported inside U_0, equal to one on a fixed neighborhood of $F=\operatorname{supp}\Phi$. Localize Q_{p_j} by the factors $\chi(X)\chi(Y)$ and $1-\chi(X)\chi(Y)$. The fixed support $F\Subset U_0$ contains every scalar-speed support. Choose the close-pair radius below smaller than both the chart margins and $\operatorname{dist}(F,\{\chi\neq1\})$. Then the cutoff factors equal one for every close pair with a nonzero test difference or potential numerator.

Choose a finite chart cover and a partition of unity in the first integration variable, with each partition element supported a fixed positive distance from its chart boundary. For close pairs on a local sheet $\Sigma_{j,a}$, use tangent coordinates $z=r\theta$ at X and put $\beta=\min\{\alpha,1\}>0$. The leading numerators are

$$(\nabla_{\Sigma_{j,a}}\phi_j(X)\cdot z)^2,\qquad 1-\nu(X)\cdot\nu(Y)=\tfrac12|A_{\Sigma_{j,a}}(X)z|^2+O(r^{2+\beta}).$$

The error in the squared test difference is also $O(r^{2+\beta})$. The fixed smooth ambient function Φ and the uniform $C^{2,\alpha}$ graph bounds make these estimates uniform over all sheets. The exterior normal differs from a consistently chosen graph normal by a constant sign on each local sheet, which cancels from its normal dot products. With a fixed small tangent radius a, the radial factors are

$$\sigma_j\int_0^a r^{-p_j}\,dr=a^{\sigma_j}\longrightarrow1,\qquad \sigma_j\int_0^a r^{\beta-p_j}\,dr=\frac{\sigma_j a^{\beta+\sigma_j}}{\beta+\sigma_j}\longrightarrow0.$$

On each fixed compact chart, the sheet parametrizations converge in C^2 uniformly over the packet to the zero-section parametrization. Their first fundamental forms therefore converge uniformly in C^0 to that of L. The limiting metric is uniformly positive definite on this compact chart, so the inverse metrics and volume densities also converge uniformly in C^0. The consistently chosen local graph normals converge in C^1, and the second fundamental form coefficients, obtained by differentiating these normals along the parametrizations, consequently converge uniformly in C^0 to those of L. Changing to the phase's exterior normal may multiply the second fundamental form on a sheet by -1, which leaves the squared curvature terms unchanged. Together with $\int_{S^{d-1}}(v\cdot\theta)^2=c_d|v|^2$, these convergences, the local normalization of 2.12 and the preceding uniform remainder estimates make the averaged same-sheet terms tend respectively to

$$\frac{c_d}{2}\int_L|\nabla_L\eta|^2,$$
$$\frac{c_d}{2}\int_L|A_L|^2\eta^2.$$

This use of a partition in the first variable introduces no derivatives of the partition functions. The estimates are uniform per sheet before averaging. It also avoids a global sheet enumeration on a one-sided domain.

For close pairs on different local sheets, let $k\ge1$ be their relative index in the local ordering. On a smaller fixed chart write the tubular-coordinate map as

$$\Psi(x,t)=Z_L(x)+t\nu_L(x),\qquad X=\Psi(x,h_{j,a}(x)),\quad Y=\Psi(y,h_{j,b}(y)).$$

The fixed tube and chart are chosen sufficiently small that Ψ and its inverse are uniformly Lipschitz there: the normal projection is nonsingular on the compact smaller tube and these coordinates are uniformly close to tangent coordinates. These constants depend on the fixed leaf chart, not on j or K_j. Let B bound the height gradients on this chart and put $r=|x-y|$. The consecutive normal-height separation, summed over the k intervening gaps, gives

$$ck\sqrt{\sigma_j} \le |h_{j,b}(x) - h_{j,a}(x)| \le |h_{j,b}(y) - h_{j,a}(x)| + Br.$$

Both r and $|h_{j,b}(y) - h_{j,a}(x)|$ are at most $C|X - Y|$ by the inverse tubular map. Consequently

$$|X - Y| \ge c_1\big(|x - y|^2 + k^2\sigma_j\big)^{1/2}.$$

The coordinate Jacobians are uniformly bounded, so the following row integral may be estimated in these base coordinates. Because Φ is ambiently Lipschitz, the Dirichlet numerator is at most $C|X - Y|^2$. The row integral for this pair is thus bounded by

$$C\int_0^a \frac{r^{d-1}}{(r^2 + k^2\sigma_j)^{(d-1+p_j)/2}}\,dr \le C\left(1 + \log_+ \frac{1}{k\sqrt{\sigma_j}}\right).$$

When $k\sqrt{\sigma_j} < a$, split at that radius: the lower part is at most $C(k\sqrt{\sigma_j})^{1-p_j}$, which is uniformly bounded on the fixed tube, and the upper part is bounded by an integral of r^{-1}. When $k\sqrt{\sigma_j} \ge a$, the whole integral is uniformly bounded. Summing both directions in each row, then averaging, gives

$$C\sigma_j \sum_{k=1}^{K_j} \left(1 + \log_+ \frac{1}{k\sqrt{\sigma_j}}\right) \le C\sigma_j K_j(1 + |\log \sigma_j|) \longrightarrow 0.$$

Any pair not accounted for in these common smaller charts has a fixed positive separation; all such Dirichlet terms, including far same-sheet terms, are bounded after averaging by $C\sigma_j K_j \to 0$. Each sheet contributes uniformly bounded area on the fixed collar. Cross-sheet potential terms inside the tube are nonpositive in Q_{p_j} and may be discarded for an upper bound. The same is true of far same-sheet potential terms. In contrast, the complementary interactions are kept combined. The regular-boundary hypotheses of 4.1 hold near F: the packet is $C^{2,\alpha}$ there and the phase is stationary. The fixed cutoff χ and support F have a positive distance as required in that lemma. It bounds the complete complementary term, with its Dirichlet and normal-potential parts combined, independently of the exterior area or number of interfaces. Its absolute normalized averaged contribution is at most

$$C\frac{\sigma_j}{K_j}\int_{\partial E_j} \phi_j^2 \le C\sigma_j \longrightarrow 0.$$

The local part has weights $\chi(X)\chi(Y)$. They equal one in the close same-sheet calculation and lie between zero and one everywhere else. They therefore preserve the computed same-sheet limits, the cross-sheet Dirichlet upper bounds and the nonpositivity of the discarded potentials. The areas of all weighted local patches are bounded by CK_j. This accounts for the entire quadratic form without using an exterior unsigned surface measure.

Consequently

$$0 \le \frac{\sigma_j}{K_j} Q_{p_j}[\phi_j] \le \frac{c_d}{2}\int_L (|\nabla_L \eta|^2 - |A_L|^2\eta^2) + o(1),$$

which proves the asserted scalar stability inequality.

To prove the assertion for covers, let $\pi_0 : \widetilde{L} \to L$ be a Riemannian covering and $f \in C_c^\infty(\widetilde{L})$. Set

$$S(x) = \sum_{y \in \pi_0^{-1}(x)} f(y)^2, \qquad g_\varepsilon(x) = \sqrt{S(x) + \varepsilon^2} - \varepsilon.$$

The sum is locally finite. Choose an evenly covered neighborhood V and a smaller $V_0 \Subset V$. The compact set $\operatorname{supp} f \cap \pi_0^{-1}(\overline{V}_0)$ is covered by the disjoint lifted sheets over V, so finitely many of those sheets cover it. Only those finitely many terms contribute on V_0. Identify all lifted gradients

with vectors in T_xL through the isometries $d\pi_0$. Thus S and g_ε are smooth, and their support is contained in the compact set $\pi_0(\operatorname{supp} f)$. Cauchy–Schwarz gives pointwise

$$|\nabla_L g_\varepsilon|^2 = \frac{|\sum_{y\mapsto x} f(y)\nabla f(y)|^2}{S+\varepsilon^2} \le \sum_{y\mapsto x} |\nabla f(y)|^2.$$

Apply the scalar base inequality to g_ε and let $\varepsilon \downarrow 0$. Dominated convergence and the covering change of variables yield

$$\int_{\widetilde{L}} |A_{\widetilde{L}}|^2 f^2 = \int_L |A_L|^2 S \le \int_{\widetilde{L}} |\nabla_{\widetilde{L}} f|^2.$$

No finiteness or growth assumption on the covering degree is needed. The base inequality uses only its metric and the scalar potential $|A_L|^2$, so it also applies when L is one-sided. Its orientable cover is two-sided in the oriented ambient space. □

5.2. **Classical graph compactness.**

Lemma 5.3 (Uniform local classical curvature and graph regularity under a curvature bound)**.** *Let $2 \le m \le 7$, $C_0 < \infty$, and $0 < r < R < \infty$. There are $p_* < 1$, $\rho > 0$ and $C < \infty$, depending only on m, C_0, r, R, with the following property. Let $p_* < p < 1$, and let E be a phase stationary and stable in $B_R \subset \mathbb{R}^m$ in the sense of 2.2. Suppose its entire boundary there is a relatively closed embedded C^2 hypersurface without hypersurface boundary, the phase occupies exactly one local side at every boundary point, and $|A_{\partial E}| \le C_0$. Then every boundary point in B_r has a graph patch over a disk of radius ρ in its tangent plane, with uniform $C^{2,1/2}$ bounds after subtracting its central height. The constants are independent of the number of neighboring interfaces. Throughout $\partial E \cap B_r$, its classical scalar mean curvature satisfies*

$$|H_{\partial E}| \le C\sqrt{1-p}.$$

Here H is the trace of the second fundamental form, and its sign will not matter. The same assertion applies on any fixed compact subset of an open set, with a uniform curvature bound on a larger compact neighborhood.

Proof. Use stationarity and unrestricted stability as defined in 2.2 and the fractional and graph conventions of 2.1, in particular $d = m-1 \le 6$, $\sigma = 1-p$ and the graph operator $\mathcal{M}(h)$. Complementation and fixed similarities preserve the phase hypotheses by 2.11. Rescale and translate about any chosen point of $\partial E \cap B_r$ so that its tangent plane is horizontal, its ambient B_2 lies in B_R, and its curvature is at most one. The scale is fixed by C_0 and $R-r$, uniformly for all the chosen points. The clipped chart of 3.3 gives a finite ordered packet over B^d_{2a} covering the entire interface in $B^d_a \times (-b,b)$, with uniformly small slopes and Hessians; here $a, b > 0$ are its fixed chart radii. These symbols denote chart radii and are independent of p and the packet cardinality.

Dilate this chart by $2/b$. Its Q_2 is the original $B^d_b \times (-b,b)$, and its required B^d_3 graph base is the original $B^d_{3b/2} \subset B^d_{2a}$. In these dilated coordinates the distinguished sheet through the chosen point, denoted by h_k, has a connected graph patch over $B^d_{1/2}$, entirely inside $B^d_{1/2} \times (-1/4, 1/4)$. We consider its points over a fixed disk $D \Subset B^d_{1/4}$. These points have a fixed positive margin from both the boundary of Q_1 and the boundary of the larger distinguished graph patch, and every graph slope is at most $L = 1/100$. Fix $\delta_0 = 1/8$ and consider $X = (x, h_k(x))$ on this patch and another interface point $Y = (y, h_i(y))$ with $|X - Y| < \delta_0$. This point lies in Q_2, so the clipped family represents it. At the common base point x, the slope bound gives $|h_i(x) - Y_m| \le L|x-y|$, hence $|h_i(x)| < 1/2$. Both this point and the distinguished point at x lie in Q_1, and so do all ordered graph points between them. The separation theorem 4.8 applies to every consecutive pair between their indices, giving $|h_i(x) - h_k(x)| \ge c\sqrt{\sigma}$ whenever $i \ne k$. The elementary comparison

$$|h_i(x) - h_k(x)| \le |Y_m - X_m| + L|x - y| \le \sqrt{1 + L^2}\,|X - Y|$$

therefore gives $|X - Y| \ge c\sqrt{\sigma}/\sqrt{1 + L^2}$. All remaining interface points are at least the fixed distance δ_0 away, including cap or remote points not in this close case. In particular the interfaces outside Q_2 have a still larger fixed distance from the chosen patch. Choose $c' > 0$ smaller than the close-case constant and, for p sufficiently near one, require $c'\sqrt{\sigma} < \delta_0$. Every ball $B_{c'\sqrt{\sigma}}(X)$ then meets no other interface. For a packet with one sheet this conclusion follows from the fixed boundary margin alone.

There is one fixed choice of the phase over the whole distinguished patch. Its upward graph normal N is continuous, as is the exterior phase normal ν_E supplied by genuine one-side regularity. The continuous function $\nu_E \cdot N$ takes values only in $\{-1, 1\}$, so it is constant on the connected patch. Complement E once if this constant is negative. Then the phase locally lies below the distinguished graph at every point of the patch. In each critical ball the boundary is exactly that graph portion, with no other interface, and the base margin keeps this portion in the same patch. This consistent local one-side description therefore identifies the phase throughout the ball with the subgraph, up to a null set: away from the graph both characteristic functions are locally constant, and near the graph they agree, so their difference is a locally constant function on the connected ball and vanishes near its center. Thus the same choice of E or E^c works in every such ball.

Subtract the central height, extend the graph from a larger base disk by a smooth cutoff, and add back the height. Choose the cutoff equal to one on a fixed neighborhood of $\overline{D}$ inside the larger disk. By increasing the near-one threshold for p, ensure that $c'\sqrt{\sigma}$ is smaller than the horizontal margin to the edge of that neighborhood. Every critical ball over D then sees precisely the original graph. The extension has uniform global Lipschitz and Hessian bounds and retains the established ball-wise phase agreement there. By 3.4 its normalized curvature has a fixed signed volume source outside those balls, and 3.5 gives uniform local Lipschitz forcing for this critical radius. For these applications choose the disk considered above as $D = B^d_{1/8}$, let $\ell = 1/64$, and make one further fixed similarity centered at the chosen point. In the resulting coordinates the graph and phase are

$$\widehat{h}(z) = \frac{h_k(\ell z) - h_k(0)}{\ell}, \qquad \widehat{E} = \ell^{-1}(E - (0, h_k(0))).$$

The graph is defined on B^d_4, since $\ell B^d_4 = B^d_{1/16} \subset D$. If K denotes the common Hessian bound in the preceding dilated chart, then $\|\nabla\widehat{h}\|_\infty \le L$ and $[\nabla\widehat{h}]_{C^{0,1}(B_4)} \le \ell K$. Thus one common $M_0 = L + \ell K + 1$ satisfies the sum bound required by the regularity corollary. Choose $c = \min\{1/4, c'/\ell\} \in (0, 1/4]$. Every base point $z \in B^d_3$ corresponds to $\ell z \in D$, and its critical ball of radius $c\sqrt{\sigma}$ corresponds under this similarity to a ball of radius $\ell c\sqrt{\sigma} \le c'\sqrt{\sigma}$. The established ball-wise phase agreement consequently holds for the same fixed phase choice throughout B^d_3. The similarity preserves stationarity and genuine regular boundary. Restricting to $p \ge 7/8$, which is harmless for the asserted near-one threshold, verifies every remaining hypothesis of 3.9. Its conclusion gives a common $C^{2,1/2}$ bound on B^d_1, containing the central point $z = 0$. The corollary 3.11 now gives $|\mathcal{M}(\widehat{h})| \le C\sqrt{\sigma}$ on $B^d_{1/2}$. Direct differentiation of the graph operator shows $\mathcal{M}(\widehat{h})(z) = \ell\mathcal{M}(h_k)(\ell z)$; undoing this last fixed scale therefore gives $|\mathcal{M}(h_k)| \le (C/\ell)\sqrt{\sigma}$ on $B^d_{\ell/2}$. On a graph the magnitude of H equals $|\mathcal{M}(h)|$, so this proves the mean-curvature estimate. Undoing the fixed scales changes only the uniform constants and provides one positive graph radius ρ and bound C at every chosen point. In an arbitrary open set the proof uses only a fixed positive distance between the inner compact set and the boundary of the larger neighborhood, so it proves the final local assertion as well. □

Proposition 5.4 (A stable classical limit)**.** *Suppose $p_j \uparrow 1$, $R_j \to \infty$, and E_j is stationary and stable in $B_{R_j} \subset \mathbb{R}^m$. Assume that ∂E_j is a closed embedded C^2 hypersurface with no hypersurface*

boundary in that ball, and E_j occupies exactly one local side at every point of ∂E_j there. Assume also $0 \in \partial E_j$ and $|A_{\partial E_j}| \le C_0$ there. If $m \le 7$, a subsequence converges locally in graph charts to a classical minimal lamination. Every leaf has a complete two-sided cover which is a stable minimal immersion. A disk selected by a convergent sequence of boundary points converges in C^2, so its second fundamental form is preserved in the limit.

The corresponding local statement holds in a fixed open set, with stability for compactly supported tests and continuation until its boundary instead of completeness in all of $\mathbb{R}^m$.

Proof. Use 2.1 for the graph and curvature conventions and 2.2 for unrestricted stationarity and stability. Apply 5.3 on any fixed compact ambient region, using a larger compact neighborhood with positive margin to the eventual domain boundary. In the expanding case that neighborhood lies in B_{R_j} for all sufficiently large j. The common curvature bound and the regular stationary-stable phase hypotheses are exactly those of the lemma. It supplies uniform graph radii and $C^{2,1/2}$ bounds, independently of packet cardinality, and

$$|H_{\partial E_j}| \le C\sqrt{\sigma_j}$$

on the smaller region. These estimates are available before extracting a classical limit.

We first place these graph families in fixed ambient charts. The functions $d_j(X) = \min\{1, \operatorname{dist}(X, \partial E_j)\}$ are bounded and 1-Lipschitz. Arzelà–Ascoli and a diagonal exhaustion give local uniform convergence, after a subsequence, to d. Put $F = \{d = 0\}$. This is a closed set containing the origin. Every $X \in F$ is approached by points $X_j \in \partial E_j$: choose a point within twice the distance, with an additional error $1/j$ if needed. Conversely, every convergent sequence of boundary points has its limit in F, by local uniform convergence of the distance functions. Points in a fixed compact set lie in the regular eventual domain for large j.

Fix a compact exhaustion K_n of the ambient domain, and choose a larger compact neighborhood of each K_n with positive margin to the domain boundary. The uniform estimates established above, together with 3.3, give common positive chart radii r_n, b_n for centers in K_n: in each center's tangent coordinates a packet of actual boundary graphs is defined on $B^d_{2r_n}$ and contains every interface in $B^d_{r_n} \times (-b_n, b_n)$. Their slopes and heights are uniformly bounded. On uniform inner disks the graph functions also have the common $C^{2,1/2}$ bounds proved above. If necessary shrink the horizontal radii once. These bounds are uniform for all packet graphs that meet the smaller cylinder: apply the local estimates at their points, and re-express their tangent graphs in the packet coordinates. Their slopes are uniformly small, so the projection derivatives have uniformly bounded inverses; the inverse derivative formula and chain rule preserve the $C^{2,1/2}$ bounds. A finite disk cover of fixed size gives a bound on each common smaller disk. Graphs that cross a cap may be retained on that disk, as they are actual boundary patches and their heights remain uniformly bounded. All these choices depend on the compact margins and the common curvature bound, not on j or the number of graphs.

Set $\eta_n = \min\{r_n/16, b_n/8\}$. Choose a finite η_n-net in the compact set $F \cap K_n$, with its centers in that set; omit empty sets. For every center X in this countable union choose $X_j \in \partial E_j$ with $X_j \to X$. Compactness of the Grassmannian and diagonal subsequences arrange that the tangent planes $T_{X_j}\partial E_j$ converge to a plane T_X, simultaneously for every chosen center. Select orthonormal coordinates tending to fixed coordinates on $T_X \oplus T_X^{\perp}$.

Here is the quantitative reparametrization of each such moving packet onto a fixed disk. Let $g_{j,i}$ be its height functions in the moving coordinates, with common height bound B and common slope bound L. Projection onto T_X, in the fixed coordinates about X, has the form

$$F_{j,i}(z) = u_j + M_j z + b_j g_{j,i}(z), \qquad u_j \to 0, \quad M_j \to I, \quad b_j \to 0.$$

For sufficiently large j, uniformly in i, one has $\|M_j^{-1}\| \le 2$, $|u_j + b_j g_{j,i}(0)| \le r_n/8$ and $2|b_j|L \le 1/4$. For $|y| < r_n/4$, the equation $F_{j,i}(z) = y$ is equivalent to

$$z = M_j^{-1}(y - u_j - b_j g_{j,i}(0) - b_j(g_{j,i}(z) - g_{j,i}(0))).$$

Its right-hand side maps $\overline{B}_{r_n}$ into itself: the constant part has norm at most $3r_n/4$, and the last part has norm at most $r_n/4$. It is a contraction with constant at most $1/4$. Thus each graph has a unique reparametrization over the same fixed $B^d_{r_n/4} \subset T_X$. The projection derivative and its inverse are uniformly bounded. Differentiating the inverse identity gives common $C^{2,1/2}$ bounds for the inverse maps; composition with the graph and the fixed linear normal projection gives common $C^{2,1/2}$ bounds for the regraphed heights. The graphs remain actual boundary disks throughout these domains.

The fixed box $B^d_{r_n/8} \times (-b_n/4, b_n/4)$ about X is contained, for large j, in the moving cylinder where the packet contains every interface. It contains the ambient ball $B_{2\eta_n}(X)$. Consequently the finite nets ensure that these countably many fixed boxes cover every point of F, and all interfaces in each box are represented by the regraphed packet.

Here is the compactness extraction when packet cardinalities are unbounded. In each fixed chart the uniform $C^{2,1/2}$ bounds on a larger disk put all graph functions in a subset whose closure is compact in C^2 on a smaller disk. The finite graph collections have subsequential Hausdorff limits as closed subsets of this compact metric graph space. Charts not meeting the limiting support may be omitted. Diagonalize over a countable cover and an exhaustion of the ambient domain. This extraction keeps all local graph limit points, rather than only finitely many enumerated sheets, and gives a closed ambient support. Each limiting graph has zero classical mean curvature by the preceding estimate.

In a common smaller chart any two limiting disks inherit ordering from the approximating packets. Transverse intersections would persist under C^1 convergence and violate that ordering. Ordered minimal graphs touching at an interior point coincide locally by the strong maximum principle. Thus distinct limiting plaques are disjoint, and overlap continuation defines compatible minimal lamination charts. Continuity of the graph family follows from compactness and uniqueness of a plaque with prescribed central height. To check transverse continuity of the tangent fields, let $u > 0$ be the difference of two distinct ordered minimal graphs. Subtracting their classical graph equations gives $\operatorname{div}(a\nabla u) = 0$, where a is the average of the derivative of $v \mapsto v/\sqrt{1+|v|^2}$ along their gradients. The uniform graph bounds give common ellipticity and Lipschitz bounds for a. The Harnack estimate of 2.13 gives $\sup u \le Cu(0)$ on an inner disk. Its interior $W^{2,q}$ estimate, followed by Sobolev embedding with $q > d$, gives $\sup|\nabla u| \le Cu(0)$ on a still smaller disk. Thus the tangent fields vary continuously, uniformly over the graph family, as the central heights approach one another. These are the ordinary local minimal-lamination charts. Elliptic regularity for the minimal graph equation makes every limiting plaque smooth. Now consider any sequence of uniform smaller disks based at boundary points converging to $X \in F$, along the subsequence already selected. Reparametrize them in the fixed tangent coordinates at X. Their uniform bounds make them precompact in C^2, and every subsequential limit is a plaque of the extracted lamination through X. Disjointness and local uniqueness give exactly one such plaque. Therefore the whole selected sequence of reparametrized disks converges to it in C^2. Choosing local normals consistently, their second fundamental forms converge as well.

Every leaf is complete in its induced metric when the ambient domain is all of $\mathbb{R}^m$. Indeed, a finite-length curve in a leaf is an ambient Cauchy curve, and its limit belongs to the closed lamination support. Choose a lamination box containing that endpoint in its interior. The final curve segment remains in this box; its plaque label is locally constant, and hence constant on the connected final parameter interval. The endpoint therefore belongs to that same plaque. On a smaller plaque its graph metric is uniformly equivalent to the Euclidean metric, so ambient convergence to the

endpoint also gives intrinsic convergence and continuation. In particular no unit-speed geodesic can terminate in finite time. Completeness lifts under Riemannian coverings, including the universal cover.

It remains to prove stability on a two-sided cover of each leaf. The external theorem of Meeks–Pérez–Ros, Limit leaves of an H lamination are stable, JDG 84 (2010), 179–189; arXiv:0801.4345v2, Theorem 1 and Remark 1, states that every two-sided cover of a limit leaf of a codimension-one minimal lamination of a Riemannian manifold is stable for its Jacobi operator, in arbitrary dimension. A limit leaf is a leaf approached by distinct local plaques; the discussion after Definition 4 includes returning plaques from the same global leaf. Plaque accumulation at one point propagates to every point of the connected leaf. Join two points by a path in the leaf and cover its compact image by a finite chain of overlapping lamination boxes. Continuing plaques through an overlap induces an injective continuous map between smaller transverse plaque parameters, fixing the parameter of the chosen leaf. Distinct parameters converging to that parameter therefore continue to distinct parameters converging to the chosen leaf at the next box. Shrinking the transverse neighborhoods through the finite chain gives accumulation at the endpoint. Thus accumulation by distinct plaques satisfies the limit-leaf characterization in that source, and the theorem applies. The same argument includes returning plaques from the same global leaf; no distinct global leaf is required. [14]

Otherwise the leaf has no accumulating plaque at any of its points. Fix compact connected collared domains $D \Subset D_1 \Subset D_2$ in it. If the leaf is compact, one may take the whole leaf for these domains, with empty lateral boundary. Compactness and plaque isolation give a fixed normal tube over D_2 which, over D_1, contains only the zero plaque of the limiting lamination. After shrinking the tube, its normal boundary over D_1 is a fixed positive distance from the limiting support. All approximating interfaces in its smaller interior converge uniformly in C^2 to the zero section: otherwise an offending sequence of graph disks would have a limit differing from that plaque. In particular normal projection on these interfaces is a local diffeomorphism, and no interface meets the normal boundary for sufficiently large j. For a compact subset of the base, its preimage is compact by this normal-boundary exclusion and relative closedness of the interfaces. Thus projection is proper over the base interior. It is nonempty by the graph convergence to the zero section; a proper local diffeomorphism onto the connected base is surjective and is a finite covering with constant cardinality.

We verify the packet's normal-coordinate spacing on this fixed compact domain. Enlarge D_1 slightly inside D_2 and repeat the preceding isolation and proper-projection argument there. Thus the complete finite normal covering is defined on an open neighborhood of $\overline{D}_1$ with a fixed positive collar in its base. Choose a fixed $\rho > 0$ and finitely many centers P in this neighborhood such that, in coordinates centered at P with its tangent plane horizontal, the limiting leaf is a graph g on $B^d_{4\rho}$, with $g(0) = 0$ and $\|Dg\|_\infty \le 1/200$. These larger leaf graph disks lie in the base of the covering, and their graph pieces over $B^d_{\rho/8}$ cover $\overline{D}_1$. Shrink ρ using the fixed collar between D_1 and D_2, the normal-tube radius and the isolation of this leaf. The cylinders $\overline{Q}_{2\rho} = \overline{B}^d_{2\rho} \times [-2\rho, 2\rho]$ are then inside the larger isolated-leaf tube and a positive distance from any portion of the limiting support outside the chosen leaf graph. These choices are possible uniformly on $\overline{D}_1$ by smoothness, embeddedness, isolation and compactness.

Over each larger leaf graph disk the finite normal covering is trivial, so all its branches are actual normal graphs. Their uniform C^2 convergence to the zero section makes their horizontal projections uniformly close to the identity in C^2. The fixed margin between $B^d_{4\rho}$ and $B^d_{3\rho}$, together with the inverse-map argument used above, gives every branch a full vertical graph $h_{j,i}$ on $B^d_{3\rho}$. The inverse derivative formula and chain rule transfer the uniform $C^{2,1/2}$ bounds to these graphs. Their slopes are at most $\ell = 1/100$ and their Hessians at most a fixed M_0, for all sufficiently large j, uniformly over all branches and the finite chart collection. The complete normal packet and the fixed exclusion of other limiting support ensure that the entire interface in $Q_{2\rho}$ is exactly the

clipped union of this ordered family. The phase occupies one local side at every such interface point, as in the original hypotheses. Furthermore, uniform convergence of every packet graph to g gives

$$|h_{j,i}(y)| < \rho/2 \qquad (|y| < \rho/4)$$

for every branch on a common tail. Rescaling by ρ^{-1} gives full graph base B_3^d, the complete clipped boundary in Q_2, slope bound $1/100$ and Hessian bound $M = \rho M_0$. Stationarity and stability hold in these cylinders, which lie in the eventual ambient domain. Choose the tail also so that $p_j > p_*(d, M)$. Theorem 4.8 therefore gives, for all adjacent vertical graphs and $|y| < \rho/4$,

$$h_{j,i+1}(y) - h_{j,i}(y) \geq \rho c(d, M)\sqrt{\sigma_j}.$$

Both comparison points lie in Q_ρ by the height bound just proved, so this checks the theorem's Q_1 hypothesis after rescaling.

Fix a point X of the leaf in one of the inner graph pieces and choose its local upward unit normal $\nu = (\nu', \nu_m)$. For two consecutive interfaces on this normal fiber write

$$X_a = X + t_a\nu = (y_a, h_{j,a}(y_a)), \qquad X_b = X + t_b\nu = (y_b, h_{j,b}(y_b)), \qquad \delta = t_b - t_a > 0.$$

Set $\delta_0 = \rho/32$. If $\delta \geq \delta_0$, then $\delta \geq \delta_0\sqrt{\sigma_j}$ since $0 < \sigma_j < 1$. For $\delta < \delta_0$, the uniform convergence of all normal heights to zero permits one common tail with $|t_i| < \rho/16$. Since the base point of X has horizontal coordinate in $B^d_{\rho/8}$, both y_a, y_b belong to $B^d_{3\rho/16} \subset B^d_{\rho/4}$. Thus the comparison at the common horizontal point y_a uses the full graph domains and the height and spacing bounds above. The normal-fiber identities give $y_b - y_a = \delta\nu'$ and

$$\Delta := h_{j,b}(y_a) - h_{j,a}(y_a) = \delta\nu_m + h_{j,b}(y_a) - h_{j,b}(y_b).$$

Because $\nu = (-Dg, 1)/\sqrt{1 + |Dg|^2}$ at X, one has $\nu_m - \ell|\nu'| \geq 1/2$. The slope bound therefore implies

$$0 < \tfrac{1}{2}\delta \leq \Delta \leq \delta(\nu_m + \ell|\nu'|) \leq (1 + \ell)\delta.$$

In particular the two distinct branches are ordered vertically in the same direction. They need not be adjacent in that vertical ordering: if there are intermediate branches, all their heights at y_a lie between the endpoint heights, hence in Q_ρ. Sum the preceding vertical spacing inequalities over these intermediate gaps. There is at least one gap, so

$$(1 + \ell)\delta \geq \Delta \geq \rho c(d, M)\sqrt{\sigma_j}.$$

Thus every consecutive normal-fiber pair is separated by $c_0\sqrt{\sigma_j}$, where $c_0 = \min\{\delta_0, \rho c(d, M)/(1+\ell)\} > 0$. The finite chart cover makes the constants and the tail uniform over $\overline{D}_1$, independently of the packet cardinality. The same argument applies to every smaller compact normal chart needed for a test. Reversing a local normal direction reverses the ordering and leaves this positive separation unchanged.

Choose a fixed ambient open set U containing the closure of this packet tube and compactly contained in the eventual ambient domain. In the expanding case $U \Subset B_{R_j}$ for all sufficiently large j; in the local case U lies inside the fixed given open set. Compact variations in U are among the original admissible variations, so the phases are stationary and stable there. The packet neighborhood lies in U, and the preceding verification supplies the spacing and uniform regularity hypotheses of Lemma 5.2. That lemma proves the scalar stability inequality for every test supported in D. As the compact domains exhaust the leaf, it proves the scalar inequality on the entire leaf and, by the covering argument there, on its orientable universal cover. The universal cover is orientable and its normal bundle in oriented Euclidean space is trivial. It is therefore a complete two-sided stable minimal immersion, whether or not its image is proper.

All arguments are local on compact ambient regions. In a fixed open set, the same construction gives the asserted minimal lamination and stability for compactly supported tests; the path continuation argument applies whenever its ambient endpoint lies inside that open set. □

5.3. **The curvature blow-up contradiction.**

Theorem 5.5 (Uniform curvature near $p = 1$)**.** *For each $3 \le m \le 7$ there are $\bar{p}_m < 1$ and $C_m < \infty$ such that every phase stationary and stable in $B_2 \subset \mathbb{R}^m$ whose boundary is a closed embedded C^2 hypersurface without hypersurface boundary there, with the phase occupying exactly one local side at every boundary point in B_2, satisfies*

$$\sup_{\partial E \cap B_1} |A_{\partial E}| \le C_m \qquad \text{if } \bar{p}_m < p < 1.$$

The smoothness assumption concerns all of B_2.

Proof. Use the perimeter conventions in 2.1 and the phase stationarity and stability of 2.2. If the asserted constants did not exist, we could choose the phases inductively so that

$$p_j > \max\{p_{j-1}, 1 - 1/j\}, \qquad \sup_{\partial E_j \cap B_1} |A_{\partial E_j}| > j,$$

with $p_0 = 0$. Thus $p_j \uparrow 1$ and the interior curvature maxima diverge. For each j, relative closedness and C^2 regularity make $\partial E_j \cap \overline{B}_{3/2}$ compact and its curvature continuous. Maximize

$$(3/2 - |X|)|A_{\partial E_j}(X)|$$

on that set. Its maximum is positive and its weight vanishes on the outer sphere, so a maximizing point X_j lies in $B_{3/2}$. Put

$$\delta_j = 3/2 - |X_j|, \qquad \lambda_j = |A_{\partial E_j}(X_j)|, \qquad T_j = \lambda_j \delta_j.$$

Then $T_j \ge \frac{1}{2} \sup_{\partial E_j \cap B_1} |A_{\partial E_j}|$ tends to infinity. If $|X - X_j| \le \delta_j/2$, the weight at X is at least $\delta_j/2$, and maximality gives $|A_{\partial E_j}(X)| \le 2\lambda_j$.

Put $\widetilde{E}_j = \lambda_j(E_j - X_j)$. By 2.11 these translations and positive dilations preserve stationarity, stability and the genuine regular-boundary hypotheses. The curvature scaling law is

$$|A_{\partial \widetilde{E}_j}|(\lambda_j(X - X_j)) = \lambda_j^{-1} |A_{\partial E_j}|(X).$$

The rescaled boundaries are closed, embedded, and without boundary in $B_{T_j/2}$; there they satisfy $|A| \le 2$ and have $|A|(0) = 1$. Their domains expand to all of $\mathbb{R}^m$.

Proposition 5.4, retaining the graph disk through the origin, gives a smooth minimal leaf whose curvature at the origin is one. Take a connected component of the complete two-sided stable cover furnished by that proposition and a lift of the origin. A connected component still covers the connected leaf; it is complete and two-sided, and stability holds for its compact tests by extension by zero to the other components. Its covering immersion preserves the second fundamental form, so its curvature at that lift is one. Hong–Li–Wang [13], *The stable Bernstein theorem in* $\mathbb{R}^7$, arXiv:2609.15720v1 (14 September 2026), Theorem 1.1 and Corollary 1.2, state that a smooth, connected, complete, two-sided stable minimal immersion of intrinsic dimension $2 \le n \le 6$ into $\mathbb{R}^{n+1}$ has vanishing second fundamental form and image an affine hyperplane. Theorem 1.1 treats $n = 6$, and Corollary 1.2 gives the full indicated range without a properness or volume-growth hypothesis. Our connected covering immersion has all these properties, with $n = m - 1 \in \{2, \dots, 6\}$, but has curvature one at the chosen lift. This contradiction proves the uniform curvature bound. □

6. Stable cone rigidity in ambient dimensions four through seven

The curvature bound gives fixed annular charts for regular cones. We first identify their limiting spherical links, then align the graphs with one plane and control their heights at scale $\sqrt{\sigma}$. Height control and separation together bound the number of sheets. The homogeneous gap inequality then excludes every remaining multiple-sheet configuration.

6.1. **Compact spherical links.**

Lemma 6.1 (Almost-minimal links in the sphere)**.** *Let $2 \le k \le 5$ and let $S_j \subset S^{k+1}$ be nonempty closed embedded hypersurfaces. Suppose they have uniform local $C^{2,\alpha}$ graph bounds with a uniform graph radius, and that $\sup_{S_j} |H_{S_j}| \to 0$. Then, after a subsequence, there is a connected compact embedded minimal hypersurface $\Sigma \subset S^{k+1}$ such that each S_j is a finite union of ordered normal graphs over Σ, and the supremum of the C^2 norms of these graphs tends to zero. Their number need not be bounded at this stage.*

Proof. Choose a radius smaller than the common graph radius and work first with all pointed graph disks of that radius, rather than with a chosen enumeration of sheets. A pointed disk is recorded by its center in the sphere, an orthonormal tangent frame there, and its graph function in the corresponding spherical coordinates. The centers and frames lie in a compact frame bundle. The uniform $C^{2,\alpha}$ bounds put the graph functions in a precompact subset of C^2 on a smaller disk. Thus the closures of these pointed-disk collections lie in one compact metric space. Hausdorff compactness of its nonempty closed subsets supplies a subsequence retaining every pointed disk limit, including limits along varying centers and frames. The union of the limiting disks is nonempty and closed: if their points converge, take the corresponding pointed disks at those points and use the same compactness. Every limiting disk is minimal because the graph mean curvatures tend uniformly to zero. In each fixed spherical chart its minimal graph equation is uniformly elliptic. Interior elliptic bootstrapping therefore makes the limiting C^2 disks smooth.

At a point of this support, two limiting disks cannot intersect transversely: their approximating disks are subsets of the same embedded S_j, and a transverse intersection persists under C^1 convergence. Moreover tangent planes of disks centered at nearby support points converge to that of a disk through the given point. Otherwise pointed-disk compactness would produce a second disk through that point with a distinct tangent plane, hence a transverse intersection. Choose a sufficiently small box about the point. All its disks can consequently be reparametrized, with a common base disk and controlled inverse projection, as graphs over the same tangent plane. Their approximations are disjoint graphs, so their limiting graphs are ordered. Two ordered minimal graphs touching at an interior point coincide by the strong maximum principle. The distinct limiting plaques are therefore disjoint. Graph compactness and this uniqueness make the plaque depend continuously in C^2 on its central height. They give the local product charts of a minimal lamination $\mathcal{L}$ and agree on overlapping boxes by the same uniqueness. No stability or bound on the number of prelimit disks is used in this extraction.

Every leaf is complete for its induced metric. A finite-length path has an ambient endpoint in the compact sphere. In a sufficiently small lamination chart about that endpoint its final segment lies in one plaque, because its transverse coordinate is locally constant and the final parameter interval is connected. The plaque metric is uniformly equivalent to its graph metric, so the path has an intrinsic endpoint there. The orientable double cover of the leaf, and hence any further Riemannian covering, is also complete.

If a leaf were noncompact, completeness supplies a sequence escaping all intrinsic compact sets. Pass to an ambient accumulation point and a smaller lamination chart whose plaques lie in compact subplaques of a larger chart. If only finitely many plaques of the larger chart contained the sequence, its terms in the smaller chart would lie in a finite union of intrinsic compact subplaques. This contradicts intrinsic escape. Thus distinct plaques accumulate. Infinitely many distinct leaves also

give an accumulation of distinct plaques in one of the finitely many charts. Accumulation at one point propagates along the connected leaf: cover a path in that leaf by a finite chain of overlapping lamination boxes. Continuing plaques through an overlap induces an injective continuous map between smaller transverse parameters, so distinct parameters converging to the chosen leaf remain distinct and converge to it in the next box. Shrinking the neighborhoods through the finite chain proves accumulation at the endpoint. In either case there is a limit leaf; returning plaques of a single leaf are included in the characterization after Definition 4 in (Meeks–Pérez–Ros, Limit leaves of an H lamination are stable, arXiv:0801.4345). The external limit-leaf theorem of Meeks–Pérez–Ros (JDG 84 (2010), 179–189; arXiv:0801.4345v2, Theorem 1 and Remark 1) states, in arbitrary dimension, that every two-sided cover of a limit leaf of a codimension-one minimal lamination is stable for the Jacobi operator. It therefore makes the complete orientable two-sided cover of our limit leaf stable. The external Catino–Mastrolia–Roncoroni theorem, Two rigidity results for stable minimal hypersurfaces, GF&A 34 (2024), 1–18, DOI:10.1007/s00039-024-00662-1, Corollary 1.3, states that a complete ambient manifold with nonnegative sectional curvature and uniformly positive Ricci curvature contains no complete orientable immersed stable minimal hypersurface of dimension at most five. Our cover has dimension $k \le 5$; its ambient unit sphere has sectional curvature one and Ricci curvature $k > 0$, so the theorem excludes it. Neither external result imposes properness or an area-growth hypothesis. Thus every leaf is compact and there are only finitely many leaves. The inclusion of a compact leaf is an injective immersion and is proper, hence is an embedding. Therefore $\mathcal{L}$ consists of finitely many compact embedded minimal hypersurfaces. [5, 14]

There can be only one such leaf. If two disjoint compact minimal hypersurfaces existed, a shortest geodesic between them would meet both orthogonally. Parallel transport an orthonormal basis of the tangent space of the first along that geodesic. At the other endpoint it spans the tangent space of the second. The sum of the second variation index forms of these fields is

$$-\int \mathrm{Ric}(\dot\gamma,\dot\gamma),$$

because the endpoint traces are the mean curvatures of the two hypersurfaces and vanish. This is negative, whereas a shortest geodesic between the hypersurfaces has nonnegative index form for these endpoint variations. This contradiction is the usual Frankel argument.

Denote the single compact leaf by Σ. All of S_j lies in an arbitrarily small tubular neighborhood of Σ for large j: otherwise graph compactness at an offending sequence of points would produce another leaf. The same argument with the uniform $C^{2,\alpha}$ bounds gives uniform convergence of tangent planes and second fundamental forms. Normal projection $\pi : S_j \to \Sigma$ is consequently a local diffeomorphism. It is proper, and its nonempty image is open and closed in connected Σ; therefore it is a surjective finite covering.

The compact embedded hypersurface Σ separates the sphere, so it has a global unit normal and a tubular neighborhood of the form $\Sigma \times (-\delta,\delta)$. Sort the K_j points of each covering fiber by their normal coordinate. Continuation around any loop permutes this finite ordered set preserving its order; the permutation is the identity. The sorted points thus define K_j global normal graph functions, strictly ordered at every point. Their local C^2 convergence to zero is uniform over all sheets by graph compactness, and a finite cover of compact Σ makes this global. Discarding finitely many initial indices makes the decomposition valid for every member of the retained subsequence. This proves the asserted decomposition and convergence. □

Lemma 6.2 (Classical stable cones with a smooth compact link)**.** *If $3 \le m \le 7$ and a classical minimal cone in $\mathbb{R}^m$ is stable away from its vertex and has a connected compact smooth embedded link Σ, then Σ is an equator.*

Proof. Classical stability here means $\int_C(|\nabla_C\phi|^2 - |A_C|^2\phi^2) \ge 0$ for every smooth compactly supported scalar test on the cone away from its vertex. Its embedded compact link in the sphere is

two-sided, since a compact embedded hypersurface in the sphere separates it. Minimality of the cone makes its link minimal in the sphere: the radial principal curvature is zero and its remaining curvatures at radius one are the link curvatures. Write $k = m-2$ and $a = |A_\Sigma|$. The metric on the cone is $dr^2 + r^2 g_\Sigma$, its area element is $r^{m-2}\, dr\, d\mathcal{H}^{m-2}_\Sigma$, and its squared curvature is $r^{-2}a^2$. For the product test $\xi(r)f(\theta)$, cone stability gives

$$\left(\int_0^\infty r^{m-4}\xi^2\, dr\right)\int_\Sigma (|\nabla_\Sigma f|^2 - a^2 f^2) + \left(\int_0^\infty r^{m-2}(\xi')^2\, dr\right)\int_\Sigma f^2 \geq 0.$$

Put $b = (m-3)/2$. Completing the square gives $\int r^{m-2}(\xi')^2 \geq b^2 \int r^{m-4}\xi^2$ for $\xi \in C_c^\infty((0,\infty))$. More explicitly, for $\xi(r) = r^{-b}\zeta(\log r)$ one has

$$\int_0^\infty r^{m-4}\xi^2\, dr = \int_{\mathbb{R}} \zeta^2\, dt, \qquad \int_0^\infty r^{m-2}(\xi')^2\, dr = b^2 \int_{\mathbb{R}} \zeta^2\, dt + \int_{\mathbb{R}} (\zeta')^2\, dt.$$

Taking a nonzero fixed smooth compactly supported ζ_0 and $\zeta(t) = \zeta_0(t/L)$, then letting $L \to \infty$, proves that the infimum of the radial quotient is exactly b^2. Passing to this infimum in the product-test inequality gives the spherical stability inequality displayed next.

Consequently

$$\int_\Sigma (|\nabla_\Sigma f|^2 - a^2 f^2) + \frac{(m-3)^2}{4}\int_\Sigma f^2 \geq 0.$$

For a minimal hypersurface of the unit sphere, the Codazzi equations and the Gauss equation give $\Delta_\Sigma A_\Sigma = (k - |A_\Sigma|^2)A_\Sigma$. Taking its scalar product with A_Σ gives the Simons identity (Simons, Minimal varieties in Riemannian manifolds, Ann. Math. 88 (1968), 62–105) [16]

$$\frac{1}{2}\Delta_\Sigma |A_\Sigma|^2 = |\nabla_\Sigma A_\Sigma|^2 + k|A_\Sigma|^2 - |A_\Sigma|^4.$$

Integrating over the compact link yields

$$\int_\Sigma a^4 = k\int_\Sigma a^2 + \int_\Sigma |\nabla_\Sigma A_\Sigma|^2.$$

The function $a = |A_\Sigma|$ belongs to $H^1(\Sigma)$ and is an admissible test in the spherical stability inequality displayed above, by using $a_\epsilon = (|A_\Sigma|^2 + \epsilon^2)^{1/2}$ and passing to the limit. These smooth functions converge uniformly to a, satisfy $|\nabla a_\epsilon| \leq |\nabla A_\Sigma|$, and have gradients converging almost everywhere to ∇a. At almost every point of the zero set of the smooth tensor A_Σ, its derivative and the weak derivative of its norm vanish. Dominated convergence therefore gives $a_\epsilon \to a$ in H^1, justifying both terms of the stability limit. Kato's inequality gives $|\nabla a| \leq |\nabla A_\Sigma|$ almost everywhere. The integrated Simons identity therefore yields

$$\begin{aligned} 0 &\leq \int_\Sigma |\nabla a|^2 - \int_\Sigma a^4 + \frac{(m-3)^2}{4}\int_\Sigma a^2 \\ &= \left(\frac{(m-3)^2}{4} - (m-2)\right)\int_\Sigma a^2 + \int_\Sigma (|\nabla a|^2 - |\nabla A_\Sigma|^2) \\ &\leq \left(\frac{(m-3)^2}{4} - (m-2)\right)\int_\Sigma a^2. \end{aligned}$$

The coefficient is strictly negative for $3 \leq m \leq 7$, so $A_\Sigma = 0$. By the two-sidedness already established, choose a global smooth spherical unit normal ν. For $\theta \in \Sigma$ and $T \in T_\theta\Sigma$, one has $\langle T, \nu\rangle = 0$ and $\langle \theta, \nu\rangle = 0$. The Euclidean and spherical covariant derivatives therefore satisfy

$$D_T\nu = \nabla_T^{S^{m-1}}\nu - \langle T, \nu\rangle\theta = -A_\Sigma(T) = 0.$$

Connectedness makes ν one constant vector ν_0. Hence $\Sigma \subset S^{m-1} \cap \nu_0^{\perp}$, a fixed equator. The inclusion into this equator is a local diffeomorphism because both have dimension $m-2$. Its image is nonempty and open, and is closed by compactness of Σ. The equator is connected for $m \geq 3$, so this image is the entire equator. □

Proposition 6.3 (Conical sheets converge to a common plane)**.** *Let $4 \leq m \leq 7$, let $p_j \uparrow 1$, and let E_j be nontrivial regular stationary stable p_j-minimal cones in $\mathbb{R}^m$. After rotations and a subsequence, their boundaries are finite unions of global conical graphs*

$$\partial E_j = \bigcup_{i=1}^{K_j} \{(x', h_{j,i}(x')) : x' \in \mathbb{R}^d\}, \qquad d = m-1,$$

where $h_{j,1} < \cdots < h_{j,K_j}$ off the origin and $h_{j,i}(rx') = r h_{j,i}(x')$. Their graph norms on every fixed annulus are uniformly $C^{2,\alpha}$ bounded for a fixed $\alpha > 0$, and the maximum of their C^2 norms tends to zero.

Proof. Use $d = m-1$ and $\sigma_j = 1-p_j$ from 2.1 and the genuine smooth-link phase class of 2.4, whose stationarity and stability away from the vertex have the meaning of 2.2. Theorem 5.5, applied to balls with fixed relative radius centered on a conical annulus, gives

$$|A_{\partial E_j}(X)| \leq C_m |X|^{-1} \qquad (X \neq 0)$$

for large j. The vertex is outside every ball where that theorem is applied. On fixed annuli 5.3 now supplies uniform $C^{2,1/2}$ graph bounds and $|H_{\partial E_j}| \leq C\sqrt{\sigma_j}$ at radius one. These estimates are uniform in the number of local branches. The radial principal curvature of a cone is zero; its other principal curvatures at radius one are exactly the principal curvatures of its link as a hypersurface of S^{m-1}. Restricting the conical patches to the sphere preserves a uniform graph radius and $C^{2,1/2}$ bounds. Indeed, the radial vector at a unit boundary point is a unit tangent vector of the cone, and the derivative of $X \mapsto |X|^2$ in that direction is exactly two. On a uniformly small patch this derivative stays bounded away from zero. The quantitative implicit-function theorem, applied to this sphere constraint in the uniformly $C^{2,1/2}$ cone coordinates, supplies link charts with a common radius and common $C^{2,1/2}$ bounds. Coordinate charts of the unit sphere have uniformly bounded smooth derivatives on these fixed radii, so they preserve those bounds. The link mean curvatures equal the cone mean curvatures at radius one by the principal-curvature relation above. Therefore the links $S_j = \partial E_j \cap S^{m-1}$ satisfy the hypotheses of Lemma 6.1 with $k = m-2$. They converge, with possibly unbounded multiplicity, to one compact embedded minimal link Σ. The normal graphs over this fixed Σ also have uniform $C^{2,1/2}$ bounds: choose a finite fixed smooth atlas of compact Σ and its normal tubular coordinates. In each chart the local graph bounds just obtained control the parameter maps in $C^{2,1/2}$. The asserted uniform C^2 convergence makes normal projection uniformly nonsingular. Its inverse parameter maps consequently have uniform $C^{2,1/2}$ bounds, by differentiating the inverse and using the common inverse-Jacobian bound; composition in the fixed tubular coordinates then gives the same bounds for every normal height function. A finite atlas makes these estimates uniform over all sheets. On any fixed conical annulus the map $(r, \omega) \mapsto r\omega$ and its coordinate inverses have bounded smooth derivatives. The coned normal graphs therefore retain these local uniform $C^{2,1/2}$ bounds.

On each fixed conical annulus the interfaces consequently form an ordered covering packet converging in C^2 to the classical cone $C(\Sigma)$. Here are fixed charts for the spacing assertion on a compact subannulus. Choose a larger compact subannulus where the coned covering packet is defined. Cover the test domain in the smooth embedded limit cone by finitely many inner graph patches centered at points P, and choose a common sufficiently small $\rho > 0$ for their larger neighborhoods. In the tangent-plane frame at each P, the limit patch is defined over the full disk $B^d_{4\rho}$, has height zero at its center and slope at most $1/200$. The closed ambient cylinder $\overline{Q}_{2\rho}$ centered at P lies in the

larger subannulus and is a positive distance from the portion of the limiting support outside this graph neighborhood. Reduce ρ further so that all these cylinders avoid zero, and take sufficiently many centers so that their inner graph pieces over $B^d_{\rho/4}$ cover the test domain.

Every local branch of the coned normal covering packet has horizontal projection uniformly close in C^2 to that of this limit graph. Its projection is therefore a local diffeomorphism with a common inverse-Jacobian bound; the margin between $B^d_{4\rho}$ and $B^d_{3\rho}$ gives each branch a full graph over $B^d_{3\rho}$. Differentiating the inverse parameter maps transfers the uniform $C^{2,1/2}$ bounds proved above to these tangent-plane graphs. In particular their Hessians are bounded by one common M_0 and their slopes are at most $1/100$ for all sufficiently large j. These estimates hold for every local branch, uniformly in the packet cardinality. All interfaces in $Q_{2\rho}$ are among these branches: the complete normal-packet description accounts for every interface in the larger subannulus, and uniform convergence together with the fixed separation just chosen excludes branches over any other part of the limiting support. Thus the entire boundary in this cylinder is the clipped union of the ordered graphs, including graphs which cross its caps. Genuine one-side phase occupancy is inherited from the regular cone phases.

On the inner disk $B^d_{\rho/4}$ the central limit graph has height at most $\rho/800$. Uniform convergence then puts every packet height there in $(-\rho/2, \rho/2)$ for sufficiently large j. Both points of every adjacent pair over this disk consequently lie in Q_ρ. Rescaling by ρ^{-1} gives exactly Q_2, full graph base B^d_3, slope bound $1/100$, and Hessian bound $M = \rho M_0$, together with the required Q_1 inclusion. Stationarity and unrestricted stability hold in these charts by 2.2 and 2.4, because they avoid zero. Since $p_j \to 1$, choose the tail also so that $p_j > p_*(d, M)$. Theorem 4.8 gives spacing at least $\rho c(d, M)\sqrt{\sigma_j}$ in the original coordinates on each inner chart. The coned packet is a complete packet in Euclidean normal coordinates as well. Indeed, write a spherical normal sheet over Σ as $\cos f(\theta)\,\theta + \sin f(\theta)\,\nu_\Sigma(\theta)$, where ν_Σ is a unit normal within the sphere. Its cone point at radial parameter $r > 0$ has the form

$$r\big(\cos f(\theta)\,\theta + \sin f(\theta)\,\nu_\Sigma(\theta)\big) = X + t\nu_\Sigma(\theta), \qquad X = (r\cos f(\theta))\theta, \qquad t = |X|\tan f(\theta).$$

The vector $\nu_\Sigma(\theta)$ is also the unit normal of $C(\Sigma)$ at X. On a common tail $|f| < \pi/4$, so $r \mapsto r\cos f(\theta)$ is a bijection of $(0, \infty)$ for every θ and every sheet. Thus the complete spherical packet gives the complete Euclidean normal packet after this radial reparametrization. On fixed cone annuli its normal heights $|X|\tan f(\theta)$ tend uniformly to zero in C^2, with uniform $C^{2,1/2}$ bounds, by the corresponding uniform spherical graph bounds and fixed annular coordinates.

We transfer the vertical lower bound to these normal-line coordinates. Take enough of the fixed centers so that the smaller limit graph pieces over $B^d_{\rho/8}$ still cover the compact test domain. The vertical spacing assertion above remains available on $B^d_{\rho/4}$. Put $\ell = 1/100$ and fix a base point $X = (X', X_m)$ in one of these smaller limit pieces. Choose its local unit normal $\nu = (\nu', \nu_m)$ with $\nu_m > 0$. The limit graph slope bound gives $\nu_m - \ell|\nu'| \geq 1/2$. For two successive packet points on its normal fiber, write

$$X_a = X + t_a\nu = (y_a, h_a(y_a)), \qquad X_b = X + t_b\nu = (y_b, h_b(y_b)), \qquad \delta = t_b - t_a > 0.$$

Here h_a, h_b are actual members of the complete tangent-plane graph family on $B^d_{3\rho}$. Uniform convergence of all normal heights to zero puts y_a, y_b in $B^d_{\rho/4}$ for all sufficiently large j, with a fixed positive horizontal margin independent of the chosen sheets and their number. Indeed $|X'| < \rho/8$ and $|y_a - X'|, |y_b - X'| \leq \max_i |t_i| \to 0$ uniformly. The height bounds already established on $B^d_{\rho/4}$ put $(y_a, h_a(y_a))$ and $(y_a, h_b(y_a))$ in Q_ρ. Since $y_b - y_a = \delta\nu'$ and both graph slopes are at most ℓ, their common-horizontal height difference satisfies

$$\Delta := h_b(y_a) - h_a(y_a) = \delta\nu_m + h_b(y_a) - h_b(y_b), \qquad 0 < \delta(\nu_m - \ell|\nu'|) \le \Delta \le \delta(\nu_m + \ell|\nu'|) \le (1+\ell)\delta.$$

In particular these are distinct vertical graphs, with h_b above h_a at y_a. They need not be consecutive in the vertical ordering. If other chart graphs lie between them, their heights at y_a lie between the two displayed endpoint heights and hence also in Q_ρ. The complete chart family contains every such intermediate interface. Summing the preceding adjacent vertical-gap bound between their indices, or using that bound directly if they are consecutive, gives $\Delta \ge \rho c(d,M)\sqrt{\sigma_j}$. Therefore

$$t_b - t_a = \delta \ge \frac{\rho c(d,M)}{1+\ell}\sqrt{\sigma_j}.$$

The finite fixed cover thus gives the normal-fiber separation required by 5.2, with one positive lower constant and one tail index independent of j and K_j on the compact test domain. For each compact test domain in such an annulus, choose its packet tube with closure in $U = \mathbb{R}^m \setminus \{0\}$. Every cone is stationary and stable in this U by 2.2 and 2.4, and the packet neighborhood is contained there. Lemma 5.2 therefore proves classical stability of $C(\Sigma)$ on every annulus. Every compactly supported test away from the vertex is contained in such an annulus, so the cone is stable away from its vertex. Lemma 6.2 shows that Σ is an equator.

Rotate this equator to $\{x_m = 0\}$. A small normal graph over its spherical link can be written as

$$\theta \longmapsto (\cos f(\theta)\,\theta, \sin f(\theta)), \qquad \theta \in S^{d-1}.$$

Its cone is the global graph $h(r\theta) = r\tan f(\theta)$. Set $h(0) = 0$. This is the continuous one-homogeneous extension, and the vertex lies in the boundary by closure. Strict ordering follows from monotonicity of tan on the common small normal-coordinate interval.

The ordered normal graphs over the equator hence give exactly the conical graph decomposition in the statement. Their uniform C^2 smallness and $C^{2,\alpha}$ bounds on fixed annuli follow from those of the link graphs and homogeneity. □

6.2. **A common plane and critical heights.**

Lemma 6.4 (Distance to a plane for a small conical graph)**.** *For each $d \ge 2$ there are $\varepsilon_d > 0$ and $C_d < \infty$ with the following property. Suppose h is one-homogeneous, C^2 away from zero, and*

$$\|h\|_{C^2(S^{d-1})} \le \varepsilon_d, \qquad \sup_{S^{d-1}} \left| \operatorname{div} \frac{\nabla h}{\sqrt{1+|\nabla h|^2}} \right| \le \varepsilon.$$

There is $a \in \mathbb{R}^d$ such that

$$\sup_{S^{d-1}} |h(x') - a\cdot x'| \le C_d\varepsilon.$$

The difference restricted to the sphere also satisfies a $W^{2,2d}$ bound by $C_d\varepsilon$.

Proof. The graph mean-curvature operator $\mathcal{M}$ is defined in 2.1. Let $a\cdot\theta$ be the $L^2(S^{d-1})$ projection of $f(\theta) = h(\theta)$ onto the first spherical harmonics, and write $h(x') = a\cdot x' + e(x')$, where e is one-homogeneous. Put $g = e|_{S^{d-1}}$ and $\mathcal{L} = \Delta_{S^{d-1}} + (d-1)$, and fix $q = 2d$. The eigenvalues of $-\Delta_{S^{d-1}}$ on harmonics of degree ℓ are $\ell(\ell+d-2)$, so the kernel of $\mathcal{L}$ consists precisely of the first spherical harmonics; g is orthogonal to this kernel. We first prove the following signed kernel-complement estimate:

$$\|g\|_{W^{2,q}(S^{d-1})} \le C_{d,q}\|\mathcal{L}g\|_{L^q(S^{d-1})}.$$

For completeness, the strong elliptic estimate with its lower-order term is

$$\|w\|_{W^{2,q}(S^{d-1})} \le C\big(\|\mathcal{L}w\|_{L^q(S^{d-1})} + \|w\|_{L^q(S^{d-1})}\big).$$

It holds for signed w: apply Gilbarg–Trudinger, *Elliptic Partial Differential Equations of Second Order*, Springer 2001, Theorem 9.11, in a finite atlas with fixed smooth cutoffs. The smooth metric gives uniformly elliptic smooth local coefficients. The commutators and the bounded zeroth-order term give only a first-derivative term and an L^q term. The interpolation inequality $\|\nabla w\|_{L^q} \le \delta\|w\|_{W^{2,q}} + C_\delta\|w\|_{L^q}$, with sufficiently small fixed δ, absorbs the former. This proves the displayed estimate on the fixed sphere without any positivity restriction. [12]

If its L^q term could not be removed on the kernel's L^2 orthogonal complement, choose w_j in that complement with $\|w_j\|_{W^{2,q}} = 1$ and $\|\mathcal{L}w_j\|_{L^q} \to 0$. Reflexivity and Rellich compactness on the fixed compact sphere give, after a subsequence, weak convergence in $W^{2,q}$ and strong convergence in L^q to a function w. The differential operator is continuous from $W^{2,q}$ to L^q, so $\mathcal{L}w = 0$ in distributions. Since $q = 2d > 2$ and the sphere has finite measure, the convergence is also strong in L^2 and the L^2 orthogonality passes to w. The spherical harmonic eigenvalue formula above applies to $w \in W^{2,q} \subset W^{2,2}$, showing that w belongs to the first-harmonic kernel. Orthogonality gives $w = 0$. But the displayed strong elliptic estimate applied to w_j would then give $1 \le C(\|\mathcal{L}w_j\|_{L^q} + \|w_j\|_{L^q}) \to 0$, a contradiction. This proves the kernel-complement inverse estimate.

Define

$$A(z) = \frac{I}{\sqrt{1+|z|^2}} - \frac{z \otimes z}{(1+|z|^2)^{3/2}}.$$

Because $D^2(a \cdot x') = 0$, the exact graph mean-curvature equation at radius one reads

$$\mathcal{M}(h) = A(\nabla h) : D^2 e = \mathcal{L}g + (A(\nabla h) - I) : D^2 e.$$

The tensor $D^2 e$ on the sphere is bounded in L^q by $C_d\|g\|_{W^{2,q}}$, while $\|A(\nabla h) - I\|_\infty \le C_d\varepsilon_d^2$. Indeed $D^2 e$ vanishes whenever one argument is radial, while for tangent vectors on the unit sphere its restriction is $\nabla^2_{S^{d-1}} g + g\, g_{S^{d-1}}$. Its trace is consequently $\Delta_{S^{d-1}} g + (d-1)g = \mathcal{L}g$.

Consequently the spherical inverse estimate displayed above gives

$$\|g\|_{W^{2,q}} \le C_{d,q}\varepsilon + C_{d,q}\varepsilon_d^2\|g\|_{W^{2,q}}.$$

Choose ε_d so that the last term is absorbed. Sobolev embedding proves the asserted best-plane sup norm estimate. This argument is the linearized spherical estimate underlying the best-plane comparison in (Chan–Dipierro–Serra–Valdinoci [6], Nonlocal approximation of minimal surfaces: optimal estimates from stability, arXiv:2308.06328v2 Lemma 4.8); the displayed calculation gives the estimate in precisely the form used here. □

Proposition 6.5 (One common frame and bounded multiplicity)**.** *Under the assumptions of Proposition 6.3, the following assertions hold for all sufficiently large j. One may choose a single rotation for each such j such that*

$$\max_i \sup_{|x'|=1} |h_{j,i}(x')| \le C_m\sqrt{\sigma_j}, \qquad \inf_{|x'|=1} (h_{j,i+1} - h_{j,i})(x') \ge c_m\sqrt{\sigma_j}.$$

In these coordinates $K_j \le C_m$ uniformly. In this same frame, on every fixed annulus the graph functions have uniform $C^{2,1/2}$ bounds, and the maximum of their C^2 norms tends to zero.

Proof. Use $\mathcal{M}$ and σ_j from 2.1 and the variation conventions of 2.2. The packet and its uniform annular graph bounds are those of 6.3, with genuine cone phases as in 2.4. We work on a tail of the furnished subsequence, so that all dimension-dependent order thresholds and graph-smallness conditions below hold. We first fix a dimensional mean-curvature bound on a larger annular neighborhood. Write

$$\varepsilon_j = \max_i \|h_{j,i}\|_{C^2(S^{d-1})} \longrightarrow 0.$$

For a one-homogeneous graph, $h(r\theta) = rh(\theta)$, $Dh(r\theta) = Dh(\theta)$ and $D^2h(r\theta) = r^{-1}D^2h(\theta)$. The tangential derivatives on the unit sphere give the Euclidean derivatives there by the one-homogeneous extension formula, so their norms are bounded by $C_d\|h\|_{C^2(S^{d-1})}$. Consequently, uniformly over the whole packet and $1/8 \le |x'| \le 4$,

$$|h_{j,i}(x')| \le \varepsilon_j|x'|, \qquad |Dh_{j,i}(x')| \le C_d\varepsilon_j, \qquad |D^2h_{j,i}(x')| \le 8C_d\varepsilon_j.$$

In graph coordinates the first fundamental form and, up to the choice of normal sign, the second fundamental form have coefficients

$$g_{ab} = \delta_{ab} + \partial_a h\,\partial_b h, \qquad b_{ab} = \frac{\partial_{ab}h}{\sqrt{1+|Dh|^2}}.$$

Since $g \ge I$, its inverse has operator norm at most one. The tensor norm therefore obeys $|A|_g^2 = g^{ac}g^{bd}b_{ab}b_{cd} \le |b|_{\mathrm{HS}}^2 \le |D^2h|_{\mathrm{HS}}^2$. In particular $|A| \le C_d|D^2h|$ for any of the equivalent fixed-coordinate Hessian norms. On a sufficiently late tail $\varepsilon_j \le 1$, and a graph point $X = (x', h_{j,i}(x'))$ satisfies

$$|x'| \le |X| \le \sqrt{2}\,|x'|.$$

Thus every boundary point in the larger ambient annulus $U = \{1/4 < |X| < 4\}$ has its base point in $1/8 < |x'| < 4$. The preceding Hessian estimate gives $|A_{\partial E_j}| \le C_d'\varepsilon_j \le 1$ throughout U for a sufficiently late tail. All graph points with $|x'| = 1$ lie in the fixed compact inner annulus $K = \{1/2 \le |X| \le 2\}$ on that tail.

For every $X \in K \cap \partial E_j$, $B_{1/8}(X) \Subset U$. The cone phase has its genuine regular boundary there and is stationary and stable there, since these balls avoid the vertex. Apply 5.3, after translating X to zero, with $C_0 = 1$, $R = 1/8$ and $r = 1/16$. The tail may be chosen so that p_j exceeds its fixed near-one threshold. Its constant $C(m, 1, 1/16, 1/8)$ is dimensional and gives $|H_{\partial E_j}(X)| \le C_m\sqrt{\sigma_j}$, uniformly at every graph point with $|x'| = 1$. The graph mean curvature $\mathcal{M}$ equals this signed geometric mean curvature up to orientation, so the same absolute bound holds on $|x'| = 1$.

The graph functions have uniformly small C^2 norms. Lemma 6.4 therefore gives

$$h_{j,i}(x') = a_{j,i}\cdot x' + e_{j,i}(x'), \qquad \sup_{|x'|=1} |e_{j,i}(x')| \le C\sqrt{\sigma_j}.$$

The slopes $a_{j,i}$ tend uniformly to zero. Since the graphs are globally ordered, evaluating $h_{j,i} - h_{j,1} \ge 0$ at both θ and $-\theta$ gives

$$|(a_{j,i} - a_{j,1})\cdot\theta| \le 2C\sqrt{\sigma_j} \quad (\theta \in S^{d-1}),$$

and hence $|a_{j,i} - a_{j,1}| \le 2C\sqrt{\sigma_j}$. All the graphs are thus within $C\sqrt{\sigma_j}|x'|$ of the single plane $x_m = a_{j,1}\cdot x'$.

By 2.11 rotations and the rescalings below preserve stationarity, stability and the genuine regular-boundary hypotheses. Choose a rotation sending the plane $x_m = a_{j,1}\cdot x'$ to the horizontal plane, whose new vertical coordinate measures signed distance with respect to the old-coordinate normal $(-a_{j,1}, 1)/\sqrt{1+|a_{j,1}|^2}$. On sheet i the new height is exactly

$$z_{j,i} = \frac{h_{j,i}(x') - a_{j,1}\cdot x'}{\sqrt{1+|a_{j,1}|^2}}.$$

The projected horizontal parameter map has the form $F_{j,i}(x') = M_j x' + b_j h_{j,i}(x')$, with M_j uniformly invertible. The perturbation is globally Lipschitz with norm smaller than the least singular

value of M_j for large j, since all graph slopes tend to zero. Hence $F_{j,i}$ is injective with a uniform lower Lipschitz bound; for every y' the equation $x' = M_j^{-1}(y' - b_j h_{j,i}(x'))$ is a contraction and has a solution. Thus $F_{j,i}$ is a global bi-Lipschitz bijection. It is one-homogeneous, so the resulting graph is one-homogeneous too, and $|F_{j,i}(x')|$ and $|x'|$ are uniformly comparable. Differentiating $F_{j,i}$ and its inverse on fixed annuli gives uniform $C^{2,\alpha}$ bounds for both parameter maps; the inverse derivatives are bounded because the smallest singular values stay uniformly positive. The new height is obtained by composing the displayed linear combination of $x \mapsto h_{j,i}(x)$ and $x \mapsto a_{j,1} \cdot x$ with this inverse. Since the old graph C^2 norms and $a_{j,1}$ tend uniformly to zero, the resulting graphs have C^2 norms tending uniformly to zero on fixed annuli, with a uniform $C^{2,\alpha}$ bound. The displayed height formula and the previous estimate prove the first bound in the two critical-height estimates in the statement in this single frame. The graphs are pairwise disjoint; on the connected punctured horizontal base their height order is constant. Relabeling once therefore gives an ordered family of global conical graphs in the new coordinates.

For the spacing assertion fix $\rho = 1/100$ and any $\theta \in S^{d-1}$. Use the cylinder $B^d_{2\rho}(\theta) \times (-2\rho, 2\rho)$ centered at $(\theta, 0)$, and rescale by ρ^{-1}. Its full required graph base $B^d_{3\rho}(\theta)$ lies in the fixed annulus $1 - 3\rho < |x'| < 1 + 3\rho$, where all translated graphs are C^2. This cylinder avoids the vertex since $|x'| \geq 1 - 2\rho > 0$, so the cone variations give stationarity and stability in it. The entire boundary there is the clipped union of the global family, and genuine phase separation is unchanged. For sufficiently large j the uniform C^2 smallness just proved gives slopes at most $1/100$ and $|D^2 h_{j,i}| \leq 1$ on every such base. The rescaled Hessian norm is at most ρ, a fixed allowed M. The critical height bound and homogeneity imply $|h_{j,i}(x')| \leq C_m \sqrt{\sigma_j} |x'|$, and therefore, again for sufficiently large j, every graph point above θ has $|h_{j,i}(\theta)| < \rho/2$. Both points of each adjacent pair above θ thus lie in the rescaled Q_1. Choose the tail also so that $p_j > p_*(d, \rho)$, the threshold of 4.8 with $M = \rho$. All its hypotheses hold with fixed d, M, ρ, and it gives an original-coordinate gap at least $\rho c(d, \rho)\sqrt{\sigma_j}$. The constants and tail index work uniformly for every θ, proving the second estimate. At any fixed $\theta \in S^{d-1}$, summing the consecutive gaps gives

$$c_m(K_j - 1)\sqrt{\sigma_j} \leq h_{j,K_j}(\theta) - h_{j,1}(\theta) \leq 2C_m\sqrt{\sigma_j}.$$

Thus K_j is bounded independently of j.

The best-plane constant is dimensional as well. The rotation and graph reparametrizations above have, on a sufficiently late tail, upper and inverse differential bounds by fixed numerical constants, because their slopes tend uniformly to zero. The resulting height bound therefore has a constant H_m depending only on m. The spacing constant is $c_m = \rho c(d, \rho) > 0$ with the fixed $\rho = 1/100$. The preceding gap sum gives $K_j \leq 1 + 2H_m/c_m$. Increase the displayed C_m, if necessary, to dominate both H_m and this last dimensional number. These choices are independent of the original sequence; only how far out its tail starts can depend on that sequence.

To obtain the conclusion for all sufficiently large indices of the original sequence, suppose instead that infinitely many indices admit no single rotated frame with the ordered global conical graph presentation and both stated estimates with these fixed C_m, c_m. Restrict the original sequence to those indices. It still satisfies all the hypotheses of the equatorial-compactness proposition cited at the start, including $p_j \uparrow 1$ and nontrivial regular stationary stable cone phases. Apply 6.3 to this restricted sequence and pass to its furnished subsequence. The argument just proved provides the required single frame and the dimensional height, gap and multiplicity bounds for every sufficiently late index of that subsequence. This contradicts its defining failure property. Hence only finitely many indices of the original sequence can fail, as claimed.

It remains to select these frames so that the annular smallness also holds along the full tail. Let $\mathcal{R}_j$ be the nonempty set of rotations admitting the ordered global graph presentation and the fixed dimensional height, gap and multiplicity bounds just proved. Define

$$b_j = \inf_{R\in\mathcal{R}_j} \max_i \|h^R_{j,i}\|_{C^2(S^{d-1})}.$$

If b_j did not tend to zero, some fixed $\varepsilon > 0$ and an infinite restricted sequence would satisfy $b_j \geq \varepsilon$. Equatorial compactness 6.3 and the common-plane construction above, applied to that restricted sequence, give rotations in $\mathcal{R}_j$ with graph C^2 norms tending to zero along a subsequence. This is a contradiction. Choose $R_j \in \mathcal{R}_j$ with graph norm at most $b_j + 1/j$. Homogeneity then gives uniform C^2 convergence to zero on every fixed annulus.

On a fixed larger annulus this smallness implies $|A| \leq 1$ for all sufficiently large j. With this fixed curvature bound and fixed annular margins, 5.3 gives tangent graph charts of uniform radius with uniform $C^{2,1/2}$ bounds, independently of the sheet number. The selected global graph slopes tend uniformly to zero. Vertical projection of each tangent chart consequently has a uniformly bounded inverse on a fixed smaller disk. Differentiating the inverse parameter map transfers its $C^{2,1/2}$ bounds to the selected vertical graph. A finite cover of a fixed base annulus, with centers and margins independent of the sheet, gives the uniform annular bound. Homogeneity extends it to every fixed annulus. Thus the selected frames satisfy all the conclusions simultaneously, with $\alpha = 1/2$. □

6.3. **Multiplicity one and cone rigidity.**

Lemma 6.6 (The homogeneous gap contradiction)**.** *For $3 \leq d \leq 6$ there is no positive one-homogeneous function $v \in C^1_{\mathrm{loc}}(\mathbb{R}^d \setminus \{0\})$ satisfying*

$$\Delta v \leq \frac{4}{v}, \qquad 4\int_{\mathbb{R}^d} \frac{\eta^2}{v^2} \leq \int_{\mathbb{R}^d} |\nabla\eta|^2 \quad (\eta \in C^\infty_c(\mathbb{R}^d \setminus \{0\})),$$

where the differential inequality is understood weakly.

Proof. Write $v(r\theta) = rV(\theta)$. The positive function $V \in C^1(S^{d-1})$ has a positive minimum. To obtain its sphere inequality from the distributional inequality for v, test with $\varphi(r\theta) = \alpha(r)\psi(\theta)$, where $\alpha \in C^\infty_c((0,\infty))$ is nonnegative and nonzero and $\psi \geq 0$ is smooth. Since $\nabla v = V\theta + \nabla_{S^{d-1}} V$, radial integration by parts and cancellation of $\int_0^\infty r^{d-2}\alpha(r)\,dr > 0$ give

$$-\int_{S^{d-1}} \nabla V\cdot\nabla\psi + (d-1)\int_{S^{d-1}} V\psi \leq 4\int_{S^{d-1}} \frac{\psi}{V}.$$

Approximate V^{-1} in C^1 by nonnegative smooth functions to use it as a test. This yields

$$(d-1)|S^{d-1}| + \int_{S^{d-1}} |\nabla \log V|^2 \leq 4\int_{S^{d-1}} V^{-2}.$$

For $\eta(x') = \xi(|x'|)$, with $0 \neq \xi \in C^\infty_c((0,\infty))$, stability yields

$$4\int_{S^{d-1}} V^{-2} \leq |S^{d-1}| \frac{\int_0^\infty r^{d-1}(\xi')^2\,dr}{\int_0^\infty r^{d-3}\xi^2\,dr}.$$

Put $b = (d-2)/2$ and $\xi(r) = r^{-b}\zeta(\log r)$. The quotient is exactly $b^2 + \int(\zeta')^2/\int\zeta^2$. Choose $0 \neq \zeta_0 \in C^\infty_c(\mathbb{R})$ and $L > 0$. The dilated logarithmic cutoffs $\zeta(t) = \zeta_0(t/L)$ have derivative quotient $L^{-2}\int(\zeta_0')^2/\int\zeta_0^2$, so the radial quotients approach b^2 as $L \to \infty$. Consequently

$$4\int_{S^{d-1}} V^{-2} \leq \frac{(d-2)^2}{4}|S^{d-1}|.$$

This contradicts the preceding lower bound, since $d - 1 > (d-2)^2/4$ for $3 \leq d \leq 6$. □

Lemma 6.7 (A stationary Lipschitz graphical cone is a plane)**.** *Let $h : \mathbb{R}^d \to \mathbb{R}$ be Lipschitz, one-homogeneous, and C^2 away from the origin. If its subgraph is stationary for the p-perimeter away from the origin, then h is linear.*

Proof. Use 2.1 for $0 < p < 1$, the subgraph B_h, graph point X_h, normalized curvature F_h and principal-value convention; stationarity is that of 2.2. The first-variation result 2.9 gives $F_h = 0$ at every graph point away from the vertex. The nonlinearity G_p is constructed in 3.1. Since h is globally Lipschitz and locally $C^{1,1}$ away from zero, 3.2 gives the exact zero-forcing equation there, namely

$$\mathrm{PV}\int_{\mathbb{R}^d} \frac{G_p((h(x)-h(y))/|x-y|)}{|x-y|^{d+p}}\,dy = 0, \qquad x \neq 0.$$

The equation also holds weakly: multiply each symmetrically truncated equation by a compact smooth test ψ supported away from zero, integrate, and interchange x, y. The limit is

$$\iint_{\mathbb{R}^d\times\mathbb{R}^d} G_p\Big(\frac{h(x)-h(y)}{|x-y|}\Big)\,\frac{\psi(x)-\psi(y)}{|x-y|^{d+p}}\,dx\,dy = 0.$$

This double integral is absolutely convergent: the bounded slope bounds G_p and the test difference is $O(|x-y|)$ at the diagonal, giving the radial majorant Cr^{-p}, while the tail is integrable for $p > 0$. Uniform local $C^{1,1}$ regularity removes the pointwise principal-value truncations on the test support. This is the zero-forcing weak graph equation in Fall's notation, up to an irrelevant factor two and a sign; his order is $t = (1+p)/2$.

Fall's fixed-order theorem (Regularity results for nonlocal equations and applications, arXiv:1806.09139v3, Theorem 1.5(ii), with weak equation (1.19)) makes a measurable graph locally smooth wherever it is locally Lipschitz and its forcing is smooth. It requires no exterior integrability. Here h is globally Lipschitz, $t \in (1/2, 1)$, the forcing is zero, and the preceding identity supplies the weak hypothesis on every ball avoiding zero. Applying that theorem at arbitrary derivative orders on nested balls therefore makes h smooth away from zero; its vertex and linear growth do not affect applicability. For a fixed unit vector e, the derivative $w = \partial_e h$ is bounded, smooth away from zero, and zero-homogeneous. Assign any value to $w(0)$; that single null point does not affect any integral below. The graph equation and a translation difference quotient give [10]

$$\mathrm{PV}\int_{\mathbb{R}^d} G_p'\Big(\frac{h(x')-h(y')}{|x'-y'|}\Big)\,\frac{w(x')-w(y')}{|x'-y'|^{d+1+p}}\,dy' = 0, \qquad x' \neq 0,$$

where $G_p'(t) = (1+t^2)^{-(d+1+p)/2} > 0$. For clarity, first subtract the graph equations at $x' + te$ and x', translate the integration variable in the first, and divide by t. The fundamental theorem of calculus gives the symmetric difference-quotient kernel

$$\frac{1}{|x'-y'|^{d+1+p}}\int_0^1 G_p'((1-s)q_0(x',y') + sq_t(x',y'))\,ds,$$

where $q_0 = (h(x')-h(y'))/|x'-y'|$ and $q_t = (h(x'+te)-h(y'+te))/|x'-y'|$. The difference quotients of h are globally bounded by its Lipschitz constant. On a small ball about $x' \neq 0$, pair the integration offsets z and $-z$. The leading graph difference quotients at these offsets have opposite signs. Since G_p' is even and smooth on the bounded slope interval, the two kernel coefficients differ by $O(|z|)$. The linear terms in the difference-quotient function thus cancel after pairing; their coefficient error and the Taylor remainders leave a numerator bounded by $C|z|^2$, uniformly for small t. After radial integration the bound is $C\int_0^a r^{-p}\,dr < \infty$. Dominated convergence applies outside that ball because the quotient functions are bounded and the kernel tail is integrable. In particular, the integration region near $y' = 0$ is separated from x' and creates no singular contribution. Letting $t \to 0$ gives the derivative Jacobi equation displayed above. This is the differentiation justified in (Farina–Valdinoci, Flatness results for nonlocal minimal cones and subgraphs, published Ann. SNS Pisa 19 (2019), 1281–1301, DOI:10.2422/2036-2145.201708_019; preprint arXiv:1706.05701 Section 3 and Appendix B), in the stationary graph setting. [11]

The continuous function w attains a maximum at a point of the unit sphere. Zero-homogeneity makes it a global maximum on the punctured space. At that point the integrand in the derivative Jacobi equation displayed above is nonnegative and the coefficient is strictly positive. The integral is convergent: near the maximum $w(x') - w(y') = O(|x' - y'|^2)$ and $1 + p < 2$, and the tail is controlled by boundedness. Its value zero forces w to be constant almost everywhere and then everywhere away from zero. This holds for each coordinate derivative. Thus $Dh = a$ on the punctured space for one constant vector a. On each connected component, $h(x) - a \cdot x$ is constant. Positive one-homogeneity makes each such constant zero, including when $d = 1$ and there are two components. Lipschitz continuity gives $h(0) = 0$, so $h(x) = a \cdot x$ globally. No equation at the vertex has been used. □

Proof of Theorem 1.3. Use 2.1 for $d = m - 1$, $\sigma_j = 1 - p_j$ and $\mathcal{M}$, 2.2 for unrestricted stationarity and stability, and 2.4 for regular cone phases and variations away from the vertex. If the conclusion failed for some fixed $m \in \{4, 5, 6, 7\}$, choose nonflat regular stationary stable cones E_j inductively with $p_j > \max\{p_{j-1}, 1 - 1/j\}$ and $p_0 = 0$. Thus $p_j \uparrow 1$. Propositions 6.3 and 6.5 describe them, in a single coordinate frame for each j, by globally ordered one-homogeneous graphs $h_{j,1}, \dots, h_{j,K_j}$. Their numbers are uniformly bounded; pass to a subsequence on which $K_j = K$ is constant.

The chosen frames also have annular C^2 smallness and uniform $C^{2,1/2}$ bounds by 6.5; these are the graph bounds used below.

If $K = 1$, the phase is the subgraph of $h_{j,1}$ or its complement, up to a null set. Indeed the complement of the graph has exactly two connected components. Away from its boundary the phase indicator is locally constant, hence constant on each component. The genuine interface condition forces opposite values on the two components; the graph itself has ambient measure zero. The graph is globally Lipschitz: its gradient is zero-homogeneous and bounded on the unit sphere by the annular C^2 bound; homogeneity gives $h(0) = 0$ and $|h(x)| \le C|x|$. Integrating the gradient bound on a line segment proves the Lipschitz estimate if the segment avoids zero, and splitting a segment at zero proves the same estimate in the remaining case. If E_j is its complement, 2.11 transfers stationarity to its subgraph. Lemma 6.7 therefore applies and makes each cone flat, a contradiction.

Suppose $K \ge 2$. Set $g_{j,i} = h_{j,i}/\sqrt{\sigma_j}$. By 6.5 and homogeneity, these functions are uniformly bounded on every compact annulus, and their consecutive gaps are at least $c|x'|$. They are also precompact in C^1 on smaller annuli. On a fixed compact annulus, apply 5.3 on a larger annulus, using the graph-curvature bounds just cited, to get $|\mathcal{M}(h_{j,i})| \le C\sqrt{\sigma_j}$. The exact divergence equation for the normalized graph is

$$\operatorname{div}(B_{j,i}\nabla g_{j,i}) = \frac{\mathcal{M}(h_{j,i})}{\sqrt{\sigma_j}}, \qquad B_{j,i} = (1 + |\nabla h_{j,i}|^2)^{-1/2} I.$$

The right side is uniformly bounded. The matrices are uniformly elliptic and have uniform Lipschitz bounds on the larger annulus, since the unnormalized graphs have uniform $C^{2,1/2}$ bounds. On each fixed larger ball avoiding zero, add a single uniform constant to all $g_{j,i}$ so that they are positive there; their common L^∞ bound permits this. The equation is unchanged. Proposition 2.13, with forcing equal to the negative of the displayed right side, gives uniform interior $W^{2,q}$ bounds for every fixed finite q. Subtracting the added constant preserves these bounds. A finite cover yields the corresponding bounds on the smaller annulus. Choosing $q > d$ and then $0 < \beta < 1 - d/q$ gives compactness in C^1 by Sobolev embedding and Arzelà–Ascoli. After a diagonal subsequence the limits g_i are one-homogeneous, $g_{i+1} - g_i \ge c|x'|$, and the convergence is local C^1 on $\mathbb{R}^d \setminus \{0\}$.

All hypotheses of Proposition 4.7 hold on that punctured space. Explicitly, take $V = \mathbb{R}^d \setminus \{0\}$, $V_j = \{1/j < |x'| < j\}$, and $T_j = j$. These cylinders avoid the vertex, so stationarity and stability hold there. The global ordered family gives the entire clipped boundary in each cylinder, with phase alternation at every genuine interface. The fixed number K, the local $C^{2,\alpha}$ bounds, the

normalized C^1 convergence, and the positive limiting gaps are exactly the remaining hypotheses of that proposition. For any adjacent pair, the proposition therefore gives

$$v = g_{i+1} - g_i > 0, \qquad \Delta v \le 4/v, \qquad 4\int v^{-2}\eta^2 \le \int |\nabla \eta|^2$$

for every test compactly supported away from the origin. Since $d = m - 1 \in \{3, 4, 5, 6\}$, Lemma 6.6 gives a contradiction. This excludes $K \ge 2$ and completes the proof. □

Corollary 6.8 (Regular stable cones near the local limit)**.** *For every $m \in \{4, 5, 6, 7\}$ there exists $p_m < 1$ such that every nontrivial regular stationary stable p-minimal cone (in the sense of 2.4 and 2.2) in $\mathbb{R}^m$ is a half-space whenever $p_m < p < 1$. Here regular means that the spherical link is an embedded C^2 hypersurface, equivalently the cone boundary is smooth away from the vertex.*

Proof. Theorem 1.3 supplies a threshold in $(0, 1)$ for the stationary stable cone class of Definitions 2.4 and 2.2. It identifies each phase with a half-space up to null sets. Under Definition 2.1, the density-one representative of an open half-space is that same half-space: its interior has density one, its exterior density zero, and its boundary density one half. Null-equivalent phases have identical volume fractions in every ball. Consequently their density-one representatives coincide, so the phase-level half-space conclusion has precisely the canonical form stated here. □

7. End profiles, directed blow-downs, and the eight-dimensional theorem

Regular stable-cone rigidity through ambient dimension seven first classifies every bounded stable solution in dimension seven. Applied to the end profiles of a monotone solution in dimension eight, this gives energy minimization. Directionality passes to minimizing blow-downs, and the classification of these directed cones provides the oriented phase trapping used to finish the equation argument.

7.1. Equation normalization and compactness.

Definition 7.1 (Fractional Allen–Cahn normalization and the eight-dimensional monotone equation)**.** Let $N \ge 1$ be an integer and $0 < p < 1$. Set $s = p/2$ and

$$a_{N,p} = \frac{p 2^{p-1}\Gamma((N+p)/2)}{\pi^{N/2}\Gamma(1-p/2)}, \qquad W(t) = \frac{1}{4}(1-t^2)^2.$$

For bounded C^2 functions define

$$(-\Delta)^{p/2}u(x) = a_{N,p}\,\mathrm{PV}\int_{\mathbb{R}^N} \frac{u(x)-u(y)}{|x-y|^{N+p}}\,dy.$$

The fractional Allen–Cahn equation is $(-\Delta)^{p/2}u = u - u^3$. A solution of the eight-dimensional monotone equation is a bounded classical solution in $\mathbb{R}^8$ with $\partial_{x_8}u > 0$ and $0 < s < 1/2$, where classical means the pointwise principal-value equation holds and interior derivatives used in it exist. For a solution in $\mathbb{R}^N$, stability means

$$\frac{a_{N,p}}{2}\iint_{\mathbb{R}^N\times\mathbb{R}^N} \frac{(\phi(x)-\phi(y))^2}{|x-y|^{N+p}}\,dx\,dy + \int_{\mathbb{R}^N}(3u^2-1)\phi^2\,dx \ge 0 \qquad (\phi \in C_c^\infty(\mathbb{R}^N)).$$

For bounded measurable functions u and bounded open U, define the local energy

$$\mathcal{E}_p(u;U) = \frac{a_{N,p}}{4}\iint_{(\mathbb{R}^N)^2\setminus(U^c)^2} \frac{(u(x)-u(y))^2}{|x-y|^{N+p}}\,dx\,dy + \int_U W(u)\,dx.$$

A global minimizer means $\mathcal{E}_p(u;U) \le \mathcal{E}_p(v;U)$ for every bounded open U and every bounded measurable competitor $v = u$ outside U; infinite energies are allowed with the usual extended-real convention. A one-dimensional increasing layer is a bounded solution $q : \mathbb{R} \to (-1, 1)$ of the one-dimensional equation with $q' > 0$ and $q(t) \to \pm 1$ at the respective ends. A phase blow-down

of u is a measurable set E, identified up to null sets, for which some sequence $R_j \to \infty$ has $u(R_j\cdot) \to \chi_E - \chi_{E^c}$ in L^1_{loc}.

Proof. For the operator just constructed, pair offsets z and $-z$ on a small ball about x. Bounded second derivatives there give $2u(x) - u(x+z) - u(x-z) = O(|z|^2)$. Its radial integral is bounded by $C\int_0^1 r^{1-p}\,dr$, finite for $p < 2$. The bounded tail has radial majorant $C\int_1^\infty r^{-1-p}\,dr$, finite for $p > 0$. Thus the pointwise principal value exists. The energy integrands are nonnegative measurable functions, so their extended-real integrals define the construction without ambiguity. For a bounded smooth u and bounded U, compact-neighborhood derivative bounds control the diagonal by the same square-difference radial estimate, and boundedness controls the tails. The energy of such a solution is consequently finite on each bounded U. The other terms are definitions of properties and require no existence assertion.

For later comparison with stability defined using compactly supported Lipschitz tests, the two test classes give the same condition. Such a test ϕ belongs to $H^1(\mathbb{R}^N)$. Its smooth mollifications ϕ_ε have supports in one fixed compact neighborhood and converge in H^1 and L^2. For any $w \in H^1$, the translation estimates $\|w(\cdot+z) - w\|_2 \le |z|\|\nabla w\|_2$ for $|z| < 1$ and $\|w(\cdot+z) - w\|_2 \le 2\|w\|_2$ for $|z| \ge 1$ give, by Tonelli,

$$\iint \frac{|w(x) - w(y)|^2}{|x-y|^{N+p}}\,dx\,dy \le |S^{N-1}|\left(\frac{\|\nabla w\|_2^2}{2-p} + \frac{4\|w\|_2^2}{p}\right).$$

Apply this to $w = \phi_\varepsilon - \phi$. The kinetic seminorm tends to zero, so the corresponding quadratic kinetic energies converge. Boundedness of u makes $3u^2 - 1$ bounded, and the potential terms converge by L^2 convergence. Passing to the limit extends the smooth-test inequality to every compactly supported Lipschitz test. The converse holds because a smooth compact test is Lipschitz. In particular, this verifies the test-class convention used in the cited stable-solution classification. □

Definition 7.2 (Relative flatness and oriented phase trapping)**.** Let $f : \mathbb{R}^N \to (-1,1)$. Its intermediate level sets have vanishing relative flatness, centered at the fixed origin, when every compact interval $I \subset (-1,1)$ admits a threshold $R_I > 0$ and widths $a_I : (R_I, \infty) \to [0,\infty)$, with $a_I(R) \to 0$, such that for every $R > R_I$ there is $e_{I,R} \in S^{N-1}$ with

$$\{x \in B_R : f(x) \in I\} \subset \{x : |e_{I,R}\cdot x| \le a_I(R)R\}.$$

For a fixed $0 < \kappa < 1$, oriented two-phase trapping means that there are $R_\kappa > 0$, a positive nonincreasing function $b_\kappa(R) \to 0$, and unit vectors $e_{\kappa,R}$, for all $R > R_\kappa$, such that inside B_R

$$\{e_{\kappa,R}\cdot x \le -b_\kappa(R)R\} \subset \{f \le -1+\kappa\} \subset \{f \le 1-\kappa\} \subset \{e_{\kappa,R}\cdot x \le b_\kappa(R)R\}.$$

These clauses define two properties; they do not require an existence result for functions satisfying them.

Proposition 7.3 (Fourier normalization, spatial rescaling and one-coordinate reduction)**.** *Use 7.1. With the unitary Fourier transform, the fractional operator has multiplier $|\xi|^p$, and its kinetic quadratic form is $\int |\xi|^p|\widehat{\phi}|^2$. If $\kappa > 0$ and $\lambda^p = a_{N,p}/\kappa$, then $v(x) = u(\lambda x)$ solves $\kappa I_p v + W'(v) = 0$ whenever $a_{N,p} I_p u + W'(u) = 0$, where I_p is the principal-value integral without its normalizing coefficient. This rescaling preserves stability, compact-perturbation minimization, strict directional monotonicity and vanishing relative flatness of level sets. In particular one may take $\kappa = 2 - p$. If $u(x',t) = v(x')$ with $N \ge 2$, then its equation reduces to the $(N-1)$-dimensional equation with coefficient $a_{N-1,p}$.*

Proof. Use 7.2 for the two precise flatness properties. The normalization of 7.1 is the standard Riesz fractional Laplacian. To identify the constant explicitly, the Gaussian Laplace identity

$$|z|^{-N-p} = \frac{1}{\Gamma((N+p)/2)}\int_0^\infty t^{(N+p)/2-1}e^{-t|z|^2}\,dt$$

and Tonelli give

$$\int_{\mathbb{R}^N} \frac{1-\cos(\xi\cdot z)}{|z|^{N+p}}\,dz = \frac{\pi^{N/2}}{\Gamma((N+p)/2)}\int_0^\infty t^{p/2-1}(1-e^{-|\xi|^2/(4t)})\,dt.$$

For $\xi \neq 0$, substitute $r = |\xi|^2/(4t)$ and integrate by parts to obtain

$$\int_0^\infty r^{-1-p/2}(1-e^{-r})\,dr = \frac{2\Gamma(1-p/2)}{p}.$$

The boundary terms vanish since $0 < p < 2$. Hence the cosine integral is $|\xi|^p/a_{N,p}$, also true at zero. Applying the paired operator to a Schwartz function and taking its Fourier transform gives the multiplier $a_{N,p}\int(1-\cos(\xi\cdot z))|z|^{-N-p}\,dz = |\xi|^p$. The distributional multiplier follows by testing against Schwartz functions. Plancherel and Tonelli applied to smooth compact tests yield

$$\frac{a_{N,p}}{2}\iint \frac{(\phi(x)-\phi(y))^2}{|x-y|^{N+p}}\,dx\,dy = \int |\xi|^p|\widehat{\phi}(\xi)|^2\,d\xi.$$

The change of variables $y = \lambda^{-1}z$ gives $I_p[u(\lambda\cdot)](x) = \lambda^p I_p u(\lambda x)$, including the principal value by the same pairing. For the energy and stability form with coefficient κ, the change of variables in both integration variables gives respectively $\mathcal{E}_\kappa(v;U) = \lambda^{-N}\mathcal{E}_{a_{N,p}}(u;\lambda U)$ and the same positive factor for the stability forms after transforming the test. Competitors correspond bijectively under this scaling. Monotonicity is multiplied by the positive factor λ. For the properties in 7.2, $v(x) = u(\lambda x)$ maps B_R bijectively to $B_{\lambda R}$ and preserves every function value. If a_I is the level-set width for u, take $a_I^v(R) = a_I(\lambda R)$ and $e_{I,R}^v = e_{I,\lambda R}$. The defining slab $|e\cdot(\lambda x)| \le a_I(\lambda R)\lambda R$ becomes $|e\cdot x| \le a_I^v(R)R$, with $R_I^v = R_I/\lambda$. The identical substitution $b_\kappa^v(R) = b_\kappa(\lambda R)$ transforms all four oriented phase inclusions; positivity, monotonicity and convergence to zero are preserved. Applying the inverse dilation proves preservation in both directions. Finally, for $r > 0$,

$$\int_{\mathbb{R}} (r^2+z^2)^{-(N+p)/2}\,dz = \frac{\sqrt{\pi}\Gamma((N+p-1)/2)}{\Gamma((N+p)/2)} r^{-(N-1)-p}.$$

Multiplication by $a_{N,p}$ gives exactly $a_{N-1,p}$. Bounded tails and paired local second-order differences justify integrating the equation in this way, or equivalently the Fourier multiplier restricted to the coordinate hyperplane is $|\xi'|^p$. □

Lemma 7.4 (Fixed-order local regularity and translated compactness)**.** *Fix $N \ge 1$, $0 < p < 1$ and $M < \infty$. Bounded classical solutions u of the equation in 7.1 with $\|u\|_\infty \le M$ have uniform local smoothness and bounded derivative estimates independent of the center. Their translates have subsequences converging locally in C^2 (indeed in every fixed finite differentiability order), and each limiting function is a bounded solution of the same equation. Stability passes to such local limits.*

Proof. Use the isotropic operator and cubic equation of 7.1. First a classical solution is distributional. Integrate a diagonally truncated equation against a compact smooth test and symmetrize. On a fixed neighborhood of its support, paired second differences of the classical solution and differences of the test give an integrable local majorant; outside that neighborhood boundedness gives the integrable tail. Letting the truncation decrease to zero supplies the weak equation required by the regularity theorem. The fixed-order regularity input is Cabré–Cinti–Serra [2], arXiv:2111.06285, Proposition C.1, which gives uniform local $C^{2,\alpha}$ estimates for bounded solutions when the nonlinearity is C^2 and the kernel has its stated isotropic ellipticity and derivative bounds. The kernel here is $a_{N,p}|z|^{-N-p}$; all its required derivative bounds follow by differentiation, and one may choose the lower order parameter in that proposition to be $p/2 < p$. For this fixed p all constants are finite and depend only on N, p, M and the prescribed nested balls. Translation invariance makes them independent of the center. If $N = 1$, lift u to $U(x,t) = u(x)$ in $\mathbb{R}^2$. The coordinate-integration identity in 7.3 shows that U solves the two-dimensional equation with its required coefficient. Apply Proposition C.1 in dimension two and restrict its estimate to $t = 0$; this gives the asserted

one-dimensional bounds as well. For higher orders use M. M. Fall [10], *Regularity results for nonlocal equations and applications*, arXiv:1806.09139v3, Theorem 1.4, with order $s = p/2$ and kernel $a_{N,p}|x-y|^{-N-p}$. This kernel belongs to every coefficient-regularity class required there: its normalized polar coefficient is a positive constant, and its global upper and local lower bounds hold with a fixed ellipticity constant for this N, p. Local C^2 regularity gives membership in $H^{p/2}$, and boundedness gives the required global bounded tail. The cubic composition inherits each local $C^{k,\beta}$ estimate of u. Theorem 1.4 then gains p derivatives on smaller balls; when an integer exponent is met, choose a slightly smaller positive Hölder exponent to avoid its excluded resonance. Iteration, with finitely many nested balls for each desired order, yields estimates in every finite differentiability order. All constants depend only on N, p, M and that order. Consequently derivatives are uniformly bounded on all translated unit balls. Arzelà–Ascoli and a diagonal extraction over balls and differentiability orders give the claimed compactness. To pass the equation, split the integral into a small ball, a fixed annulus and a tail. On the small ball the symmetric second-order difference is uniformly $O(|z|^2)$ and has integrable radial majorant Cr^{1-p}. On the fixed annulus local uniform convergence suffices; on the tail boundedness gives an error bounded uniformly by CR^{-p}. Dominated convergence followed by $R \to \infty$ proves the limiting equation. In the stability inequality each fixed compact test has the same kinetic term while its potential converges uniformly on the support; hence stability passes. No estimate uniform in p is asserted. □

7.2. **Stable solutions in dimension seven.**

Lemma 7.5 (Range and stability)**.** *Let $N \geq 1$ be an integer, $0 < p < 1$, and use 7.1. Let u be a bounded nonconstant solution of $(-\Delta)^{p/2}u = u - u^3$ in $\mathbb{R}^N$. Then $|u| < 1$. If in addition $\partial_e u > 0$ for some fixed direction e, then u is stable. In every dimension N, a constant solution of this equation is stable if and only if it equals 1 or -1.*

Proof. Use the equation and stability construction in 7.1 and the compactness result 7.4. For translates based at points approaching the supremum M, 7.4 gives a locally smooth subsequential limit U that solves the same equation and satisfies $U(0) = M = \sup U$. If $M > 1$, the integral defining $(-\Delta)^{p/2}U(0)$ is nonnegative whereas $M - M^3 < 0$, a contradiction. The corresponding argument at a translated minimum excludes $\inf u < -1$. Thus $-1 \leq u \leq 1$. If $u(x_0) = 1$, the equation at x_0 gives

$$0 = a_{N,p} \int_{\mathbb{R}^N} \frac{1 - u(y)}{|x_0 - y|^{N+p}}\, dy.$$

The integrand is nonnegative and the kernel positive, so $u = 1$ everywhere. The same argument applies to -1. Nonconstancy therefore yields $|u| < 1$.

If $w = \partial_e u > 0$, the global smoothness and derivative bounds from 7.4 justify differentiation. On a fixed small ball the differentiated numerator is bounded by $C|x-y|$, with radial majorant Cr^{-p}; outside that ball w is bounded and the tail has majorant Cr^{-1-p}. Thus differentiating the equation gives $(-\Delta)^{p/2}w + (3u^2 - 1)w = 0$. For $\phi \in C_c^\infty(\mathbb{R}^N)$, the function ϕ^2/w is a smooth compactly supported test function, since w has a positive minimum on its support. Testing the equation for w and symmetrizing shows that the left-hand side of the stability form in 7.1 equals

$$\frac{a_{N,p}}{2} \iint_{\mathbb{R}^N \times \mathbb{R}^N} \frac{w(x)w(y)}{|x-y|^{N+p}} \left(\frac{\phi(x)}{w(x)} - \frac{\phi(y)}{w(y)} \right)^2 dx\, dy \geq 0.$$

Extend ϕ/w by zero outside the compact support of ϕ. On a fixed compact neighborhood of that support, w has a positive minimum and bounded derivatives. The squared difference in the displayed identity is then $O(|x-y|^2)$, which is integrable at the diagonal. In the tail only one compactly supported factor can be nonzero; boundedness of w controls the remaining factor. These estimates justify the symmetrization and removal of the diagonal truncation. This proves stability.

Finally a constant solution c satisfies $c - c^3 = 0$, so $c \in \{-1, 0, 1\}$. The constants ± 1 are stable because their potential coefficient $3c^2 - 1$ is two and the kinetic form is nonnegative. To exclude zero, fix $0 \neq \phi \in C_c^\infty(\mathbb{R}^N)$ and set $\phi_R(x) = \phi(x/R)$. The Fourier normalization in 7.3 and a change of variables give its stability form at zero as

$$R^{N-p} \int_{\mathbb{R}^N} |\xi|^p |\widehat{\phi}(\xi)|^2 \, d\xi - R^N \int_{\mathbb{R}^N} \phi^2 \, dx.$$

It is negative for sufficiently large R, since $p > 0$. Hence the stable constant solutions are exactly the two wells. □

Theorem 7.6 (Stable-solution reduction, Cabré–Cinti–Serra)**.** *Fix $p \in (0, 1)$ and $n \geq 3$. Assume that for every $3 \leq m \leq n$, every nontrivial stable p-minimal cone in $\mathbb{R}^m \setminus \{0\}$ whose entire boundary is smooth away from the origin is a half-space. Then every stable solution $v : \mathbb{R}^n \to (-1, 1)$ of $(-\Delta)^{p/2} v = v - v^3$ is an increasing one-dimensional layer: $v(x) = q(e \cdot x)$ for a unit vector e, where $q' > 0$ and $q(t) \to \pm 1$ as $t \to \pm\infty$.*

Proof. Use 7.1 for the equation, layers, stability and energy, and 7.3 for rescaling. Use 2.1 for the phase perimeter and boundary convention, and 2.4 and 2.2 for the regular cone hypothesis. This is (Cabré–Cinti–Serra [2], Stable solutions to the fractional Allen–Cahn equation in the nonlocal perimeter regime, arXiv:2111.06285 Theorem 1.8 and Definition 1.6), with their operator order equal to p and with the spatial rescaling of 7.3. Their definition requires the regular-cone hypothesis in every ambient dimension $3 \leq m \leq n$, exactly as assumed here; no hypothesis on all singular stable cones is needed. Their theorem gives a one-dimensional increasing profile. We verify the strict derivative and well limits in the present layer convention.

By 7.4 the resulting profile q is smooth with bounded derivatives. Successive coordinate reduction in 7.3 gives its one-dimensional equation, and monotonicity gives $q' \geq 0$. Differentiating, justified by the bounded derivative estimates and $p < 1$, gives

$$(-\Delta_{\mathbb{R}})^{p/2} q' = (1 - 3q^2) q'.$$

If $q'(t_0) = 0$, its nonnegative minimum and this equation imply

$$0 = -a_{1,p} \int_{\mathbb{R}} \frac{q'(y)}{|t_0 - y|^{1+p}} \, dy.$$

The integral converges: near its minimum the smooth function q' is $O(|y - t_0|^2)$, and boundedness controls its tail. The positive kernel forces $q' \equiv 0$. Then the original stable solution would be a constant strictly between the wells, contradicting the constant classification in 7.5. Thus $q' > 0$ everywhere.

The two monotone end limits $\ell_- < \ell_+$ exist. Translating the stable solution $v(x) = q(e \cdot x)$ along te makes it converge locally uniformly to $\ell_\pm$ as $t \to \pm\infty$. By 7.4 each limit is a constant solution of the same n-dimensional equation and is stable. Lemma 7.5 makes these limits wells; their strict ordering gives $\ell_- = -1$ and $\ell_+ = 1$. □

Proof of Theorem 1.2. Put $p = 2s$ and use the equation, stability and layer conventions of 7.1. The Fourier identity in 7.3 identifies the kinetic stability form with $\int_{\mathbb{R}^7} |\xi|^p |\widehat{\phi}(\xi)|^2 \, d\xi$, where the Fourier transform is unitary.

Let $p_3 \in (0, 1)$ be the threshold from Cabré–Cinti–Serra [1], *Stable s-minimal cones in $\mathbb{R}^3$ are flat for $s \sim 1$*, arXiv:1710.08722, Theorem 1.2 and Definition 1.1. Its perimeter order is our p, and its stability tests are arbitrary smooth ambient flows supported away from the vertex. The perimeter of 2.1 is their localized perimeter, with each cross-phase interaction counted once. Let $p_4, p_5, p_6, p_7 \in (0, 1)$ be supplied by 1.3 and set

$$p_* = \max\{1/2, p_3, p_4, p_5, p_6, p_7\} < 1, \qquad s_0 = p_*/2 \in (0, 1/2).$$

For $p > p_*$, we check the cone hypothesis of 7.6 in every ambient dimension $3 \le m \le 7$. Take one of its nontrivial stable cones whose entire boundary is smooth away from zero, using the canonical phase representative of 2.1. In a small smooth graph chart, each connected side contains no boundary and hence is either entirely occupied or entirely unoccupied. Both sides cannot be occupied, since then each point on the graph has density one and is interior to the canonical phase. Both sides cannot be unoccupied, since the phase has measure zero in the chart and its density-one representative is empty there. Thus the boundary locally separates exactly the two phase sides. Dilation invariance places the radial vector in its tangent space, so the sphere meets it transversely. The spherical link is consequently an embedded smooth hypersurface and is compact by relative closedness of the boundary away from zero. This gives the geometric class of 2.4. We next compare the stability conventions before applying the cone classification.

For the stability comparison, take a regular cone $C \subset \mathbb{R}^m$, $m \ge 2$, in the sense of 2.4. Its boundary is genuine and C^2 on a neighborhood of the closed annulus $\overline{A}$, where $A = \{1/2 < |X| < 1\}$. Proposition 2.9 therefore gives $\mathrm{Per}_p(C; A) < \infty$. Put $A_k = 2^{-k}A$. Conicality and the change of variables in the perimeter of 2.1 give $\mathrm{Per}_p(C; A_k) = 2^{-k(m-p)} \mathrm{Per}_p(C; A)$. Up to the vertex and countably many spheres, which have zero ambient measure, these annuli cover B_1. The nonnegative localized integrand hence gives

$$\mathrm{Per}_p(C; B_1) \le \sum_{k=0}^{\infty} \mathrm{Per}_p(C; A_k) = \frac{\mathrm{Per}_p(C; A)}{1 - 2^{-(m-p)}} < \infty,$$

since $m - p > 0$.

Now let Ψ_t be a smooth flow supported in $K \Subset B_1 \setminus \{0\}$, and choose a bounded smooth W with $K \Subset W \Subset B_1 \setminus \{0\}$. It preserves both B_1 and W. Its uniform Jacobian and bi-Lipschitz bounds imply $\mathrm{Per}_p(\Psi_t C; B_1) \le C \mathrm{Per}_p(C; B_1) < \infty$. The full-ball and W-localized differences agree exactly: after pulling back the two phase variables, each localization mask is one wherever the transformed kernel minus the original kernel is nonzero, since at least one argument then lies in K. Both differences are therefore the same finite function $D(t)$ with $D(0) = 0$.

Suppose first that the cone satisfies the full-ball liminf stability condition in Cabré–Cinti–Serra [2], arXiv:2111.06285v1, Definition 2.8, or Definition 1.1 in the three-dimensional cone paper. For every autonomous smooth field supported in $B_1 \setminus \{0\}$ this condition is

$$\liminf_{t \to 0} \frac{D(t)}{t^2} \ge 0.$$

Proposition 2.9 gives $D(t) = tD'(0) + o(t)$. If $D'(0) \ne 0$, approaching zero with the opposite sign makes $D(t)/t^2 \to -\infty$, contradicting the two-sided liminf. Thus $D'(0) = 0$ for every such autonomous field. A positive dilation moves any compact support away from zero into $B_1 \setminus \{0\}$. Conjugating its flow, conicality and the change of variables in the perimeter of 2.1 multiply its localized perimeter difference by the positive factor λ^{m-p}, so the same vanishing holds for every compact autonomous field away from zero.

For a time-dependent compact smooth flow, choose the autonomous field equal to its initial velocity. The first-variation formula 2.9 depends only on that initial normal speed and gives the same first derivative. Hence every such derivative vanishes: the cone is stationary in the sense of 2.2.

We may now apply 2.10. Together with first variation it supplies the second-order Taylor expansion

$$D(t) = \tfrac{1}{2} D''(0) t^2 + o(t^2), \qquad D''(0) = 2Q_p[\phi].$$

For autonomous fields supported in $B_1 \setminus \{0\}$ the source's liminf therefore gives $D''(0) \ge 0$. Dilation extends this to every compact autonomous field away from zero. The same second-variation

proposition says that at a stationary phase the second derivative depends only on the initial normal speed. Thus every time-dependent compact flow has the same second derivative as its autonomous counterpart, and it is nonnegative. This is stability under all compact flows in 2.2.

Conversely, stationarity and stability in that definition, with the first and second variation propositions just cited, give the same Taylor expansion and

$$\lim_{t\to 0}\frac{D(t)}{t^2}=\tfrac{1}{2}D''(0)\geq 0.$$

This is exactly the source's full-ball liminf condition. The two stability conventions consequently agree for these regular cones; no second-variation formula was used before stationarity was established.

The three-dimensional result just cited covers $m=3$, while 1.3 covers $4\leq m\leq 7$.

If v is constant, its stability and 7.5 make it one of the two wells. If v is nonconstant, 7.5 gives $|v|<1$ throughout $\mathbb{R}^7$. Applying 7.6 with $n=7$ and the cone hypotheses established above gives a unit $e\in S^6$ and an increasing one-dimensional layer q_s with the two well limits such that $v(x)=q_s(e\cdot x)$. The theorem's displayed representation follows with $c=0$; translations of the layer allow the stated arbitrary-shift notation. The layer equation has the one-dimensional Fourier normalization by successive coordinate reduction in 7.3 after rotating e to a coordinate direction. Every threshold used above depends only on a fixed dimension and the fixed potential. No monotonicity or end-limit condition on v has been used. □

7.3. **End profiles and minimization.**

Lemma 7.7 (The translated limits are stable seven-dimensional solutions)**.** *Let u satisfy the eight-dimensional monotone-equation hypotheses of 7.1. Then*

$$v^{\pm}(x')=\lim_{t\to\pm\infty}u(x',t),\qquad x'\in\mathbb{R}^7,$$

exist, solve

$$(-\Delta_{\mathbb{R}^7})^{p/2}v=v-v^3,$$

and are stable in $\mathbb{R}^7$. Each is either 1, -1, or has strict range $(-1,1)$.

Proof. Use 7.1 for the monotone equation and stability, and 7.3 for Fourier kinetic energy and coordinate reduction. Monotonicity and boundedness give the pointwise limits. By 7.4, uniform interior estimates make every family of translates $u(x',t+t_j)$ precompact locally smoothly. For $t_j\to\pm\infty$ and each fixed t, also $t+t_j\to\pm\infty$; the pointwise limit therefore identifies every such subsequential limit as $V^{\pm}(x',t)=v^{\pm}(x')$. Uniqueness of this subsequential limit yields convergence of the whole translated family on every compact set. The nonlocal equation passes to the limit: local smoothness controls the principal-value part and the uniform bound controls the integrable tails.

The limiting function $V^{\pm}(x',t)=v^{\pm}(x')$ is independent of its last coordinate. The exact coordinate reduction in 7.3 therefore gives the stated equation with coefficient $a_{7,p}$.

The original u is stable by 7.5 with 7.1. First pass the stability inequality of the translates to $V^{\pm}$ with each fixed compactly supported test function. The kinetic term is unchanged and the potential term converges locally uniformly, so $V^{\pm}$ is stable in $\mathbb{R}^8$. Now take $\phi_R(x',t)=\zeta(x')\eta(t/R)$ and divide by $R\|\eta\|_2^2$, where $\zeta\in C_c^\infty(\mathbb{R}^7)$ and $0\neq\eta\in C_c^\infty(\mathbb{R})$. With the unitary Fourier transform, the resulting kinetic term is

$$\frac{1}{\|\eta\|_2^2}\int_{\mathbb{R}^7}\int_{\mathbb{R}}(|\xi|^2+|\tau|^2/R^2)^{p/2}|\widehat{\zeta}(\xi)|^2|\widehat{\eta}(\tau)|^2\,d\tau\,d\xi.$$

For $R \geq 1$ this is dominated by the integrable function obtained by replacing the multiplier by $|\xi|^p + |\tau|^p$. Dominated convergence gives the seven-dimensional kinetic form. The divided potential term is already $\int_{\mathbb{R}^7}(3(v^\pm)^2 - 1)\zeta^2$ because $V^\pm$ is independent of t. Hence $v^\pm$ is stable in $\mathbb{R}^7$. If an end profile is nonconstant, 7.5 gives strict range $(-1, 1)$. If it is constant, its seven-dimensional stability and the constant classification in that same lemma make it one of the two wells. This proves the asserted alternative. □

For the common near-half interval, Theorem 1.2 now classifies both stable end profiles as constant wells or increasing layers. The following criterion turns that classification into minimization.

Proposition 7.8 (Monotone end profiles imply minimization)**.** *Let $N \geq 2$ and let $u : \mathbb{R}^N \to (-1, 1)$ solve the fractional Allen–Cahn equation and satisfy $\partial_{x_N} u > 0$. If both limits $u^\pm(x') = \lim_{t\to\pm\infty} u(x', t)$ are either constant wells or one-dimensional increasing layers, then u is a global minimizer of the fractional Allen–Cahn energy under compact perturbations.*

Proof. Use 7.1 for the equation, layers and energy, and 7.3 for the operator and energy rescaling. Put

$$\lambda^p = \frac{a_{N,p}}{2-p}, \qquad U(x) = u(\lambda x).$$

Then U solves the equation with kernel coefficient $2 - p$, exactly as in Cabré–Cinti–Serra [2], *Stable solutions to the fractional Allen–Cahn equation in the nonlocal perimeter regime*, arXiv:2111.06285v1, Proposition 7.9. Its end profiles are $u^\pm(\lambda x')$. A well remains a well. An increasing one-dimensional end profile, lifted to be independent of the last coordinate, solves the full N-dimensional equation by the coordinate-integration identity in 7.3; rescaling makes its lift solve that same equation with coefficient $2 - p$. Thus the source's exact end-profile hypothesis holds, with strict vertical monotonicity and strict range. Proposition 7.9 gives minimization, and scaling back preserves compact-perturbation minimization for the energy with coefficient $a_{N,p}/4$.

The source's minimizer definition following its equation (1.10) allows arbitrary competitors with the prescribed exterior data. To cover every bounded open U_0 in 7.1, enclose $\overline{U_0}$ in a ball B. A bounded measurable competitor equal to u outside U_0 is admissible in B. If its U_0-energy is infinite there is nothing to prove. For a finite-energy competitor, first verify that u has finite energy in B. Let $M = \|u\|_\infty$ and let L_B bound $|\nabla u|$ on the closed unit neighborhood of B; 7.4 supplies this finite local bound. For $x \in B$, use $|u(x) - u(y)| \leq L_B|x - y|$ when $|x - y| < 1$ and the bound $2M$ otherwise. Symmetry and polar integration give

$$\iint_{(\mathbb{R}^N)^2\setminus(B^c)^2} \frac{|u(x) - u(y)|^2}{|x - y|^{N+p}}\, dx\, dy \leq 2|B||S^{N-1}|\left(\frac{L_B^2}{2-p} + \frac{4M^2}{p}\right) < \infty.$$

The potential is bounded on $[-M, M]$, so its integral over B is finite too. Therefore the energy terms in B not counted in U_0 are finite, since they equal the corresponding terms for u. These additional interactions involve two points outside U_0; the additional potential also lies outside U_0. They are identical for both functions. Canceling this finite common contribution from the ball inequality proves the required U_0 comparison. □

Corollary 7.9 (The eight-dimensional monotone solution is a minimizer)**.** *There exists $p_{\text{end}} < 1$ such that if $p_{\text{end}} < p < 1$ and u satisfies the eight-dimensional monotone-equation hypotheses of 7.1, then u is a global minimizer.*

Proof. Use 7.1 for the equation, layers and global minimization. Let $s_7 \in (0, 1/2)$ be a threshold supplied by 1.2 and put

$$p_{\text{end}} = 2s_7 \in (0, 1).$$

If $p_{\mathrm{end}} < p < 1$, then $s = p/2 \in (s_7, 1/2)$. By 7.7, the profiles

$$v^{\pm}(x') = \lim_{t \to \pm\infty} u(x', t)$$

are bounded classical stable solutions of the seven-dimensional equation at this exponent. The stable classification 1.2 therefore makes each profile a constant well or an increasing one-dimensional layer. Lemma 7.5 supplies $-1 < u < 1$, since strict vertical monotonicity makes u nonconstant. The hypotheses of 7.8 now hold: the equation is classical, the range is strict, vertical monotonicity is strict, and both end profiles are wells or increasing layers. Its conclusion gives global minimization for the energy of 7.1. The threshold depends only on the seven-dimensional classification. □

7.4. **Directed minimizing blow-downs.**

Theorem 7.10 (Near-local regularity in dimension eight, Caffarelli–Valdinoci)**.** *There exists $\varepsilon_8 > 0$ such that if $1 - \varepsilon_8 < p < 1$, every p-minimal set in $\mathbb{R}^8$ is locally $C^{1,\alpha}$, and hence smooth, outside at most countably many isolated singular points. In dimensions at most seven every nontrivial p-minimal cone is a half-space for p sufficiently close to one.*

Proof. Use 2.1 for the fractional phase perimeter and canonical boundary, with $0 < p < 1$, and 2.5 for minimizing phases and cones.

First we establish local finiteness of the perimeter. Fix a ball $B = B_R(a)$ in the ambient dimension m and use the competitor $F = E \setminus B$, which agrees with E outside B. The minimizing inequality of 2.5 and the nonnegative interactions in 2.1 give

$$\mathrm{Per}_p(E; B) \le \mathrm{Per}_p(F; B) = L_p(E \setminus B, B) \le L_p(B^c, B).$$

For $X \in B$, put $\delta(X) = R - |X - a| > 0$. Every $Y \notin B$ satisfies $|X - Y| \ge \delta(X)$, so polar integration gives

$$\begin{aligned} L_p(B, B^c) &\le \frac{|S^{m-1}|}{p} \int_B \delta(X)^{-p}\, dX \\ &= \frac{|S^{m-1}|^2}{p} \int_0^R r^{m-1}(R - r)^{-p}\, dr \le \frac{|S^{m-1}|^2 R^{m-p}}{p(1-p)} < \infty. \end{aligned}$$

Thus the ball comparison proves local finiteness. Every bounded open U lies in a ball B and has $\mathrm{Per}_p(E; U) \le \mathrm{Per}_p(E; B)$ by the nonnegative localized integrand. This verifies the finite-perimeter minimization hypothesis of the cited regularity theorems.

The cone assertion is restricted to nontrivial phases, meaning both phases have positive measure; trivial minimizing phases are excluded. In dimension one, conicality is literal for the canonical representative by its compatibility with dilations: each open half-axis is either entirely included or entirely excluded, since any two of its points differ by a positive dilation. Nontriviality therefore leaves exactly one half-axis, giving the one-dimensional assertion directly. The low-dimensional cone assertion is (Caffarelli–Valdinoci [4], Regularity properties of nonlocal minimal surfaces via limiting arguments, arXiv:1105.1158 Theorem 2); regularity of minimizing sets in dimensions at most seven is their Theorem 3; the dimension-eight statement with at most countably many isolated singular points is their Theorem 4. Their perimeter parameter s is the present p, rather than the PDE exponent $p/2$. Smoothness of regular nonlocal minimizing interfaces follows from the regularity bootstrap recorded in (Farina–Valdinoci [11], Flatness results for nonlocal minimal cones and subgraphs, published Ann. SNS Pisa 19 (2019), 1281–1301, DOI:10.2422/2036-2145.201708_019; preprint arXiv:1706.05701 Section 2); for the cones needed here, (Farina–Valdinoci, Flatness results for nonlocal minimal cones and subgraphs, published Ann. SNS Pisa 19 (2019), 1281–1301, DOI:10.2422/2036-2145.201708_019; published Corollary 2.2 (the corresponding statement follows Lemma 8 in arXiv:1706.05701)) directly gives smoothness outside the vertex once singular minimizing cones in $\mathbb{R}^7$ have been excluded. □

Theorem 7.11 (Conical subgraph rigidity, Farina–Valdinoci)**.** *Let $n \geq 2$, and let $E \subset \mathbb{R}^{n+1}$ be a p-minimal cone which is the subgraph of a function on $\mathbb{R}^n$. If that function is locally Lipschitz on $\mathbb{R}^n \setminus \{0\}$, then E is a half-space.*

Proof. Use 2.1 for the phase perimeter and canonical boundary, with $0 < p < 1$, and 2.5 for minimizing phases and cones. The localized perimeter has the same cross-phase interactions as the source; a positive normalization does not change minimizers. Local finiteness follows directly from the competitor $E \setminus B$ in any ball B: its interaction with B is bounded by $L_p(B, B^c) < \infty$, since $\int_B \operatorname{dist}(X, B^c)^{-p}\, dX < \infty$ for $p < 1$.

Write $E = \{x_{n+1} < h(x')\}$ up to null sets. For each fixed $\lambda > 0$, conicality and Fubini give $h(\lambda x') = \lambda h(x')$ for almost every x'. Both sides are continuous on the punctured base by local Lipschitz continuity, so this identity holds at every $x' \neq 0$. Set $h(0) = 0$; this changes only a null vertical fiber and gives an exactly dilation-invariant subgraph representative for the same minimizing phase.

Apply Farina–Valdinoci [11], *Flatness results for nonlocal minimal cones and subgraphs*, arXiv:1706.05701, Theorem 1, in its preprint formulation. That theorem assumes $n > 1$, a minimizing conical subgraph, and precisely local Lipschitz continuity of its graphing function on $\mathbb{R}^n \setminus \{0\}$. These are the stated hypotheses here, so it gives a half-space. This direct application uses the punctured formulation of the cited version; no global Lipschitz extension is required. □

Lemma 7.12 (Directed minimizing cones in $\mathbb{R}^8$ are flat near $p = 1$)**.** *There exists $p_{\mathrm{dir}} < 1$ such that the following holds for $p_{\mathrm{dir}} < p < 1$. Let $E \subset \mathbb{R}^8$ be a nontrivial cone globally minimizing the p-perimeter and satisfying*

$$E + te_8 \subset E \qquad (t > 0)$$

up to null sets. Then E is a half-space.

Proof. Choose $p_{\mathrm{dir}} < 1$ from Theorem 7.10 so that every nontrivial minimizing cone in $\mathbb{R}^7$ is a half-space for $p > p_{\mathrm{dir}}$. The absence of singular minimizing cones there allows Corollary 2.2 of Farina–Valdinoci [11] to give a smooth boundary for E away from zero.

Pass to the density-one representative from Definition 2.1. Almost-everywhere inclusion $A \subset B$ implies $A^{(1)} \subset B^{(1)}$, because it compares the volume fractions in every ball. Conicality and upward directedness are therefore exact for this representative. Away from zero, regularity makes both phase interiors open and gives density $1/2$ on the boundary. At zero, conicality makes the volume fraction equal to $|E \cap B_1|/|B_1|$ for every radius. Nontriviality places this fraction strictly between zero and one, so zero is on the boundary and is not in either density-one phase. Thus E is open globally.

Suppose a vertical pole belongs to ∂E. For $\tau > 0$ write $E_\tau = E + \tau e_8 \subset E$. If e_8 is a boundary pole, use $q = 2\tau e_8$; if $-e_8$ is a boundary pole, use $q = -\tau e_8$. In either case,

$$q \in \partial E \cap \partial E_\tau, \qquad q \neq 0, \qquad q - \tau e_8 \neq 0.$$

The two boundaries are smooth near q and tangent there by nesting. Their graph difference over the common tangent hyperplane is $O(|z|^2)$. Hence the volume of $E \setminus E_\tau$ in $B_r(q)$ is $O(r^9)$, and the singular integral of that difference is finite:

$$\int_{E \setminus E_\tau} |q - y|^{-8-p}\, dy < \infty, \qquad \sum_{k \geq 0} O\big((2^{-k})^{1-p}\big) < \infty.$$

The series controls the small dyadic annuli, and $p > 0$ controls the tail. Minimization in Definition 2.5 gives stationarity: both signs of every compact flow are admissible competitors. The first variation formula, Proposition 2.9, yields

$$H_p^A(q) = \mathrm{PV} \int_{\mathbb{R}^8} \frac{\chi_{A^c}(y) - \chi_A(y)}{|q - y|^{8+p}}\, dy = 0 \qquad (A = E, E_\tau).$$

The finite difference integral permits subtraction without ambiguity:

$$0 = H_p^{E_\tau}(q) - H_p^E(q) = 2\int_{E\setminus E_\tau} \frac{dy}{|q-y|^{8+p}}.$$

Thus $E = E_\tau$ almost everywhere by kernel positivity, and exactly after canonicalization. Dilation gives $E = E + te_8$ for every $t > 0$, hence for all real t. Translating the boundary point zero shows that e_8 is a boundary point. Smoothness there and translation invariance then give smoothness at zero. A cone with a C^1 boundary at zero equals its tangent half-space: every rescaling equals the cone, while the local tangent-plane description makes those rescalings converge in measure to that half-space.

Now suppose neither pole lies on the boundary. The lower pole cannot belong to E: a ball $B_\delta(-e_8) \subset E$ would dilate to $B_{T\delta}(-Te_8) \subset E$ for all $T > 0$. Given (x', t), take T with $|x'| < T\delta$ and $t > -T$. Then $(x', -T) \in E$, and directedness places (x', t) in E, contradicting nontriviality. Complementing the argument, using $E^c - te_8 \subset E^c$, excludes an upper pole in the exterior. We therefore have balls

$$B_\delta(e_8) \subset E, \qquad B_\delta(-e_8) \subset E^c.$$

Their dilates show that each vertical fiber contains sufficiently high points and excludes sufficiently low ones. Since E is open and upward closed, every fiber is an open upper ray with a finite endpoint. Write

$$E = \{(x', x_8) : x_8 > h(x')\}$$

for its finite endpoint function. For every real a, the set $\{h < a\} = \{x' : (x', a) \in E\}$ is open, so h is Borel measurable. Its graph is null by Fubini, as each vertical section is a singleton. The complementary phase therefore agrees almost everywhere with the subgraph $\{x_8 < h(x')\}$ and remains minimizing under complementation. The measurable-subgraph theorem of Farina–Valdinoci [11, Theorem 1.3], using the already established absence of singular minimizing cones in $\mathbb{R}^7$, makes that subgraph a half-space. The same conclusion follows for E. □

Lemma 7.13 (Every blow-down of the monotone minimizer is a half-space)**.** *Let u solve the eight-dimensional monotone equation of 7.1 with $\max\{p_{\mathrm{end}}, p_{\mathrm{dir}}\} < p < 1$, where the thresholds are those in 7.9 and 7.12. Then every phase blow-down of u is a half-space.*

Proof. Use 7.1 for phase blow-downs and energy minimization, 2.5 for minimizing phases, and 2.1 for the perimeter. Take an arbitrary existing phase blow-down E and its defining sequence $R_j \to \infty$, so that

$$u(R_j\,\cdot) \longrightarrow \chi_E - \chi_{E^c} \quad \text{in } L^1_{\mathrm{loc}}(\mathbb{R}^8).$$

Pass to an increasing subsequence of R_j; the defining convergence is preserved. The solution is stable with strict range by 7.5. To apply Cabré–Cinti–Serra [2], arXiv:2111.06285v1, Proposition 6.1, use 7.3 and set

$$\lambda^p = \frac{a_{8,p}}{2-p}, \qquad v(x) = u(\lambda x), \qquad r_j = R_j/\lambda.$$

The rescaled solution has their kernel coefficient $2-p$, is stable, and has strict range. Proposition 6.1 applied to the prescribed scales r_j gives a further subsequence whose limit is a nontrivial cone. The identity $v(r_j x) = u(R_j x)$ and uniqueness of the already specified L^1_{loc} limit identify that cone with E up to null sets. This verifies conicality and nontriviality for the particular blow-down being considered.

By 7.9 the original u is a global energy minimizer. In Dipierro–Serra–Valdinoci [9], arXiv:1611.10105v3, equations (1.9)–(1.10), take their order to be p and their kinetic constant $C_{8,p} = a_{8,p}$. Their displayed energy is exactly $4\mathcal{E}_p$ of 7.1. Multiplication by four preserves every minimizing inequality. Hence their Lemma 8.1 applies to u at these same prescribed scales,

with $\varepsilon_j = 1/R_j$. Passing to a further subsequence gives a phase minimizing the p-perimeter in every ball, with finite perimeter there. Boundedness converts its almost-everywhere convergence to local L^1 convergence, so uniqueness again identifies the phase with E.

The characteristic-function kinetic energy in that source is $2a_{8,p}\,\mathrm{Per}_p(E;B)$, since its symmetric integral counts each cross-phase interaction twice. This positive factor does not change minimization. To obtain the full bounded-open-set convention of 2.5, let $U \Subset B$ and let $F = E$ outside U. If $\mathrm{Per}_p(F;U) = \infty$ its comparison is automatic. Otherwise all interactions counted in B but not in U involve two points outside U and are identical for E, F. Their common sum is finite because E has finite perimeter in B. Canceling it proves the minimizing inequality in U. Every bounded open set can be enclosed in such a ball, so E is globally minimizing in the required sense.

For each fixed $t > 0$, vertical monotonicity gives $u(R_j(x + te_8)) \geq u(R_j x)$. Local L^1 convergence also holds on the translated compact sets, so passing to the limit gives $\chi_E(x + te_8) \geq \chi_E(x)$ almost everywhere. Equivalently, $E + te_8 \subset E$ up to null sets. Lemma 7.12 now applies and makes E a half-space. Since we began with an arbitrary phase blow-down, the conclusion holds for every one of them. □

7.5. **Oriented trapping and the symmetry theorem.**

Theorem 7.14 (Improvement of flatness for nonlocal phase transitions, Dipierro–Serra–Valdinoci)**.** *Let $0 < p < 1$ and let $u : \mathbb{R}^N \to (-1, 1)$ solve $(-\Delta)^{p/2}u = u - u^3$. Fix $0 < \kappa < 1 - 1/\sqrt{3}$. Suppose that there are $R_0 > 1$ and a function $a : (R_0, \infty) \to (0, 1]$ decreasing to zero such that, for every $R > R_0$, some $e_R \in S^{N-1}$ satisfies*

$$\{e_R \cdot x \leq -a(R)R\} \subset \{u \leq -1 + \kappa\} \subset \{u \leq 1 - \kappa\} \subset \{e_R \cdot x \leq a(R)R\} \qquad \text{in } B_R.$$

Then u is one-dimensional.

Proof. We verify the hypotheses of Theorem 1.2 in the arXiv:1611.10105v3 version of Dipierro–Serra–Valdinoci [9]. Their operator is

$$Lv(x) = \int (v(x) - v(x + y))\|y\|_K^{-N-p}\, dy.$$

Take $K = \overline{B}_{1/2}$, so $\|y\|_K = 2|y|$ and its kernel coefficient is $c_K = 2^{-N-p}$. By Proposition 7.3, $v(x) = u(\lambda x)$ with $\lambda^p = a_{N,p}/c_K$ solves their equation. The body K is even, convex and contained in B_1. For every $z \in \partial K$, the ball of radius $1/4$ centered at $z/2$ lies in K and is tangent at z. This checks their (H1) with $r_K = 1/4$.

For the cubic $f(t) = t - t^3$ in Definition 7.1, $f(\pm 1) = 0$ and, on the two well neighborhoods,

$$f'(t) = 1 - 3t^2 \leq 1 - 3(1 - \kappa)^2 < 0.$$

The choice $c_\kappa = (3(1-\kappa)^2 - 1)/2 > 0$ gives the strict bound in their (H2). Their (H3), an increasing one-dimensional heteroclinic, holds for the quartic double well, as recorded immediately after that condition in the cited paper.

The scaling preserves the required oriented trapping exactly. Put $a_v(R) = a(\lambda R)$ and $e_R^v = e_{\lambda R}$. The dilation maps B_R to $B_{\lambda R}$, preserves all solution values and divides each slab width by λ. Thus the four inclusions have relative width $a_v(R)$ for v. This function remains positive and nonincreasing and tends to zero. The theorem gives one-dimensionality of v, and scaling back gives one-dimensionality of u. □

Lemma 7.15 (Planar blow-downs imply asymptotically flat level sets)**.** *Under the hypotheses of Lemma 7.13, u satisfies the oriented two-phase trapping property in Theorem 7.14, for every $0 < \kappa < 1 - 1/\sqrt{3}$. In particular, its zero level set and every fixed intermediate level set are asymptotically flat.*

Proof. Fix a well-neighborhood parameter $0 < \kappa < 1 - 1/\sqrt{3}$. For $R > 1$, let $\alpha(R)$ be the infimum of relative widths $a \in [0, 1]$ for which the oriented inclusions of Theorem 7.14 hold in B_R with some unit normal. Width one is admissible: in the open ball its two outer half-spaces are respectively empty and the whole ball. We prove $\alpha(R) \to 0$ by ruling out a prescribed sequence of bad scales.

Suppose $\alpha(R_j) \geq \varepsilon > 0$ for $R_j \to \infty$, and take an increasing subsequence. Lemma 7.5 gives stability and strict range for the equation in Definition 7.1. Proposition 6.1 and Theorem 2.10 of Cabré–Cinti–Serra [2] apply in dimension eight, at fixed $p \in (0, 1)$, with the quartic potential. Any normalization change is a fixed dilation $v(x) = u(\lambda x)$ from Proposition 7.3. Use the prescribed scales $r_j = R_j/\lambda$ in their results; then $v(r_j x) = u(R_j x)$, so the resulting convergence concerns exactly the bad sequence in the original coordinates.

Proposition 6.1 yields a nontrivial conical phase limit E in L^1_{loc}. Lemma 7.13 makes E a half-space. If $E = \{e \cdot x > c\}$, dilation invariance compares thresholds c and λc for every $\lambda > 0$. Distinct thresholds differ on a positive-measure slab, so $c = 0$. Pass to the further subsequence of Theorem 2.10. Its cone agrees with E almost everywhere in B_1 by uniqueness of the L^1 limit; their common conicality extends the agreement to every ball. Its Hausdorff convergence on B_2 consequently gives convergence of $\{u(R_j \cdot) \geq -1 + \kappa\} \cap B_2$ to $E \cap B_2$. Apply the same result to $-u$, which satisfies the odd cubic equation and has the same stability form and strict range. After another subsequence, its phase limit is the complementary phase by uniqueness in B_1 and dilation. Thus $\{u(R_j \cdot) \leq 1 - \kappa\} \cap B_2$ converges to $E^c \cap B_2$. The density-based representative in that theorem has, for a half-space, exactly the plane $e \cdot x = 0$ as its boundary. Null changes of the phase cannot change this conclusion.

With Hausdorff error less than $\delta/2$ on B_2, the corresponding one-sided inclusions on B_1 are

$$\{u(R_j x) \geq -1 + \kappa\} \cap B_1 \subset \{e \cdot x > -\delta\}, \qquad \{u(R_j x) \leq 1 - \kappa\} \cap B_1 \subset \{e \cdot x < \delta\}.$$

Indeed the first set lies within that error of points with $e \cdot x \geq 0$, and the second within that error of points with $e \cdot x \leq 0$. Taking complements and returning to scale R_j gives the oriented well trapping at relative width δ. Choosing $0 < \delta < \varepsilon$ contradicts the bad-scale condition.

For sufficiently large R, choose the positive monotone envelope

$$a(R) = \sup_{\rho \geq R} \alpha(\rho) + \frac{1}{R}.$$

It tends to zero and is at most one after enlarging R_0. Since $a(R) > \alpha(R)$, there is an admissible width below $a(R)$; widening its slab retains all inclusions. This is the oriented trapping of Definition 7.2 and the exact input for Theorem 7.14. It confines all levels in $[-1 + \kappa, 1 - \kappa]$ to that slab. For a general compact interval $I = [a, b] \subset (-1, 1)$, repeat the argument with $0 < \kappa' < \min\{1 - 1/\sqrt{3}, 1 + a, 1 - b\}$. Its slab contains every level in I. This proves the full relative-flatness assertion, including each fixed intermediate level and the zero level. □

Proof of Theorem 1.1. Use 7.2 for the flatness property and 7.1 for the exact bounded monotone equation. By 7.5 the solution is stable and has strict range. By 7.4 its translated limits have the stated regularity. Let $p_{\mathrm{end}} < 1$ be a threshold supplied by 7.9, and let $p_{\mathrm{dir}} < 1$ be the threshold of 7.12. Set

$$p_0 = \max\{1/2, p_{\mathrm{end}}, p_{\mathrm{dir}}\} < 1, \qquad s_0 = \frac{p_0}{2} < \frac{1}{2}.$$

Take $s_0 < s < 1/2$, so $p = 2s > p_0$. By Corollary 7.9, u is a global minimizer. Lemmas 7.13 and 7.15 show that its level sets are asymptotically flat. Theorem 7.14 gives

$$u(x) = q_s(e \cdot x - c)$$

for a unit vector e. Differentiating and using $\partial_{x_8} u > 0$ gives $q_s' e_8 > 0$. After replacing $(e, c, q_s(t))$ by $(-e, -c, q_s(-t))$ if necessary, we have $e_8 > 0$ and $q_s' > 0$.

Strict monotonicity and boundedness give ordered limits $\ell_- < \ell_+$ for q_s. Since $e_8 > 0$, the vertical end profiles are the constants $v^{\pm} = \ell_{\pm}$. Lemma 7.7 makes both constants stable solutions of the seven-dimensional equation, and 7.5 classifies each as a well. Their ordering therefore gives $\ell_- = -1$ and $\ell_+ = 1$. The reduced one-dimensional equation has the coefficient in 7.1 by successive coordinate reduction in 7.3, after a rotation of the unit direction. The thresholds involve only the fixed dimension and cubic potential. The choice $p_0 \geq 1/2$ gives $s_0 > 0$. This proves the theorem. □

References

[1] X. Cabré, E. Cinti, and J. Serra, *Stable s-minimal cones in $\mathbb{R}^3$ are flat for $s \sim 1$*, J. Reine Angew. Math. **764** (2020), 157–180, https://arxiv.org/abs/1710.08722.

[2] X. Cabré, E. Cinti, and J. Serra, *Stable solutions to the fractional Allen–Cahn equation in the nonlocal perimeter regime*, Amer. J. Math. **147** (2025), 957–1024, https://arxiv.org/abs/2111.06285.

[3] L. Caffarelli, J.-M. Roquejoffre, and O. Savin, *Nonlocal minimal surfaces*, Comm. Pure Appl. Math. **63** (2010), 1111–1144, https://arxiv.org/abs/0905.1183.

[4] L. Caffarelli and E. Valdinoci, *Regularity properties of nonlocal minimal surfaces via limiting arguments*, Adv. Math. **248** (2013), 843–871, https://arxiv.org/abs/1105.1158.

[5] G. Catino, P. Mastrolia, and A. Roncoroni, *Two rigidity results for stable minimal hypersurfaces*, Geom. Funct. Anal. **34** (2024), 1–18, https://doi.org/10.1007/s00039-024-00662-1.

[6] H. Chan, S. Dipierro, J. Serra, and E. Valdinoci, *Nonlocal approximation of minimal surfaces: optimal estimates from stability*, Comm. Pure Appl. Math. (2026), Early View, article e70063, https://doi.org/10.1002/cpa.70063; also arXiv:2308.06328v2, https://arxiv.org/abs/2308.06328v2.

[7] J. Dávila, M. del Pino, and J. Wei, *Nonlocal s-minimal surfaces and Lawson cones*, J. Differential Geom. **109** (2018), 111–175, https://arxiv.org/abs/1402.4173.

[8] M. del Pino, M. Kowalczyk, and J. Wei, *On De Giorgi's conjecture in dimension $N \geq 9$*, Ann. of Math. (2) **174** (2011), 1485–1569.

[9] S. Dipierro, J. Serra, and E. Valdinoci, *Improvement of flatness for nonlocal phase transitions*, Amer. J. Math. **142** (2020), 1083–1160, https://arxiv.org/abs/1611.10105.

[10] M. M. Fall, *Regularity results for nonlocal equations and applications*, Calc. Var. Partial Differential Equations **59** (2020), article 181, https://arxiv.org/abs/1806.09139.

[11] A. Farina and E. Valdinoci, *Flatness results for nonlocal minimal cones and subgraphs*, Ann. Sc. Norm. Super. Pisa Cl. Sci. (5) **19** (2019), no. 4, 1281–1301, https://doi.org/10.2422/2036-2145.201708_019; also arXiv:1706.05701, https://arxiv.org/abs/1706.05701.

[12] D. Gilbarg and N. S. Trudinger, *Elliptic Partial Differential Equations of Second Order*, Classics in Mathematics, Springer, Berlin, 2001.

[13] H. Hong, H. Li, and G. Wang, *The stable Bernstein theorem in $\mathbb{R}^7$*, preprint, arXiv:2609.15720v1, 14 September 2026, https://arxiv.org/abs/2609.15720v1.

[14] W. H. Meeks III, J. Pérez, and A. Ros, *Limit leaves of an H lamination are stable*, J. Differential Geom. **84** (2010), 179–189, https://arxiv.org/abs/0801.4345.

[15] O. Savin, *Regularity of flat level sets in phase transitions*, Ann. of Math. (2) **169** (2009), 41–78.

[16] J. Simons, *Minimal varieties in Riemannian manifolds*, Ann. of Math. (2) **88** (1968), 62–105.

[17] M. Zhao and collaborators, *Stable nonlocal cones and the fractional De Giorgi conjecture in dimension four below one half*, research manuscript, 7 October 2026 (candidate proof under verification), GitHub preprint (PDF).